# Critical Invariant Polygons and the Farey–Ito Boundary of Stochastic Spectra

Boundary Contacts, First Returns, and the Karpelevič–Ito Theorem

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**Abstract**

Let $T$ be an orientation-preserving real-linear contraction of an oriented plane with nonreal eigenvalues, and let its polygonal complexity be the least number of vertices of a nondegenerate convex polygon $P$ satisfying $TP \subseteq P$. We first develop an intrinsic contact theory for radially critical maps. For polygonal complexity $N \geq 4$, contact on every side and at every image vertex, a half-open side-contact assignment, exact vertex replacements, a finite cyclic reduction, and the unimodular lattice sail lead to a first-return normal form. A projective argument then proves that, when some cyclic-shift orbit contributes more than one relative-interior contact, the first return reaches the next such contact in cyclic order. In an adapted complex coordinate, the resulting monodromy yields a heterogeneous Ito product and an exact lifted phase identity carried by Farey neighbors. Together with a direct treatment of orders at most three, we then apply this structure to determine, for every $n$, the union $\Theta_n$ of the spectra of all real row-stochastic $n \times n$ matrices. A strictly convex log-sine potential equalizes the monodromy factors and gives the sharp radial equation. Sparse stochastic matrices attain every equality point, while scalar Farey refinement proves nesting from order $n-1$ to order $n$. Together with filling each radial segment by convex combinations with the identity matrix and the unit-circle analysis, this establishes the full Farey–Ito description of the Karpelevič region from critical invariant polygons.

**Keywords.** critical invariant polygons; elliptic contractions; vertex replacement; projective holonomy; Farey sequences; stochastic matrices; eigenvalue regions; Karpelevič–Ito theorem.

## Contents

# Part I: Critical Invariant Polygons and Farey Return Normal Forms

## 1 The intrinsic theorem

Let $V$ be a real plane. For every real-linear map $A : V \to V$, define its polygonal complexity by

$$\nu_{\mathrm{poly}}(A) = \min\Big\{|\mathrm{Ext}\, P| : P \subset V \text{ is a nondegenerate compact convex polygon and } AP \subseteq P\Big\}, \tag{1.1}$$

with the value $\infty$ when the class is empty. An invertible map $T : V \to V$ is *elliptic* if

$$(\operatorname{tr} T)^2 < 4 \det T,$$

and it is an *elliptic contraction* when, in addition, its spectral radius belongs to $(0, 1)$. Defining (1.1) for all linear maps is essential: the radial comparison maps $tT$ in definition 1.1 need not remain contractions.

**Definition 1.1** (Radial polygonal criticality)**.** For $N \geq 3$, an elliptic contraction $T$ is $N$-*critical* if

$$\nu_{\text{poly}}(T) = N, \qquad \nu_{\text{poly}}(tT) > N \quad \text{for every } t > 1. \tag{1.2}$$

The second inequality allows the value $\infty$.

Both conditions are invariant under real-linear conjugacy. They are therefore properties of the planar dynamics, not of a chosen Euclidean metric, complex coordinate, polygon, or vertex numbering.

We write $\operatorname{Ext} K$ for the set of extreme points of a compact convex set $K$, $\partial K$ for its boundary, $\operatorname{relint} F$ for relative interior, and $\operatorname{aff} S$ for affine hull.

**Definition 1.2** (Strict polygon)**.** A planar convex polygon is called *strict* when it has nonempty interior and its displayed cyclic vertex list contains every point of $\operatorname{Ext} P$ exactly once and contains no other points. Here "strict" is custom polygonal terminology and does not mean strictly convex in the usual convex-geometric sense. Thus adjacent sides are not collinear, every side is a maximal boundary segment, and the interior of the normal cone at every vertex is nonempty. A supporting line through a vertex is *strict* when its contact face is that vertex alone.

For a strict polygon $P$ with an oriented boundary, let $\mathcal{E}(P)$ be its cyclic set of sides. For $e \in \mathcal{E}(P)$ write $\operatorname{tail}(e)$ and $\operatorname{head}(e)$ for its tail and head, let $\operatorname{succ}(e)$ be the next side, and put

$$e^{\triangleright} = (\operatorname{tail}(e), \operatorname{head}(e)].$$

A *half-open contact bijection* for $TP \subseteq P$ is a cyclic-order-preserving bijection

$$\chi : \operatorname{Ext} P \longrightarrow \mathcal{E}(P)$$

such that $Tv \in \chi(v)^{\triangleright}$ for every vertex $v$. The sides whose assigned image points lie in their relative interiors form the set $I$, and the induced cyclic shift of the side labels is denoted by $\sigma$:

$$I = \Big\{ e \in \mathcal{E}(P) : T\chi^{-1}(e) \in \operatorname{relint} e \Big\}, \qquad \sigma(e) = \chi(\operatorname{head}(e)). \tag{1.3}$$

Both maps $e \mapsto \operatorname{head}(e)$ and $\chi$ preserve cyclic order. Their composition $\sigma$ is therefore an automorphism of the finite cyclic order; after identifying $\mathcal{E}(P)$ with $\mathbb{Z}/N\mathbb{Z}$, it is a translation. When $e \in I$ and $\operatorname{succ}(e) \notin I$, the permitted vertex replacement at $e$ replaces $\operatorname{head}(e)$ by the image point $T\chi^{-1}(e) \in \operatorname{relint} e$. Whenever this replacement produces a strict polygon $P'$, put $v' = T\chi^{-1}(e)$ and let

$$b_e : \mathcal{E}(P) \longrightarrow \mathcal{E}(P')$$

be the *label-preserving bijection between the old and new side sets* defined by

$$b_e(f) = \begin{cases} [\operatorname{tail}(e), v'], & f = e, \\ [v', \operatorname{head}(\operatorname{succ}(e))], & f = \operatorname{succ}(e), \\ f, & f \notin \{e, \operatorname{succ}(e)\}, \end{cases}$$

where a side in the last line is identified with the geometrically unchanged boundary segment of $P'$. This bijection preserves cyclic order. If $\operatorname{succ}'$ is the successor map on $\mathcal{E}(P')$, then $b_e \circ \operatorname{succ} = \operatorname{succ}' \circ b_e$.

**Theorem 1.3** (Contact and first-return structure of an $N$-critical invariant polygon)**.** *Let $N \geq 4$ and let $T$ be an $N$-critical elliptic contraction. One of the two orientations of $V$ admits a strict invariant $N$-gon $P$ and a half-open contact bijection with the following properties.*

(i) Contact on every side and at every image vertex. *Every strict polygon $Q$ with at most $N$ vertices and $TQ \subseteq Q$ has exactly $N$ vertices; every side of $Q$ intersects $TQ$, and every vertex of $TQ$ lies on $\partial Q$.*

(ii) Equivariant vertex-replacement update. *Every permitted vertex replacement at $e$ produces another strict invariant $N$-gon $P'$. Let $b = b_e$ be the label-preserving bijection between the old and new side sets, and let $I'$ be the set of sides with relative-interior contact and $\sigma'$ the induced cyclic permutation of the primed side labels. Then*

$$\begin{aligned} I' &= b\big((I \setminus \{e\}) \cup \{\sigma(e)\}\big), \\ \sigma' &= b \circ \sigma \circ b^{-1}. \end{aligned} \tag{1.4}$$

(iii) Interval reduction. *Among the finitely many sets of sides with relative-interior contact realized by finite permitted vertex-replacement sequences, there is a representative for which $I$ is a nonempty cyclic interval. If $\delta$ is the number of $\sigma$-orbits and $\varphi = |I|$, then $\varphi \geq \delta$.*

(iv) First-return cases. *If $\sigma$ is the identity, every contact lies in the relative interior of its assigned side. Assume $\sigma$ is not the identity. If $\varphi = \delta$, then $I$ contains exactly one side from each $\sigma$-orbit (equivalently, it is a complete orbit transversal), and every base has return time $N/\delta$. If $\varphi > \delta$, the first-return map of $\sigma$ to $I$ is the cyclic successor on $I$. There exist integers $q > 0$ and $h \geq 0$ such that every first-return orbit segment has length $q$ or $q + h$, and these orbit segments form a disjoint partition of the $N$ vertex labels.*

*All assertions are invariant under invertible real-linear conjugacy.*

The orientation asserted in the theorem is the one in which the right-half-open side assignment holds; it selects an intermediate eigenvalue for the first-return analysis. The later complex monodromy theorem may select the conjugate eigenvalue after reflecting the completed closed return-recurrence chain.

The restriction $N \geq 4$ is used in the proper-boundary-chain and common-argument-interval steps; no assertion about the $N = 3$ case is made here.

These first-return cases are the substantive conclusion. Before the projective argument is used, the first-return map is an arbitrary positive cyclic shift of $I$. The theorem says that, when some orbit contributes more than one relative-interior contact, the first return reaches the cyclic successor.

For a nonzero complex number $z$ that is not positive real, write

$$\arg_+(z) \in (0, 2\pi)$$

for its positive lifted argument. The orientation used to construct the right-half-open contact assignment is an intermediate choice. The complex monodromy theorem is allowed to use either that orientation or the opposite orientation, and it makes this choice only after the return case is known. The exact reflection of the final closed return-recurrence chain is proved in lemmas 8.2 and 8.3; no vertex-replacement or cyclic-successor argument is transported across that reflection.

**Theorem 1.4** (Complex monodromy and Farey product data)**.** *Let $N \geq 4$ and let $T$ be an $N$-critical elliptic contraction. There is an adapted complex coordinate in one of the two orientations of $V$, hence one of the two conjugate eigenvalues $\mu$ of $T$, such that*

$$Tz = \mu z, \qquad 0 < |\mu| < 1.$$

*Let $\vartheta = \arg_+(\mu)$ and put $y = \vartheta/(2\pi)$. There are consecutive reduced fractions*

$$\frac{p}{q} < y < \frac{r}{s}$$

*in the Farey sequence $F_N$ whose ordered denominators satisfy $q \le s$. Put*

$$d = \lfloor N/q \rfloor, \qquad e = s - dq \in \mathbb{Z}.$$

*Then there are parameters*

$$0 \le \beta_j < 1, \qquad \alpha_j = 1 - \beta_j > 0 \qquad (1 \le j \le d) \tag{1.5}$$

*for which the following polynomial identity holds:*

$$\mu^s \prod_{j=1}^{d} (\mu^q - \beta_j) = \mu^{dq} \prod_{j=1}^{d} \alpha_j. \tag{1.6}$$

*Since $\mu \neq 0$, this is equivalent to the Laurent identity*

$$\mu^e \prod_{j=1}^{d} (\mu^q - \beta_j) = \prod_{j=1}^{d} \alpha_j. \tag{1.7}$$

*The signed closing exponent $e$ is not asserted to be nonnegative. In the reflected one-contact-per-orbit case it is negative; in that case (1.6), rather than the Laurent notation (1.7), is the primary identity.*

*Put*

$$A = q\vartheta - 2\pi p, \qquad M = \arg_+(\mu^q - 1).$$

*Then $0 < A < M < \pi$, and for each $j$ there is a unique $u_j \in [A, M)$ such that*

$$e^{iu_j} = \frac{\mu^q - \beta_j}{|\mu^q - \beta_j|}. \tag{1.8}$$

*These arguments satisfy the phase identity*

$$e\vartheta + \sum_{j=1}^{d} u_j = 2\pi(r - dp). \tag{1.9}$$

*The factors with $\beta_j = 0$ that are introduced only to complete a closed return-recurrence chain are zero-factor padding and are not claimed to be additional relative-interior contacts.*

The stochastic statement is deferred to section 9. Its proof will consist only of identifying $N$-criticality through the invariant-polygon criterion and applying theorem 1.4.

### 1.1 Relation with the stochastic eigenvalue literature

The final application belongs to the classical study of eigenvalue regions of stochastic matrices initiated by Dmitriev–Dynkin and Karpelevič; see [1, 4]. Farey descriptions and explicit matrix realizations appear in several later forms [2, 3, 5, 6]. The result proved here is organized in the opposite direction: the parent statement is intrinsic to a critical planar linear map, while stochastic matrices enter only through the invariant-polygon criterion in section 9.

### 1.2 Logical architecture

The proof is divided into five modules with one-way dependencies:

$$\boxed{\text{criticality}} \Longrightarrow \boxed{\text{contact on every side and at every image vertex}}$$

$$\Longrightarrow \boxed{\text{half-open contacts and vertex replacements}} \Longrightarrow \boxed{\text{first returns of a cyclic permutation}}$$

$$\Longrightarrow \boxed{\begin{array}{c}\text{adjacent first return when an orbit}\\ \text{contains multiple contacts, for } N\geq 4\end{array}}.$$

The Farey product is then the scalar monodromy of the final closed return-recurrence chain. The finite-rotation module is proved through consecutive lattice-sail vectors; it does not use a parity-indexed Euclidean remainder chain. The projective module contains no modular arithmetic. Detailed determinant and endpoint audits occur only inside the vertex-replacement theorem and the boundary-chain deformation where they are logically necessary.

For review purposes, each module exports one invariant-level contract:

| module | input | exported conclusion |
|---|---|---|
| contact on every side and at every image vertex | $N$-criticality | every invariant polygon with at most $N$ vertices has all sides and all image vertices in the stated boundary contact |
| half-open contact assignment | the preceding contact theorem | a cyclic half-open contact bijection and an exact, geometrically realized vertex-replacement update |
| first returns of a cyclic permutation | one reduced strict interval | a bijective two-height tower partition with unimodular return data |
| projective adjacent-return proof | $N \geq 4$, tower partition, exact boundary-edge classification, and validity after each vertex replacement | a globally defined invariant deformation with one escaped image vertex; hence, when an orbit contains multiple contacts, the first return is the adjacent successor |
| complex monodromy | closed recurrence along adjacent returns | the Farey product data, heterogeneous product, and lifted phase identity |

No later module uses the internal indices of an earlier proof; it uses only the theorem displayed in the last column. This is the stability mechanism of the argument.

### 1.3 Foundational tools

The proof uses only finite-dimensional linear algebra, elementary planar convexity, and the index formula for a rank-two sublattice of $\mathbb{Z}^2$. For completeness, the precise forms of Perron–Frobenius, strict separation, polarity, Hausdorff continuity of polygonal area, and the lattice parallelogram count used below are proved in appendix A. Thus every nonstandard dependency of the main argument is available inside the paper. We use in their standard forms Cayley–Hamilton and Jordan normal form, elementary compactness and subsequence arguments, dominated convergence, Smith normal form for rank-two integer lattices, and elementary real projective geometry.

## 2 Elliptic coordinates and convex preliminaries

**Proposition 2.1** (Adapted complex structures)**.** *Let $T$ be elliptic. Then* $\det T > 0$. *Put*

$$\rho = \sqrt{\det T}, \qquad \cos\theta = \frac{\operatorname{tr} T}{2\rho}, \qquad 0 < \theta < \pi,$$

*and define*

$$J_+ = \frac{\rho^{-1}T - \cos\theta\, I}{\sin\theta}, \qquad J_- = -J_+. \tag{2.1}$$

*Then $J_+^2 = J_-^2 = -I$, and the two complex structures induce the two opposite orientations of $V$. There is an inner product for which both are isometries and*

$$T = \rho(\cos\theta\, I + \sin\theta\, J_+) = \rho(\cos\theta\, I - \sin\theta\, J_-). \tag{2.2}$$

*Thus $T$ is multiplication by $\rho e^{i\theta}$ in the $J_+$ complex coordinate and by $\rho e^{-i\theta}$ in the $J_-$ coordinate. In the adapted inner product,*

$$T^* = \rho(\cos\theta\, I - \sin\theta\, J_+),$$

*and $T^*$ is conjugate to $T$ by an orientation-reversing reflection.*

*Proof.* The characteristic polynomial of $\rho^{-1}T$ is

$$X^2 - 2\cos\theta\, X + 1.$$

Cayley–Hamilton gives

$$(\rho^{-1}T - \cos\theta I)^2 = -\sin^2\theta I,$$

so $J_+^2 = -I$, and the same is true of $J_- = -J_+$. A complex structure on a real plane determines an orientation by declaring $(v, Jv)$ positive; replacing $J$ by $-J$ reverses that orientation. Hence $J_+$ and $J_-$ induce precisely the two orientations of $V$.

Starting from any inner product $g_0$, set

$$g(x, y) = g_0(x, y) + g_0(J_+x, J_+y).$$

Then $g(J_+x, J_+y) = g(x, y)$, so $J_+$, and therefore also $J_-$, is orthogonal. Formula (2.2) follows directly from (2.1); its two complex interpretations follow from the rule that multiplication by $a + ib$ acts as $aI + bJ$. Since $J_+^* = -J_+$, the formula for $T^*$ follows. In an orthonormal $J_+$-adapted basis, reflection in the first coordinate satisfies $SJ_+S^{-1} = -J_+$, and consequently $STS^{-1} = T^*$. □

**Proposition 2.2** (Real-linear invariance of polygonal complexity)**.** *For every invertible real-linear map $A$,*

$$\nu_{\mathrm{poly}}(ATA^{-1}) = \nu_{\mathrm{poly}}(T).$$

*Consequently $N$-criticality is invariant under conjugacy, and $\nu_{\mathrm{poly}}(tT^*) = \nu_{\mathrm{poly}}(tT)$ for every $t > 0$ in the adapted inner product.*

*Proof.* The correspondence $P \mapsto AP$ preserves convexity, nondegeneracy, the number of extreme points, and inclusion. Moreover

$$TP \subseteq P \iff (ATA^{-1})(AP) \subseteq AP.$$

The first assertion follows. The second uses the reflection conjugacy in proposition 2.1. □

**Proposition 2.3** (Real-linear covariance of contact geometry)**.** *Let $A : V \to V$ be invertible, put*

$$\widetilde{T} = ATA^{-1}, \qquad \widetilde{P} = AP,$$

*and suppose that $P$ is a strict polygon. Then the following statements hold.*

(i) *$A$ maps* $\operatorname{Ext} P$ *bijectively onto* $\operatorname{Ext} \widetilde{P}$*, maps every side of $P$ onto a side of $\widetilde{P}$, and preserves relative interiors and proper inclusions.*

(ii) *Every face of a polygon is exposed by a nonzero linear functional. For every face $F$ of $P$, the image $AF$ is the corresponding face of $\widetilde{P}$. In particular, side contacts, vertex contacts, strict support contacts, and the assertions "every side is touched" and "every image vertex lies on the boundary" are transported exactly by $A$.*

(iii) *If $TP \subseteq P$, then $\widetilde{T}\widetilde{P} \subseteq \widetilde{P}$. If a polygon $P'$ is obtained from $P$ by clipping with a closed half-plane, replacing a vertex by a contact point, or performing a finite sequence of such operations, then $AP'$ is obtained from $AP$ by the corresponding transformed operations. All vertex counts, incidence equalities, strictness conditions, and invariance statements are equivalent before and after the transformation.*

(iv) *Give $\widetilde{P}$ the boundary orientation transported by $A$. If $\chi : \operatorname{Ext} P \to \mathcal{E}(P)$ is a half-open contact bijection, then*

$$\widetilde{\chi}(Av) = A\chi(v)$$

*is a half-open contact bijection for $\widetilde{T}\widetilde{P} \subseteq \widetilde{P}$. If $A_{\mathcal{E}} : \mathcal{E}(P) \to \mathcal{E}(\widetilde{P})$ is the induced side bijection, then*

$$\widetilde{I} = A_{\mathcal{E}}(I), \qquad \widetilde{s} = A_{\mathcal{E}} s A_{\mathcal{E}}^{-1}, \qquad \widetilde{\sigma} = A_{\mathcal{E}} \sigma A_{\mathcal{E}}^{-1}. \tag{2.3}$$

*If $A$ reverses the originally displayed ambient orientation, then the transported orientation on the target is the opposite displayed orientation. Re-expressing the target in the original displayed orientation reverses cyclic order and may change which half-open side an endpoint is assigned to; this orientation and endpoint-assignment issue is recorded in lemma 2.4.*

*Proof.* An invertible real-linear map is a homeomorphism that maps line segments to line segments and preserves convex combinations. A point $x \in P$ is extreme precisely when an equality $x = tu + (1-t)v$ with $u, v \in P$ and $0 < t < 1$ forces $u = v = x$; applying $A$ and $A^{-1}$ proves the first assertion about extreme points. Maximal boundary segments, relative interiors, and proper inclusions are therefore preserved as well.

Every face of a polygon is exposed. If a nonzero functional $\ell$ supports $P$ on the face $F = \{x \in P : \ell(x) = \max_P \ell\}$, then $\widetilde{\ell} = \ell \circ A^{-1}$ supports $AP$ on

$$AF = \{y \in AP : \widetilde{\ell}(y) = \max_{AP} \widetilde{\ell}\}.$$

Thus all face and strict-support assertions are equivalent. The identity

$$\widetilde{T}(AP) = A(TP)$$

proves the invariance and image-contact claims. Finally, $A(P \cap H) = AP \cap AH$ for every half-plane $H$, and replacing a point $x$ by a point $c$ in a displayed convex hull commutes with applying $A$. Iteration proves the statement for finite sequences of these clipping and vertex-replacement operations.

With the transported orientation, $A$ preserves tail, head, cyclic successor, and every right-half-open side:

$$A\big((\operatorname{tail}(e), \operatorname{head}(e)]\big) = (A\operatorname{tail}(e), A\operatorname{head}(e)].$$

Thus $\widetilde{\chi}$ satisfies the required half-open-side membership condition, and the set $I$ and cyclic shift $\sigma$ satisfy (2.3). The condition for the permitted vertex replacement is expressed only through membership in $I$ and the successor map, so it is preserved. The final orientation statement follows because the sign of every oriented area is multiplied by $\det A$. □

**Consequences used in later topics.** Any later construction that uses only $(\mathcal{E}, I, s, \sigma)$ and is unchanged by relabelling is transported by $A_{\mathcal{E}}$. Each such construction is defined and checked when it is introduced.

Whenever an adapted complex coordinate is fixed, we identify $V$ with $\mathbb{C}$ and use

$$\langle z, w\rangle = \operatorname{Re}(\overline{z}w), \qquad \det(z, w) = \operatorname{Im}(\overline{z}w). \tag{2.4}$$

The determinant is positive on positively oriented ordered bases. For a nonempty compact convex set $K$, its support function is

$$h_K(u) = \max_{z\in K}\langle u, z\rangle. \tag{2.5}$$

Until a consistent assignment to half-open sides is selected, we use the $J_+$ coordinate and write

$$Tz = \lambda z, \qquad \lambda = \rho e^{i\theta}, \qquad 0 < \rho < 1, \qquad 0 < \theta < \pi.$$

The opposite orientation labels the same real map by $\overline{\lambda} = \rho e^{-i\theta}$, whose positive lifted argument is $2\pi - \theta$. No geometric property of the real-linear dynamics depends on this choice.

**Lemma 2.4** (Coordinate reversal, orientation, and endpoint assignment)**.** *Let $C : \mathbb{C} \to \mathbb{C}$ be complex conjugation. For every $\lambda \in \mathbb{C}\setminus\mathbb{R}$ and every compact convex polygon $P$,*

$$C(\lambda P) = \overline{\lambda}\, C(P), \qquad \lambda P \subseteq P \iff \overline{\lambda}\, C(P) \subseteq C(P). \tag{2.6}$$

*Conjugation preserves the number of extreme points, strictness, relative interiors, supporting incidences, side-intersection properties, image-vertex-on-boundary properties, and Hausdorff-continuous deformation families.*

*If $x_i$ is a positively indexed boundary list for $P$ and*

$$\widetilde{x}_i = \overline{x}_{-i},$$

*then $\widetilde{x}_i$ is a positively indexed boundary list for $C(P)$. However, conjugation reverses the half-open convention: the image of the right-half-open side $(x_{j-1}, x_j]$ is the left-half-open side*

$$[\widetilde{x}_{-j}, \widetilde{x}_{1-j}). \tag{2.7}$$

*Consequently a contact assignment to right-half-open sides is not, in general, carried to another such assignment by conjugation when endpoint contacts are present. Thus the endpoint convention must be checked again after conjugation.*

*Proof.* The identity

$$C(\lambda z) = \overline{\lambda}\, C(z)$$

gives (2.6). Since $C$ is an invertible real-affine map, it preserves convex combinations, extreme points, faces, relative interiors, and inclusions. A line supports $P$ if and only if its conjugate supports $C(P)$, with the conjugate contact face; this proves all incidence, side-intersection, and image-vertex-on-boundary assertions. Applying $C$ pointwise to a deformation family preserves Hausdorff convergence because $C$ is an isometry in the adapted metric.

Complex conjugation reverses the cyclic order on the boundary. Hence $\widetilde{x}_i = \overline{x}_{-i}$ restores positive order. The side of $C(P)$ from $\widetilde{x}_{-j} = \overline{x}_j$ to $\widetilde{x}_{1-j} = \overline{x}_{j-1}$ is the conjugate of the unoriented segment $[x_{j-1}, x_j]$. The old half-open set excludes $\overline{x}_{j-1} = \widetilde{x}_{1-j}$ and includes $\overline{x}_j = \widetilde{x}_{-j}$, which is exactly (2.7). Thus conjugation swaps which endpoint is included, proving the final assertion. □

**Roadmap for later topics.** Later arguments must reconstruct the half-open-side assignment after conjugation rather than assume that it is unchanged.

**Lemma 2.5** (Interior origin for a nonreal contraction)**.** *Let $P$ be a non-singleton compact convex hull of finitely many points, possibly a segment, and let $\lambda = \rho e^{i\theta}$ with $0 < \rho < 1$ and $\theta \not\equiv 0, \pi \pmod{2\pi}$. If $\lambda P \subseteq P$, then $P$ has nonempty interior and $0 \in \operatorname{int} P$.*

*Proof.* For any $z \in P$, invariance gives $\lambda^k z \in P$, while $\lambda^k z \to 0$; since $P$ is closed, $0 \in P$. If $P$ were a segment, its affine line would pass through 0 and would have to contain the nondegenerate segment $\lambda P$; hence multiplication by $e^{i\theta}$ would preserve a real line, forcing $\theta \equiv 0$ or $\pi$, a contradiction. Thus $P$ has nonempty interior.

Suppose $0 \in \partial P$. Choose a nonzero real linear functional $\ell$ such that $\ell \geq 0$ on $P$ and choose $z \in P$ with $\ell(z) > 0$. Invariance would give

$$\lambda^k z = \rho^k e^{ik\theta} z \in P, \qquad \rho^k \ell(e^{ik\theta} z) \geq 0.$$

Because $\rho^k > 0$, it follows that $\ell(e^{ik\theta} z) \geq 0$ for every $k \geq 0$. If $e^{i\theta}$ has finite order $m$, then $m \geq 3$ and $\sum_{k=0}^{m-1} e^{ik\theta} z = 0$; summing the displayed nonnegative quantities forces them all to vanish, contrary to $\ell(z) > 0$. If $e^{i\theta}$ has infinite order, its powers are dense in the unit circle, so some $e^{ik\theta} z$ lies in the open half-plane where $\ell < 0$, again a contradiction. Hence 0 is interior. □

**Lemma 2.6** (Oriented order on a convex boundary)**.** *Let $K \subset \mathbb{R}^2$ be a compact convex set with nonempty interior, with $\partial K$ given its positive orientation. For three distinct noncollinear points $a, b, c \in \partial K$,*

$$\det(b - a, c - a) > 0 \tag{2.8}$$

*if and only if $a, b, c$ occur in that positive cyclic order on $\partial K$. Consequently, if $Q$ is a strict polygon all of whose vertices lie on $\partial K$, then the positive cyclic order of the vertices of $Q$ is exactly the order induced from $\partial K$.*

*Proof.* Choose $o \in \operatorname{int} \operatorname{conv}\{a, b, c\}$. Since the triangle $\operatorname{conv}\{a, b, c\}$ is contained in $K$, one has $o \in \operatorname{int} K$. Every ray from $o$ meets $\partial K$ in exactly one point, so the radial map

$$\pi_o : \partial K \longrightarrow \mathbb{S}^1, \qquad \pi_o(z) = \frac{z - o}{|z - o|},$$

is a homeomorphism. With the positive boundary orientation it has degree $+1$: every ray from $o$ meets the boundary exactly once, and traversing the positively oriented boundary once makes the direction $z - o$ traverse the unit circle once counterclockwise. Hence $\pi_o$ preserves cyclic order. Because $o$ lies in the interior of the triangle $abc$, the three rays from $o$ to $a, b, c$ occur counterclockwise in the order $a, b, c$ exactly when the oriented area of that triangle is positive, which is (2.8).

No three vertices of $Q$ are collinear: a middle point of three collinear vertices would be a convex combination of the other two and therefore would not be extreme. For the final assertion, apply the first part both to $K$ and to $Q$. For every triple of vertices of $Q$, the sign of the same determinant characterizes its positive cyclic order on $\partial K$ and on $\partial Q$. Thus the two ternary cyclic-order relations coincide, and therefore so do the full cyclic orders. □

**Lemma 2.7** (Triple-sign criterion for strict convex position)**.** *Let $z_0, \ldots, z_{m-1}$ be a cyclic list of points, $m \geq 3$. The list is exactly the cyclic vertex list of a strict convex $m$-gon, in the displayed orientation, if and only if there is a sign $\varepsilon \in \{1, -1\}$ such that*

$$\varepsilon \det(z_j - z_i, z_k - z_i) > 0 \tag{2.9}$$

*for every triple of distinct indices $i, j, k$ occurring in that cyclic order. The displayed orientation is positive precisely when $\varepsilon = 1$. Here and below, indices attached to a cyclic vertex, side, or normal list are read modulo the length of that list. More explicitly, three distinct classes $i, j, k \in \mathbb{Z}/m\mathbb{Z}$ occur in that cyclic order when they have integer representatives $\widetilde{i}, \widetilde{j}, \widetilde{k}$ with*

$$\widetilde{i} < \widetilde{j} < \widetilde{k} < \widetilde{i} + m.$$

*Proof.* If the list is the cyclic vertex list of a strict polygon, apply lemma 2.6 in the displayed orientation; reversing orientation changes all determinant signs simultaneously.

Conversely, first take $\varepsilon = 1$. The inequalities force the points to be distinct and no three of them to be collinear. Fix $i$. For every $k \notin \{i, i+1\}$, the indices $i, i+1, k$ occur in positive cyclic order, so

$$\det(z_{i+1} - z_i, z_k - z_i) > 0.$$

Thus every other point lies strictly in the left half-plane of the directed line from $z_i$ to $z_{i+1}$. Hence $[z_i, z_{i+1}]$ is an exposed edge of $\operatorname{conv}\{z_0, \dots, z_{m-1}\}$. This holds for every $i$, so all displayed points are extreme, the displayed consecutive segments are exactly the boundary edges, and the resulting polygon is strict and positively oriented. The case $\varepsilon = -1$ is identical with left replaced by right, or follows by reversing the displayed orientation. □

**Lemma 2.8** (Simultaneous preservation of convexity and side conditions under perturbation)**.** *Let $I_0 \subset \mathbb{R}$ be an open interval containing $0$, and let*

$$z_0(\tau), \dots, z_{N-1}(\tau) \qquad (\tau \in I_0)$$

*be continuous maps from $I_0$ to $V$ such that $z_0(0), \dots, z_{N-1}(0)$ are the positively ordered vertices of a strict $N$-gon. Let $\mathcal{A}$ and $\mathcal{B}$ be finite sets. For every $a \in \mathcal{A}$, fix a side index $k(a)$ and a continuous map $w_a : I_0 \to V$ such that, throughout $I_0$,*

$$w_a(\tau) \in \operatorname{aff}\big(z_{k(a)-1}(\tau), z_{k(a)}(\tau)\big), \qquad w_a(0) \in \operatorname{relint}[z_{k(a)-1}(0), z_{k(a)}(0)].$$

*For every $b \in \mathcal{B}$, fix a side index $k(b)$ and a continuous map $u_b : I_0 \to V$ such that*

$$\det\big(z_{k(b)}(0) - z_{k(b)-1}(0), u_b(0) - z_{k(b)-1}(0)\big) > 0. \tag{2.10}$$

*Then there is an open interval $I \subseteq I_0$ containing $0$ on which all of the following hold simultaneously:*

- (i) Convexity is preserved: *the list $z_0(\tau), \dots, z_{N-1}(\tau)$ consists of distinct extreme points in the same positive cyclic order and therefore defines a strict $N$-gon $P_\tau$;*
- (ii) Relative-interior membership is preserved: *$w_a(\tau) \in \operatorname{relint}[z_{k(a)-1}(\tau), z_{k(a)}(\tau)]$ for every $a \in \mathcal{A}$;*
- (iii) The prescribed strict determinant inequalities are preserved: *the strict inequality obtained from (2.10) by replacing $0$ with $\tau$ holds for every $b \in \mathcal{B}$.*

*For each such $P_\tau$, its interior is the intersection of the $N$ open left half-planes determined by its positively oriented sides. In particular, a continuous test point satisfying all $N$ strict side inequalities belongs to $\operatorname{int} P_\tau$.*

*At the unperturbed polygon, a point in the relative interior of one side satisfies the strict side inequality for every other side. Equivalently, if $w \in \operatorname{relint}[z_{c-1}(0), z_c(0)]$, then*

$$\det\big(z_k(0) - z_{k-1}(0), w - z_{k-1}(0)\big) > 0 \qquad (k \neq c). \tag{2.11}$$

*Proof.* For a positively oriented strict polygon, the directed line of side $[z_{k-1}(0), z_k(0)]$ supports the polygon, the polygon lies in its closed left half-plane, and the contact face is exactly that side. If $w \in \operatorname{relint}[z_{c-1}(0), z_c(0)]$ and $k \neq c$, then $w$ does not belong to the contact face of side $k$: distinct maximal sides of a strict polygon have disjoint relative interiors and adjacent sides meet only at their common endpoint. The supporting inequality is therefore strict, proving (2.11). The same support description shows that the polygon is the intersection of the closed left half-planes and its interior is the intersection of their open counterparts.

For every cyclically ordered triple of distinct labels, the corresponding oriented determinant is positive at $\tau = 0$. There are finitely many such determinants, so continuity gives a common neighborhood on which they all remain positive. The triple-sign criterion lemma 2.7 proves that convexity is preserved, item (i).

Fix $a \in \mathcal{A}$. Choose a real linear functional $f_a$ satisfying

$$f_a\big(z_{k(a)}(0) - z_{k(a)-1}(0)\big) \neq 0.$$

Continuity gives an interval about 0 on which the corresponding denominator remains nonzero. Intersecting these finitely many intervals with the interval already chosen gives one common interval, and the collinearity hypothesis gives the unique scalar

$$\alpha_a(\tau) = \frac{f_a\big(w_a(\tau) - z_{k(a)-1}(\tau)\big)}{f_a\big(z_{k(a)}(\tau) - z_{k(a)-1}(\tau)\big)}$$

with

$$w_a(\tau) = (1 - \alpha_a(\tau))z_{k(a)-1}(\tau) + \alpha_a(\tau)z_{k(a)}(\tau).$$

This scalar is continuous and belongs to $(0, 1)$ at 0; finiteness of $\mathcal{A}$ shows that relative-interior membership is preserved, item (ii), on one common interval. Finally, the determinants in item (iii) are finitely many continuous functions that are positive at 0. Intersecting their positivity intervals with the interval already chosen preserves all their signs and proves item (iii). The interior assertion follows from the open half-plane description proved in the first paragraph. □

**Lemma 2.9** (Support-face test for strict polygons)**.** *Let $P$ be a strict polygon with cyclic vertices $x_i$, and let $L$ be a supporting line of $P$ through $x_i$. Then $L$ is a strict supporting line at $x_i$ if and only if neither adjacent vertex $x_{i-1}$ nor $x_{i+1}$ lies on $L$. Equivalently, once support is known, strictness is characterized by the two nonzero determinants*

$$\det(v, x_{i-1} - x_i) \neq 0, \qquad \det(v, x_{i+1} - x_i) \neq 0,$$

*where $v$ is any nonzero direction vector of $L$.*

*Proof.* The contact set $P \cap L$ is a face of the polygon. If it is more than the single point $x_i$, it contains a nondegenerate boundary segment having $x_i$ as one endpoint. In a strict polygon the maximal nondegenerate faces are exactly the displayed sides, so this segment lies in either $[x_{i-1}, x_i]$ or $[x_i, x_{i+1}]$, and the corresponding adjacent vertex is on $L$. Conversely, if an adjacent vertex lies on $L$, then the contact face contains the side joining it to $x_i$ and the support is not strict. The determinant formulation is the line-membership test for the two adjacent vertices. □

**Lemma 2.10** (Angular monotonicity along a strict polygon)**.** *Let $P$ be a strict polygon with $0 \in \operatorname{int} P$, and list its vertices $x_i$ in positive cyclic order. There are lifted arguments $\Theta_i$ with $\Theta_{i+N} = \Theta_i + 2\pi$ such that*

$$0 < \Theta_i - \Theta_{i-1} < \pi \qquad (i \in \mathbb{Z}). \tag{2.12}$$

*Moreover the polar argument is strictly increasing along every positively oriented side $[x_{i-1}, x_i]$.*

*Proof.* The interior of a positively oriented polygon lies strictly to the left of every oriented side. Applying this to the interior point 0 gives

$$\det(x_{i-1}, x_i) > 0.$$

Lift the polar argument continuously along the positively oriented boundary. By the radial-map argument in lemma 2.6, a positively oriented convex boundary has winding number $+1$ about every interior point. One circuit therefore increases this lift by $2\pi$. The displayed determinant sign places each consecutive angular increment in $(0, \pi)$, giving the lifted sequence and (2.12). For $z(t) = (1-t)x_{i-1} + tx_i$, $0 \le t \le 1$, the segment avoids 0 and

$$\frac{d}{dt} \operatorname{Arg} z(t) = \frac{\det(z(t), x_i - x_{i-1})}{|z(t)|^2} = \frac{\det(x_{i-1}, x_i)}{|z(t)|^2} > 0.$$

This proves the sidewise assertion. □

# 3 Contact on every side and at every image vertex

Let $P$ be a strict invariant polygon containing 0 in its interior. Write its outward unit normals in cyclic order as

$$u_i = e^{i\phi_i}, \qquad \phi_1 < \cdots < \phi_m < \phi_1 + 2\pi,$$

and denote the resulting cyclic normal fan by

$$\Phi = (\mathbb{R}_{\ge 0} u_i)_{i=1}^m,$$

regarded with the cyclic indexing convention above. Let $h_i = h_P(u_i) > 0$ be the support numbers. For each $i$, express $e^{-i\theta}u_i$ in the cone generated by the two adjacent fan rays that contain it:

$$e^{-i\theta}u_i = a_i u_j + b_i u_{j+1}, \qquad a_i, b_i \ge 0. \tag{3.1}$$

The resulting full coefficient vector is unique and defines a nonnegative matrix $B_\Phi(\theta)$. If the direction lies on a fan ray, either adjacent cone gives that same vector: the coefficient of the unit generator of that ray is 1 and the other displayed coefficient is 0.

**Proposition 3.1** (Support-function criterion in a fixed normal fan)**.** *With $h = (h_i)$,*

$$\lambda P \subseteq P \iff \rho B_\Phi(\theta) h \le h. \tag{3.2}$$

*Moreover, for the complex vector $u = (u_i)^T \in \mathbb{C}^m$,*

$$B_\Phi(\theta) u = e^{-i\theta} u. \tag{3.3}$$

*The support vectors defining strict polygons with this fixed normal fan form an open subset of $\mathbb{R}^m$.*

*Proof.* If $v = au_j + bu_{j+1}$ belongs to the cone generated by two adjacent normal rays, with $a, b \ge 0$, the two adjacent support inequalities are maximized at their common polygon vertex. For every $z \in P$ one therefore has

$$\langle v, z \rangle \le a h_P(u_j) + b h_P(u_{j+1}),$$

with equality at that common vertex. Hence $h_P(v) = a h_P(u_j) + b h_P(u_{j+1})$. Applying this to $v = e^{-i\theta}u_i$ gives

$$h_P(e^{-i\theta}u_i) = a_i h_j + b_i h_{j+1} = (B_\Phi h)_i.$$

The support function of $\lambda P$ in direction $u_i$ is $\rho h_P(e^{-i\theta}u_i)$, proving the equivalence. The second identity is the vector decomposition itself.

For openness, the two boundary lines with normals $u_i, u_{i+1}$ intersect at a vertex depending linearly on $(h_i, h_{i+1})$. At a strict polygon each such vertex satisfies all nonincident side inequalities strictly, and all cyclic turn determinants have the convex nonzero sign. These are finitely many strict inequalities in continuous functions of $h$ and therefore persist under a sufficiently small perturbation. □

**Theorem 3.2** (Contact on every side and at every image vertex)**.** *Let $T$ be $N$-critical and let $R$ be a nondegenerate polygon with at most $N$ vertices such that $TR \subseteq R$. Then $R$ has exactly $N$ vertices and is strict. Every side of $R$ intersects $TR$, and every vertex of $TR$ lies on $\partial R$.*

*Proof.* By $\nu_{\text{poly}}(T) = N$, the polygon cannot have fewer than $N$ vertices. Hence it has exactly $N$. It has nonempty interior by the nondegeneracy assumption, and lemma 2.5 gives $0 \in \operatorname{int} R$; listing its extreme points makes it a strict $N$-gon.

Write $R$ in its fixed normal fan and put $B = B_\Phi(\theta)$. From (3.2) and $h > 0$, the weighted-norm bound in lemma A.1 gives $\operatorname{spr}(\rho B) \leq 1$. Suppose the inequality were strict. Then

$$g = (I - \rho B)^{-1}\mathbf{1} = \sum_{k\geq 0} (\rho B)^k \mathbf{1} > 0$$

satisfies $g - \rho B g = \mathbf{1}$. For sufficiently small $s > 0$, the support vector $h_s = (1 - s)h + sg$ defines a strict polygon with the same fan, and

$$h_s - \rho B h_s = (1 - s)(h - \rho B h) + s\mathbf{1} > 0.$$

Put $d = h_s - \rho B h_s > 0$. Every row of $B$ expresses a nonzero unit direction as a nonnegative combination, so $Bh_s > 0$ coordinatewise. Choose

$$0 < \eta < \min_i \frac{d_i}{\rho(Bh_s)_i}.$$

Then

$$h_s - (1 + \eta)\rho B h_s = d - \eta\rho B h_s > 0,$$

so the same strict $N$-gon is invariant for $(1+\eta)T$. This contradicts radial criticality. Therefore

$$\operatorname{spr}(\rho B) = 1. \tag{3.4}$$

The nonnegative Perron theorem in lemma A.1 supplies a nonzero $w \geq 0$ such that

$$B^T w = \rho^{-1} w. \tag{3.5}$$

Since $h - \rho B h \geq 0$,

$$w^T(h - \rho B h) = w^T h - \rho(B^T w)^T h = 0.$$

Every summand is nonnegative, so

$$w_i(h_i - \rho(Bh)_i) = 0 \qquad (1 \leq i \leq N). \tag{3.6}$$

*Planar positive-cone fact.* If nonzero vectors $v_1, \ldots, v_r \in \mathbb{R}^2$ and positive coefficients $c_1, \ldots, c_r$ satisfy

$$\sum_{j=1}^{r} c_j v_j = 0,$$

then either all the $v_j$ lie in one line through 0, or their positive cone is all of $\mathbb{R}^2$.

*Proof of the fact.* After normalizing the coefficients, 0 belongs to $\mathrm{conv}\{v_1, \dots, v_r\}$. If the vectors are not collinear but 0 were on the boundary of this convex hull, a supporting functional at 0, chosen nonnegative on the hull, would have nonnegative values on all $v_j$. Their positive weighted sum is zero, so every one of those values would have to vanish, forcing all $v_j$ into the supporting line, a contradiction. Thus 0 is interior to their convex hull. A small positive multiple of every vector in $\mathbb{R}^2$ then belongs to that convex hull, proving that the positive cone is the whole plane.

We prove that every coordinate of $w$ is positive. Let $S = \{i : w_i > 0\}$. Pairing (3.5) with (3.3) gives

$$w^T u = 0,$$

because $\rho^{-1} \neq e^{-i\theta}$. Because every coefficient $w_i$ with $i \in S$ is positive, the planar fact gives the dichotomy: either the normals indexed by $S$ are contained in one line, or their positive cone is all of $\mathbb{R}^2$. We exclude the collinear alternative. If $S$ is the full index set and the normals were contained in $\{v, -v\}$, then the intersection of the side half-planes would be a strip, a half-plane, or the whole plane, never the bounded polygon $R$. Thus collinearity is impossible when $S = \{1, \dots, N\}$. Suppose instead that $S$ is proper. If its normals were contained in $\{v, -v\}$, then for every $k \notin S$ the equality

$$0 = (B^T w)_k = \sum_{i \in S} B_{ik} w_i$$

would force $B_{ik} = 0$ for every $i \in S$, because all summands are nonnegative and every $w_i$ with $i \in S$ is positive. Each row indexed by $S$ would then express $(Bu)_i$ as a real multiple of $v$, whereas $(Bu)_i = e^{-i\theta} u_i$ is not collinear with $v$. This contradiction excludes the remaining case. The normals indexed by $S$ are therefore not collinear, and the positive relation $w^T u = 0$ now implies that they positively span the plane.

Consider the intersection of the half-planes whose indices lie in $S$:

$$R_S = \bigcap_{i \in S} \{z : \langle u_i, z \rangle \le h_i\}.$$

Positive spanning makes its recession cone trivial, so $R_S$ is bounded. It is an intersection of finitely many closed half-planes and is therefore closed and compact. Because every $h_i$ is positive, a sufficiently small disk about 0 satisfies all these inequalities strictly; hence $0 \in \operatorname{int} R_S$, so $R_S$ is nondegenerate. After redundant inequalities are deleted, each remaining boundary line carries one maximal side. Thus $R_S$ has at most $|S|$ sides and, being a nondegenerate planar polygon, at most $|S|$ extreme points. For $i \in S$ and $k \notin S$, the equality $0 = (B^T w)_k = \sum_{j \in S} B_{jk} w_j$ and positivity of every $w_j$ with $j \in S$ give $B_{ik} = 0$. Hence row $i$ of $B$ uses only columns in $S$. Thus, for $z \in R_S$,

$$\langle u_i, \lambda z \rangle \le \rho \sum_{j \in S} B_{ij} h_j = \rho (Bh)_i = h_i,$$

where the last equality is complementarity. Hence $\lambda R_S \subseteq R_S$. If $S$ were proper, this would contradict $\nu_{\mathrm{poly}}(T) = N$. Consequently $w_i > 0$ for every $i$, and (3.6) gives

$$h = \rho B h. \tag{3.7}$$

For each side, equality of support values means that the side intersects $TR$. Thus $TR$ intersects every side of $R$.

It remains to prove that every vertex of $TR$ lies on $\partial R$. The polar polygon

$$R^\circ = \{y : \langle y, z \rangle \le 1 \text{ for all } z \in R\}$$

is a strict $N$-gon; its vertices correspond to the maximal sides of $R$ and its maximal sides correspond to the vertices of $R$. Polarity and the real adjoint give

$$TR \subseteq R \quad \Longrightarrow \quad T^*R^\circ \subseteq R^\circ.$$

By proposition 2.2, $T^*$ is also $N$-critical. The side argument just proved therefore applies to $R^\circ$: every side of $R^\circ$ intersects $T^*R^\circ$.

Fix a vertex $x$ of $R$ and its dual side

$$F_x = \{y \in R^\circ : \langle y, x\rangle = 1\}.$$

Choose $y \in R^\circ$ with $T^*y \in F_x$. Then

$$1 = \langle T^*y, x\rangle = \langle y, Tx\rangle.$$

In particular, $y \neq 0$. Since $y \in R^\circ$ and $Tx \in R$, equality for the nonzero supporting functional $y$ is possible only on the boundary. Thus $Tx \in \partial R$. Invertibility of $T$ identifies the vertices of $R$ with the vertices of $TR$, completing the proof. □

*Remark* 3.3. The theorem can be applied again after a polygon has been modified: once the modified polygon is independently shown to remain a strict invariant $N$-gon, every vertex of its image must again lie on its boundary.

# 4 Half-open side-contact selection

Write the vertices of a strict polygon $P$ in positive cyclic order as $x_i$ ($i \in \mathbb{Z}/N\mathbb{Z}$), and put

$$E_i = [x_{i-1}, x_i], \qquad E_i^+ = (x_{i-1}, x_i], \qquad E_i^- = [x_{i-1}, x_i).$$

All side, vertex, and image-vertex indices in this section are read modulo $N$. For $a, b \in \partial P$, write

$$[a,b]_{\partial P}, \qquad (a,b)_{\partial P}, \qquad (a,b]_{\partial P}, \qquad [a,b)_{\partial P}$$

for the closed, open, and half-open boundary arcs obtained by travelling from $a$ to $b$ in the positive orientation. We shall repeatedly use the following elementary consequence of the fact that the image polygon intersects every side.

**Lemma 4.1** (A side intersecting a polygon contains one of its vertices)**.** *Let $Q \subseteq P$ be polygons. If a side $E$ of $P$ intersects $Q$, then $E$ contains a vertex of $Q$.*

*Proof.* Let $\ell$ be a supporting functional of $P$ whose maximum face is $E$. If $E \cap Q \neq \varnothing$, then the maximum of $\ell$ on $Q$ equals its maximum on $P$. A linear functional on a polygon attains its maximum at a vertex, so a vertex of $Q$ belongs to $E$. □

## 4.1 Half-open boundary assignments

The following definition fixes the endpoint convention used in every cut and vertex-replacement argument. It is not an additional assumption; it is the right-half-open side convention written explicitly.

**Definition 4.2** (Assignment to half-open sides)**.** For a positively oriented strict polygon, side index $i$ refers to the half-open side

$$E_i^+ = (x_{i-1}, x_i].$$

Thus a polygon vertex $x_i$ belongs to the incoming half-open side $E_i^+$, not to the outgoing half-open side $E_{i+1}^+$. More generally, assigning a boundary point $z \in \partial P$ to side index $i$ means precisely that

$$z \in E_i^+.$$

In particular, assigning an image vertex $\xi_i$ to side index $i$ means $\xi_i \in E_i^+$. Either $\xi_i \in \operatorname{relint} E_i$, in which case $x_{i-1}, \xi_i, x_i$ occur in that positive boundary order, or $\xi_i = x_i$. If two consecutive image vertices $y_j, y_{j+1}$ both lie on $\partial P$, the intervening boundary arc follows the same endpoint convention: the positive half-open arc $(y_j, y_{j+1}]_{\partial P}$ includes its right endpoint and excludes its left endpoint.

**Lemma 4.3** (Half-open sides partition the boundary)**.** *Let $P$ be a positively oriented strict $N$-gon with vertices $x_i$ and oriented sides $E_i = [x_{i-1}, x_i]$. For*

$$D_i(z) = \det(x_i - x_{i-1}, z - x_{i-1}),$$

*the following statements hold.*

(i) *The half-open sides $E_i^+ = (x_{i-1}, x_i]$ are pairwise disjoint and form a partition of $\partial P$.*

(ii) *If $z \in E_i^+$, then the vanishing-index set*

$$Z(z) = \{r : D_r(z) = 0\}$$

*is*

$$Z(z) = \{i\} \quad \textit{for } z \in \operatorname{relint} E_i, \qquad Z(x_i) = \{i, i+1\}.$$

(iii) *If $z \in P$ and $D_i(z) = 0$, then $z \in E_i$.*

(iv) *If $z \in E_i^+$ and $z \in E_j^+$, then $i = j$.*

*Proof.* The positively oriented side inequalities give the half-plane and face representations

$$P = \bigcap_i \{z : D_i(z) \geq 0\}, \qquad P \cap \{z : D_i(z) = 0\} = E_i.$$

Since the polygon is strict, the nondegenerate boundary faces are precisely the displayed sides and the only sides incident with $x_i$ are $E_i$ and $E_{i+1}$. Thus a boundary point lies either in one relative side interior or at one displayed vertex, giving the vanishing-index list in (ii) and the implication in (iii). The half-open convention assigns every relative side interior to its side and assigns the vertex $x_i$ only to the incoming side $E_i^+$, proving the partition and disjointness statements. ☐

**Lemma 4.4** (A strict convex combination on the boundary lies in one side)**.** *Let $R$ be a convex polygon with nonempty interior, let $A, B \in R$ with $A \neq B$, and let $0 < s < 1$. If*

$$(1-s)A + sB \in \partial R,$$

*then $A$ and $B$ lie on one common side of $R$, and the whole segment $[A, B]$ is contained in that side.*

*Proof.* Choose a nonconstant supporting affine functional $f \leq c$ for $R$ at $z = (1-s)A + sB$. Since

$$c = f(z) = (1-s)f(A) + sf(B), \qquad f(A), f(B) \leq c,$$

positivity of both coefficients forces $f(A) = f(B) = c$. Thus $A$ and $B$ belong to the exposed face $R \cap \{f = c\}$. This face is nondegenerate because $A \neq B$, so it is a side of the polygon and contains $[A, B]$. ☐

**Lemma 4.5** (Locating the common side from adjacent half-open memberships)**.** *Let $P$ be a positively oriented strict polygon. Suppose*

$$A \in E^+_{j-1}, \qquad B \in E^+_j, \qquad A \neq B,$$

*and suppose that a strict convex combination $(1-s)A + sB$, $0 < s < 1$, lies on $\partial P$. Then*

$$A = x_{j-1}, \qquad [A, B] \subseteq E_j.$$

*In particular there is a unique $t \in (0, 1]$ such that*

$$B = (1-t)x_{j-1} + tx_j,$$

*and every strict convex combination of $A$ and $B$ lies in* $\operatorname{relint} E_j$*.*

*Proof.* By lemma 4.4, the two endpoints $A, B$ lie on one common side of $P$. Hence $Z(A) \cap Z(B)$ is nonempty. The half-open boundary partition gives

$$Z(A) \subseteq \{j-1, j\}, \qquad Z(B) \subseteq \{j, j+1\},$$

and the common index can only be $j$. The index $j$ belongs to $Z(A)$ only when $A$ is the right endpoint of $E^+_{j-1}$, namely $A = x_{j-1}$. The common side is therefore $E_j$, so $B \in E_j$ as well. Since $B \in E^+_j$ and $B \neq A = x_{j-1}$, the displayed representation with $t \in (0, 1]$ is unique. Since $A \neq B$, every strict convex combination of $A$ and $B$ lies in $\operatorname{relint}[A, B] \subseteq \operatorname{relint} E_j$. □

**Lemma 4.6** (Determinant test for side membership and containment)**.** *Let $P$ be a positively oriented strict $N$-gon with displayed vertices $x_i$ and half-open sides $E^+_i$. Let $\eta_j$ be $N$ points such that, for every $j$, either*

$$\eta_j = x_j,$$

*or*

$$\eta_j = (1-t_j)x_{j-1} + t_j x_j, \qquad 0 < t_j < 1.$$

*Put*

$$C_{r,k} = \det(x_r - x_{r-1}, x_k - x_{r-1}),$$

*so $C_{r,r-1} = C_{r,r} = 0$, and put*

$$D_{r,j} = \det(x_r - x_{r-1}, \eta_j - x_{r-1}).$$

*Then $\eta_j \in E^+_j$ for every $j$ and*

$$D_{r,j} \geq 0 \qquad (0 \leq r, j < N).$$

*More precisely, if $\eta_j = x_j$, then $D_{r,j} = C_{r,j}$; if $\eta_j = (1-t_j)x_{j-1} + t_j x_j$, then*

$$D_{r,j} = (1-t_j)C_{r,j-1} + t_j C_{r,j}.$$

*In the second case $D_{j,j} = 0$ and $D_{r,j} > 0$ for every $r \neq j$. Moreover, the points $\eta_j$ are pairwise distinct, and* $\operatorname{conv}\{\eta_j\} \subseteq P$*.*

*Proof.* The endpoint $x_j$ belongs to $E^+_j$ and to no other half-open side, while a strict convex combination of $x_{j-1}$ and $x_j$ belongs to $\operatorname{relint} E_j \subset E^+_j$. This proves the asserted side-membership statements. The determinant identities follow by bilinearity. Because the displayed vertices are exactly the extreme points of $P$ and every displayed side is a maximal boundary segment, a nonincident vertex $x_k$ lies strictly in the inward half-plane determined by $E_r$. Thus $C_{r,k} > 0$ for every nonincident vertex, while $C_{r,r-1} = C_{r,r} = 0$ for the two incident vertices. Hence all entries of the displayed matrix are nonnegative, with strict positivity in the stated nonincident cases. Because $\eta_j \in E^+_j$ and the half-open sides are pairwise disjoint, $\eta_j \neq \eta_k$ whenever $j \neq k$. The inequalities $D_{r,j} \geq 0$ are exactly the oriented side inequalities of $P$, so every $\eta_j$ belongs to $P$ and therefore $\operatorname{conv}\{\eta_j\} \subseteq P$. □

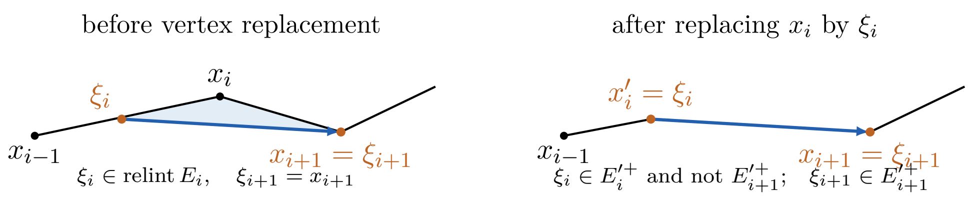


Figure 1: Half-open side membership for the unchanged image vertices in the elementary vertex-replacement pattern. The shaded region is the only part discarded from $P$; the blue edge is an actual edge of $Q = \lambda P$.

**Lemma 4.7** (Side membership after replacing one vertex)**.** *Let $P$ be a positively oriented strict $N$-gon with vertices $x_r$ and sides $E_r = [x_{r-1}, x_r]$. Choose $\xi_i \in \operatorname{relint} E_i$ and define*

$$P' = \operatorname{conv}\Big((\operatorname{Ext} P \setminus \{x_i\}) \cup \{\xi_i\}\Big).$$

*Then $P'$ is a strict $N$-gon whose positive cyclic vertex list is obtained from that of $P$ by replacing $x_i$ by $x_i' = \xi_i$. In particular, the points $x_{i-1}, x_i' = \xi_i, x_{i+1}$ occur in that positive cyclic order. For the two new sides*

$$E_i' = [x_{i-1}, \xi_i], \qquad E_{i+1}' = [\xi_i, x_{i+1}],$$

*one has*

$$\xi_i \in E_i'^{+}, \qquad x_{i+1} \in E_{i+1}'^{+}.$$

*Proof.* There is a unique $t \in (0,1)$ such that

$$\xi_i = (1-t)x_{i-1} + t x_i.$$

Let $L = \operatorname{aff}(x_{i-1}, x_{i+1})$. Convexity and strictness imply that $x_i$ lies in one open half-plane bounded by $L$, whereas every other vertex except $x_{i-1}, x_{i+1}$ lies in the opposite open half-plane. Since $\xi_i = (1-t)x_{i-1} + t x_i$ with $t > 0$, the point $\xi_i$ lies in the same open half-plane as $x_i$. It therefore does not belong to the convex hull of the unchanged vertices, so it is an extreme point of $P'$.

Each unchanged vertex remains extreme: a linear functional that strictly exposes it on $P$ still strictly exposes it among the displayed points defining $P'$ (its value at $\xi_i \in P$ is also strictly smaller). Thus all $N$ displayed points are extreme points of $P'$.

The supporting line of the old side $E_i$ meets these displayed points only at $x_{i-1}$ and $\xi_i$, so $[x_{i-1}, \xi_i]$ is an exposed side of $P'$. For every unchanged vertex $x_k \neq x_{i+1}$, affine linearity of the oriented area in its first entry gives

$$\det(x_{i+1} - \xi_i, x_k - \xi_i) = (1-t)\det(x_{i+1} - x_{i-1}, x_k - x_{i-1}) + t\det(x_{i+1} - x_i, x_k - x_i) > 0.$$

The triple-sign criterion makes both terms nonnegative and at least one strictly positive (when $x_k = x_{i-1}$, the first term is zero and the second is positive). Hence all the other displayed points lie strictly in the left half-plane of the directed line from $\xi_i$ to $x_{i+1}$, so $[\xi_i, x_{i+1}]$ is also an exposed side of $P'$. Every old side not incident with $x_i$ remains exposed by its old supporting line. These sides join all $N$ extreme points in the displayed positive cyclic order. Consequently that list is exactly the positive cyclic vertex list of $P'$; in particular, $P'$ is a strict $N$-gon and the two sides incident with $\xi_i$ are $E_i' = [x_{i-1}, \xi_i]$ and $E_{i+1}' = [\xi_i, x_{i+1}]$.

The point $\xi_i$ is the right endpoint of $E_i'$ and the left endpoint of $E_{i+1}'$, so the right-half-open convention assigns it to $E_i'^{+}$. The point $x_{i+1}$ is the right endpoint of $E_{i+1}'$, so it belongs to $E_{i+1}'^{+}$. □

## 4.2 Clipping along image edges

The cutting argument below uses only actual edges of $Q = \lambda P$. This makes the two endpoints distinct and specifies unambiguously which closed half-plane is retained, including the case in which the same line also supports $P$.

**Lemma 4.8** (Clipping along an image edge and bounding the number of vertices)**.** *Let $P$ be a strict $N$-gon, let $\lambda \neq 0$, and put $Q = \lambda P$. Assume $Q \subseteq P$ and that every vertex of $Q$ lies on $\partial P$. List the vertices of $Q$ in their positive cyclic order as $y_0, \ldots, y_{N-1}$. Their order on $\partial P$ is the same cyclic order.*

*For the edge $[y_j, y_{j+1}]$ of $Q$, let $H_j$ be the closed half-plane bounded by its line and containing $Q$, and put*

$$P_j = P \cap H_j. \tag{4.1}$$

*For every $j$, define the closed and open positive boundary arcs*

$$A_j = [y_j, y_{j+1}]_{\partial P}, \qquad A_j^\circ = (y_j, y_{j+1})_{\partial P}.$$

*Also put*

$$k_j = \#(\operatorname{Ext} P \cap A_j),$$

*so an endpoint is counted exactly when it is already a vertex of $P$.*

*If $P_j \neq P$, the set removed by the cut is $P \setminus H_j = P \setminus P_j$ and has nonempty interior, while the part of the boundary that is removed is exactly*

$$\partial P \setminus P_j = A_j^\circ.$$

*In particular, $A_j^\circ$ contains no vertex of $Q$. In this case*

$$Q \subseteq P_j \subsetneq P, \qquad \lambda P_j \subseteq P_j, \qquad |\operatorname{Ext} P_j| \leq N + 2 - k_j. \tag{4.2}$$

*If $P_j = P$, the line through $[y_j, y_{j+1}]$ also supports $P$. In that case the positive half-open boundary arc $(y_j, y_{j+1}]_{\partial P}$ lies in one side of $P$ and contains at most one vertex of $P$.*

*Proof.* Because multiplication by $\lambda \neq 0$ is an invertible real-linear map with determinant $|\lambda|^2 > 0$, it preserves orientation, and $Q = \lambda P$ is a strict polygon. All of its vertices lie on $\partial P$, so lemma 2.6 shows that the positive cyclic order on $\partial Q$ is exactly the order induced from $\partial P$. Thus consecutive $Q$-vertices bound a unique positive boundary arc of $P$ containing no other $Q$-vertex.

The line of a $Q$-edge supports $Q$, so $Q \subseteq H_j$. Hence $Q \subseteq P_j \subseteq P$. Since $P_j \subseteq P$,

$$\lambda P_j \subseteq \lambda P = Q \subseteq P_j.$$

Suppose $P_j \neq P$, and write $L_j = \operatorname{aff}(y_j, y_{j+1})$. The full-dimensional polygon $Q$ has interior points in the open half-plane bounded by $L_j$ and contained in $H_j$, while $P_j \neq P$ supplies a point of $P$ in the opposite open half-plane. Thus $P$ has points on both sides of $L_j$. Since the interior of a full-dimensional convex set is dense in that set, the discarded open half-plane also contains an interior point of $P$. Therefore $P \setminus H_j$ has nonempty interior.

We claim that every point of $\operatorname{relint} I$, where $I = P \cap L_j$, lies in $\operatorname{int} P$. Suppose instead that $z \in \operatorname{relint} I \cap \partial P$. Choose a nonzero linear functional $\ell$ and a number $c$ such that

$$\ell \leq c \quad \text{on } P, \qquad \ell(z) = c.$$

With $d = y_{j+1} - y_j$, relative interior membership gives $z \pm \varepsilon d \in I$ for all sufficiently small $\varepsilon > 0$. Applying the two support inequalities at these points forces $\ell(d) = 0$. Hence the supporting line $\{\ell = c\}$ contains $z$ and has direction $d$, so it is exactly $L_j$. This would place all of $P$ in one closed half-plane bounded by $L_j$, contradicting the points of $P$ on both sides. The claim follows.

The points $y_j, y_{j+1}$ belong to $\partial P \cap I$, so the claim forces them to be the two endpoints of the compact segment $I$. If $z \in A_j^\circ$, then $y_j, z, y_{j+1}$ occur in positive cyclic order on $\partial P$. By lemma 2.6,

$$\det(z - y_j, y_{j+1} - y_j) > 0,$$

or equivalently $\det(y_{j+1} - y_j, z - y_j) < 0$. On the other hand, because $[y_j, y_{j+1}]$ is a positively oriented edge of $Q$, the polygon $Q$ lies in the closed half-plane where this last determinant is nonnegative; this is $H_j$. Thus $A_j^\circ$ lies in the discarded open half-plane.

For a point $z$ on the complementary open boundary arc $(y_{j+1}, y_j)_{\partial P}$, the points $y_{j+1}, z, y_j$ occur in positive cyclic order. The same determinant test, now with the endpoints reversed, gives

$$\det(y_{j+1} - y_j, z - y_j) > 0,$$

so this complementary arc lies in the interior of $H_j$. Moreover, the preceding description of $I = P \cap L_j$ gives $\partial P \cap L_j = \{y_j, y_{j+1}\}$. The endpoints themselves belong to $P_j$, and the two open arcs account for all other boundary points. Consequently

$$\partial P \setminus P_j = A_j^\circ.$$

This open arc contains no $Q$-vertex because $Q \subseteq H_j$.

Clipping removes every $P$-vertex on $A_j^\circ$. Each endpoint $y_j, y_{j+1}$ is retained, and it is a new candidate vertex precisely when it was not already a $P$-vertex. Thus the number of candidate vertices is

$$N - \#(\operatorname{Ext} P \cap A_j^\circ) + \#\{\text{endpoints not already } P\text{-vertices}\} = N + 2 - k_j.$$

Deleting any listed point that lies in the interior of a straight boundary segment can only lower the number of extreme vertices.

If $P_j = P$, the edge line supports $P$ as well. Its intersection with $P$ is a nondegenerate face and therefore one displayed side. The positive arc from $y_j$ to $y_{j+1}$ lies in that side, so after its left endpoint is excluded it contains at most one vertex of $P$. ☐

**Lemma 4.9** (Old-vertex bound on discarded boundary arcs)**.** *Let $T(z) = \lambda z$, where $0 < |\lambda| < 1$ and $\lambda \notin \mathbb{R}$, and assume that $T$ is $N$-critical. Among the strict $N$-gons satisfying $\lambda P \subseteq P$, impose the unit maximal-radius normalization*

$$\max_{z \in P} |z| = 1. \tag{4.3}$$

*The normalized class is nonempty and contains a member of least area; choose such a polygon $P$. Let $Q = \lambda P$ and use the notation of lemma 4.8. Then every proper image-edge cut, meaning $P_j \neq P$, satisfies*

$$k_j \leq 2, \tag{4.4}$$

*and at most one proper image-edge cut has $k_j = 2$.*

*Proof.* Because $\nu_{\text{poly}}(T) = N$, an invariant polygon with exactly $N$ extreme vertices exists. Its maximal distance from the origin is positive, and rescaling it to make that maximum equal to 1 preserves both invariance and the number of extreme vertices. Thus the normalized class is nonempty. Write

$$\overline{\mathbb{D}} = \{z \in \mathbb{R}^2 : |z| \leq 1\}.$$

We next prove that the normalized class is compact in the Hausdorff metric. Take an arbitrary sequence in this class and display each member as the convex hull of its $N$ vertices, all of which lie in $\overline{\mathbb{D}}$. Compactness of $(\overline{\mathbb{D}})^N$ gives a subsequence for which the $N$-tuples converge. By lemma A.4, their convex hulls converge in the Hausdorff metric to a convex hull $P_\infty$ with at most $N$ extreme points; the same lemma preserves the maximal radius and the inclusion $\lambda P_\infty \subseteq P_\infty$.

The limit cannot be a singleton: if $P_\infty = \{v\}$, normalization gives $|v| = 1$, while invariance gives $\lambda v = v$, impossible because $0 < |\lambda| < 1$. It cannot be a nondegenerate segment either. Indeed, if $a \neq b$ are its endpoints, then $\lambda a, \lambda b \in P_\infty$, so the nonzero vectors $b - a$ and $\lambda(b-a)$ are parallel, which is impossible for nonreal $\lambda$. Thus $P_\infty$ has nonempty interior. If it had fewer than $N$ extreme points, it would be an invariant polygon with fewer than $N$ vertices, contrary to $\nu_{\mathrm{poly}}(T) = N$. Hence it has exactly $N$ extreme points and belongs to the normalized class. We have shown that every sequence has a subsequence converging within the class, so the class is compact. Area is continuous by lemma A.4; therefore a least-area normalized representative exists.

By theorem 3.2, every vertex of $Q = \lambda P$ lies on $\partial P$. Thus all hypotheses of lemma 4.8 hold for this least-area polygon, and its clipping conclusions apply.

Every point of $P$ is a convex combination of its vertices, so convexity of the norm gives

$$\max_{z \in P} |z| = \max_{x \in \operatorname{Ext} P} |x|.$$

Choose a vertex $v$ with $|v| = 1$. Since $Q \subseteq \rho\overline{\mathbb{D}}$, where $\rho = |\lambda| < 1$, the point $v$ is not a vertex of $Q$. If a proper image-edge cut had $k_j \geq 3$, then $P_j$ would contain the full-dimensional polygon $Q$, would satisfy $\lambda P_j \subseteq P_j$, and would have at most $N - 1$ vertices by (4.2). This contradicts $\nu_{\mathrm{poly}}(T) = N$.

Suppose a proper image-edge cut has $k_j = 2$ and $v \notin A_j^\circ$. By the discarded-boundary equality in lemma 4.8, $v \in P_j$. Hence $P_j$ remains normalized. It contains the full-dimensional polygon $Q$, is invariant, and has at most $N$ vertices. Criticality rules out fewer than $N$ vertices, so $P_j$ has exactly $N$ extreme vertices and is another strict invariant $N$-gon. But $P_j \subsetneq P$ and both polygons have nonempty interior, so lemma A.5 gives $\operatorname{area}(P_j) < \operatorname{area}(P)$. This contradicts the choice of $P$.

The open positive arcs $A_j^\circ = (y_j, y_{j+1})_{\partial P}$ are pairwise disjoint and together cover $\partial P \setminus \operatorname{Ext} Q$. Since $v$ is not a $Q$-vertex, it belongs to exactly one of them. Therefore at most one proper image-edge cut with $k_j = 2$ can escape the preceding contradiction. $\square$

**Lemma 4.10** (Finite cyclic endpoint count)**.** *Let $(r_j)_{j \in \mathbb{Z}/N\mathbb{Z}}$ take values in $\{0, 1, 2\}$, satisfy $\sum_j r_j = N$, and contain at most one entry equal to 2. Let $(c_j)_{j \in \mathbb{Z}/N\mathbb{Z}}$ be a binary cyclic word. If an entry $r_s = 2$ occurs, assume*

$$c_s = 0, \qquad c_{s+1} = 1, \tag{4.5}$$

*and assume that every transition $c_j = 1$, $c_{j+1} = 0$ can occur only at an index $j$ for which $r_j = 0$. Then either $r_j = 1$ for every $j$, or there are unique indices $s, t$ such that*

$$r_s = 2, \qquad r_t = 0, \qquad r_j = 1 \quad (j \notin \{s, t\}),$$

*and in the latter case*

$$\ell_j := r_j + c_j - c_{j+1} = 1 \qquad (j \in \mathbb{Z}/N\mathbb{Z}). \tag{4.6}$$

*Proof.* If no entry is 2, then every $r_j \leq 1$ and the sum of the $N$ entries is $N$, so all entries equal 1. Suppose $r_s = 2$. Its surplus over 1 is one. Since no second entry can exceed 1, the equality $\sum_j r_j = N$ forces a unique zero $r_t = 0$ and makes every remaining entry equal to 1.

The transition $0 \to 1$ at $s$ in (4.5) forces at least one transition $1 \to 0$ in the cyclic binary word. By hypothesis it can occur only at the unique zero index $t$. In a cyclic binary word the number of $0 \to 1$ transitions equals the number of $1 \to 0$ transitions, because

$$\sum_j (c_{j+1} - c_j) = 0.$$

Consequently the transitions at $s$ and $t$ are the unique transitions of their respective types. Hence $c_j = c_{j+1}$ for $j \notin \{s, t\}$, while

$$(c_s, c_{s+1}) = (0, 1), \qquad (c_t, c_{t+1}) = (1, 0).$$

Substitution gives $\ell_s = 2 + 0 - 1 = 1$, $\ell_t = 0 + 1 - 0 = 1$, and $\ell_j = 1$ at every remaining index. $\square$

**Lemma 4.11** (Cyclic interlacing with endpoint bookkeeping)**.** *Let $P$ be the least-area representative of lemma 4.9. Then at least one of the following two alternatives holds:*
(i) *every arc $(y_j, y_{j+1}]_{\partial P}$ contains exactly one vertex of $P$;*
(ii) *every arc $[y_j, y_{j+1})_{\partial P}$ contains exactly one vertex of $P$.*
*In alternative* (ii)*, every right-half-open side of $P$ contains exactly one vertex of $Q$. In alternative* (i)*, the conjugate configuration*

$$\widetilde{P} = \overline{P}, \qquad \widetilde{Q} = \overline{Q} = \overline{\lambda}\, \widetilde{P}, \qquad \widetilde{x}_i = \overline{x}_{-i} \tag{4.7}$$

*has the same property when its vertices are read in positive cyclic order. Thus a right-half-open convention is obtained without ever reading a convex polygon in the wrong orientation.*

*Proof.* Put

$$c_j = \begin{cases} 1, & y_j \in \operatorname{Ext} P, \\ 0, & y_j \notin \operatorname{Ext} P, \end{cases} \qquad r_j = \#(\operatorname{Ext} P \cap (y_j, y_{j+1}]_{\partial P}).$$

The half-open arcs partition the $N$ vertices of $P$, so

$$\sum_{j=0}^{N-1} r_j = N. \tag{4.8}$$

The closed arc $A_j = [y_j, y_{j+1}]_{\partial P}$ is obtained from the right-half-open arc $(y_j, y_{j+1}]_{\partial P}$ by adding its left endpoint. Consequently, for every $j$—whether or not the image-edge line also supports $P$—one has

$$k_j = r_j + c_j. \tag{4.9}$$

When $P_j = P$, lemma 4.8 gives $r_j \le 1$. When $P_j \ne P$, lemma 4.9 and (4.9) give $r_j \le k_j \le 2$. Hence

$$0 \le r_j \le 2, \qquad \text{and at most one } r_j \text{ equals } 2. \tag{4.10}$$

Indeed, $r_j = 2$ is impossible when $P_j = P$. Thus any such index gives a proper image-edge cut; then (4.9) and $k_j \le 2$ force $c_j = 0$ and $k_j = 2$. By lemma 4.9, at most one proper image-edge cut has this value.

If no $r_j$ equals 2, the combination of (4.8) and (4.10) forces $r_j = 1$ for every $j$, which is alternative (i). Otherwise there are unique indices $s, t$ with

$$r_s = 2, \qquad r_t = 0, \qquad r_j = 1 \quad (j \ne s, t). \tag{4.11}$$

We determine whether the two exceptional endpoints are vertices of $P$.

First, $c_s = 0$ by (4.9). Also $c_{s+1} = 1$. Indeed, suppose that $y_{s+1}$ were not a vertex of $P$. Since $c_s = 0$, neither endpoint of the boundary arc from $y_s$ to $y_{s+1}$ would then be a vertex of $P$. Because $r_s = 2$, its open boundary arc would contain exactly two vertices of $P$ and no other vertex between them. They would therefore be consecutive vertices of $P$, while their joining side would contain no vertex of $Q$: the points $y_s$ and $y_{s+1}$ are consecutive vertices of $Q$. This contradicts lemma 4.1 and the fact that every side of $P$ intersects $Q$. Thus the binary cyclic word $(c_j)$ has a transition $0 \to 1$ at $s$.

Now consider a transition $c_j = 1$, $c_{j+1} = 0$ with $j \neq t$. Then $r_j = 1$, so (4.9) gives $k_j = 2$. The image-edge cut is proper. Indeed, if its line also supported $P$, the positive arc starts at the $P$-vertex $y_j$ and ends at a nonvertex $y_{j+1}$, so the arc must stop before the next $P$-vertex, so it would have $r_j = 0$, contrary to $r_j = 1$. It would therefore be a second proper image-edge cut with $k = 2$, which is impossible. Hence every transition $1 \to 0$ can occur only at the unique zero index $t$. The hypotheses of lemma 4.10 are now verified: (4.11) gives the unique two and zero, while the preceding paragraph gives the required $0 \to 1$ transition at $s$. Therefore the opposite half-open counts

$$\ell_j = \#(\operatorname{Ext} P \cap [y_j, y_{j+1})_{\partial P}) = r_j + c_j - c_{j+1}$$

all equal 1. This is alternative (ii).

It remains to pass from gaps between $Q$-vertices to sides between $P$-vertices. In alternative (ii), let $x_j$ be the unique $P$-vertex in $[y_j, y_{j+1})$. In the positive boundary order, $y_j$ lies after $x_{j-1}$ and no later than $x_j$; equivalently,

$$y_j \in (x_{j-1}, x_j]_{\partial P} = E_j^+.$$

No other $Q$-vertex lies there.

In alternative (i), let $x_j$ be the unique vertex in $(y_j, y_{j+1}]$. Then $y_{j+1} \in [x_j, x_{j+1})$. Apply complex conjugation and index the conjugate polygon positively by $\widetilde{x}_i = \overline{x}_{-i}$. The conjugate of $[x_j, x_{j+1})$ is, with its order reversed,

$$(\widetilde{x}_{-j-1}, \widetilde{x}_{-j}];$$

therefore it is a right-half-open side of $\widetilde{P}$. Conjugation is a bijection on vertices and takes $Q$ to $\widetilde{Q} = \overline{\lambda}\,\widetilde{P}$, so every such side contains exactly one vertex of $\widetilde{Q}$. This proves the final assertion with all endpoint inclusions explicit. $\square$

**Corollary 4.12** (Bijective global half-open assignment)**.** *In either orientation supplied by lemma 4.11, the contact assignment is a genuine labeled bijection*

$$\{\text{vertices of } Q\} \longleftrightarrow \{E_i^+ : i \in \mathbb{Z}/N\mathbb{Z}\}.$$

*Thus every image vertex belongs to exactly one half-open side, and every half-open side contains exactly one image vertex. After the possible conjugation in (4.7), the labels may be chosen so that the unique image assigned to $E_i^+$ is denoted by $\xi_i$.*

*Proof.* The half-open sides $E_i^+ = (x_{i-1}, x_i]$ are pairwise disjoint and their union is $\partial P$. In alternative (ii) of lemma 4.11, each interval $[y_j, y_{j+1})_{\partial P}$ contains exactly one vertex of $P$, and the proof there gives $y_j \in E_j^+$. Thus every side contains at least one image vertex. There are exactly $N$ sides and exactly $N$ image vertices, while the disjoint half-open partition prevents one image vertex from belonging to two sides; hence the assignment is a bijection. In alternative (i), the same argument applies after the conjugate reindexing (4.7). This proves that the labeling is bijective. $\square$

The next step has two logically distinct parts. The half-open assignment and cyclic order make the vertex-to-side assignment an orientation-preserving degree-one correspondence of two cyclic $N$-point sets, hence a shift modulo $N$. Passing that correspondence to the universal cover of the circle introduces a possible extra full turn at each side index. The interval estimate in the proof rules out those turns separately; summation then gives a strict rational bracket for the multiplier angle.

**Lemma 4.13** (Order-preserving half-open contact assignment)**.** *After replacing $(P, \lambda)$ by the conjugate configuration (4.7) if required, there is an invariant $N$-gon representative, a positive cyclic indexing, and an integer $\kappa$ with $1 \le \kappa \le N$ such that*

$$\xi_i := \mu x_{i-\kappa} = \beta_i x_{i-1} + \alpha_i x_i, \qquad \alpha_i > 0, \quad \beta_i \ge 0, \quad \alpha_i + \beta_i = 1. \tag{4.12}$$

*Here $\mu$ is either $\lambda$ or $\overline{\lambda}$, indices are modulo $N$, and $\vartheta$ is the unique number in $(0, 2\pi)$ such that $\mu = |\mu| e^{i\vartheta}$. If $y = \vartheta/(2\pi)$, then*

$$\frac{\kappa - 1}{N} < y < \frac{\kappa}{N}. \tag{4.13}$$

*If $\kappa = N$, every contact lies in the relative interior of its assigned side.*

*Proof.* Choose the least-area normalized invariant $N$-gon from lemma 4.9. In alternative (ii) of lemma 4.11, put $\mu = \lambda$. In alternative (i), replace the configuration by (4.7) and put $\mu = \overline{\lambda}$. In either case the vertices are now in positive cyclic order and every $E_i^+$ contains exactly one vertex $\xi_i$ of $Q = \mu P$.

Multiplication by the nonzero complex number $\mu$ preserves cyclic order. Hence the order-preserving incidence bijection between source vertices and right-half-open sides is a single cyclic shift. Indeed, after identifying both cyclic $N$-sets with $\mathbb{Z}/N\mathbb{Z}$, a cyclic-order-preserving bijection sends each successor to the successor of its image and therefore has the form $j \mapsto j + \kappa$ modulo $N$. Choose the representative $\kappa \in \{1, \dots, N\}$ of this cyclic shift and write $\xi_i = \mu x_{i-\kappa}$. Because $\xi_i \in (x_{i-1}, x_i]$, it has the unique barycentric form (4.12) with $\alpha_i > 0$ and $\beta_i \ge 0$.

Choose lifted polar angles $\Theta_i$ with $\Theta_{i+N} = \Theta_i + 2\pi$, and choose the lifted multiplier argument $\vartheta \in (0, 2\pi)$. By lemma 2.5, $0 \in \operatorname{int} P$. Hence lemma 2.10 applies: after choosing the increasing lift, polar angle is strictly increasing along every positively oriented side. The side incidence first gives, for some integer $m_i$,

$$\Theta_{i-1} < \Theta_{i-\kappa} + \vartheta + 2\pi m_i \le \Theta_i. \tag{4.14}$$

The integer $m_i$, which accounts for the possible addition of $2\pi m_i$ to the chosen lift, is necessarily zero. If $1 \le \kappa < N$, then

$$0 \le \Theta_{i-1} - \Theta_{i-\kappa} < \Theta_i - \Theta_{i-\kappa} < 2\pi,$$

because this interval spans $\kappa$ consecutive angular gaps and omits at least one positive gap. If $\kappa = N$, then

$$\Theta_{i-1} - \Theta_{i-N} = 2\pi - (\Theta_i - \Theta_{i-1}), \qquad \Theta_i - \Theta_{i-N} = 2\pi.$$

Thus the possible difference interval in (4.14) is contained in $[0, 2\pi)$ in the first case and in $(0, 2\pi]$ in the second. Since $\vartheta \in (0, 2\pi)$, the only possible integer is $m_i = 0$. Hence (4.12) gives the genuine lifted inequality

$$\Theta_{i-1} < \Theta_{i-\kappa} + \vartheta \le \Theta_i.$$

Summing for one complete period, say over $i = 1, \dots, N$, and using $\sum_{i=1}^N \Theta_{i-\kappa} = \sum_{i=1}^N \Theta_i - 2\pi\kappa$ and $\sum_{i=1}^N \Theta_{i-1} = \sum_{i=1}^N \Theta_i - 2\pi$ yields

$$2\pi(\kappa - 1) < N\vartheta \le 2\pi\kappa.$$

The final equality cannot occur. Equality in the sum would force equality in every right-hand inequality, so every contact would be an endpoint and $\mu x_{i-\kappa} = x_i$ for all $i$. Iteration around a shift orbit gives $x_i = \mu^{N/\gcd(N,\kappa)} x_i$, impossible because $|\mu| < 1$ and no vertex is 0. This proves (4.13).

If $\kappa = N$, then the source of contact $i$ is $x_i$ itself. An endpoint contact would give $\mu x_i = x_i$, again impossible. Hence every contact lies in the relative interior of its assigned side. $\square$

From this point through the end of the first-return analysis, fix the contact data supplied by lemma 4.13 and denote its complex multiplier by $\lambda$. Put

$$\theta = \arg_+(\lambda) \in (0, 2\pi), \qquad x = \frac{\theta}{2\pi}.$$

The change from the symbol $\mu$ in lemma 4.13 to $\lambda$ is only a local renaming of the multiplier in these already fixed data; it is never undone. The chosen orientation, half-open assignment, cyclic side permutation, vertex-replacement class, first-return data, and projective deformation are not changed anywhere in the vertex-replacement, finite-rotation, or adjacent-return arguments. If the final closed return-recurrence chain is more naturally read in the opposite orientation, that reversal is performed only after those arguments have ended, by the explicit reflection lemma lemma 8.3; the resulting output eigenvalue then receives the new symbol $\mu$.

More precisely, the *standing contact data* are a tuple

$$(P, \lambda, \kappa; (\alpha_i, \beta_i)_{i \in \mathbb{Z}/N\mathbb{Z}})$$

satisfying the following five assumptions:
(A0) multiplication by $\lambda$ is an $N$-critical elliptic contraction;
(A1) $P$ is a positively indexed strict invariant $N$-gon for this multiplication;
(A2) every side of $P$ intersects $\lambda P$, and every vertex of $\lambda P$ lies on $\partial P$;
(A3) the fixed representative $\kappa \in \{1, \ldots, N\}$ and the coefficients $(\alpha_i, \beta_i)$ satisfy (4.12), so every right-half-open side $E_i^+$ contains its assigned image vertex; and
(A4) $S = \{i : \beta_i > 0\}$ is exactly the set of indices whose assigned image vertices lie in the relative interiors of their sides.
Later references to the standing assumptions mean precisely (A0)–(A4).

**Lemma 4.14** (Iteration of endpoint equalities)**.** *For contact data satisfying* (A0)–(A4)*, extend the vertex angles by* $\Theta_{i+N} = \Theta_i + 2\pi$*. For every integer side index* $j$*,*

$$\Theta_{j-1} < \Theta_{j-\kappa} + \theta \le \Theta_j. \tag{4.15}$$

*If the contact at index* $j$ *is an endpoint contact, equality holds on the right. Hence, if the side indices* $a+\kappa, \ldots, a+t\kappa$ *are all endpoint contacts, then*

$$\Theta_a + t\theta = \Theta_{a+t\kappa}. \tag{4.16}$$

*If the first* $t-1$ *destination indices are endpoint contacts and the contact at index* $a+t\kappa$ *lies in the relative interior of its assigned side, then instead*

$$\Theta_{a+t\kappa-1} < \Theta_a + t\theta < \Theta_{a+t\kappa}. \tag{4.17}$$

*Proof.* The side incidence gives an integer $m_j$ such that

$$\Theta_{j-1} < \Theta_{j-\kappa} + \theta + 2\pi m_j \le \Theta_j.$$

Exactly as in the lift check in lemma 4.13, the interval of possible differences is contained in $[0, 2\pi)$ when $1 \le \kappa < N$ and in $(0, 2\pi]$ when $\kappa = N$. Since the chosen multiplier argument satisfies $0 < \theta < 2\pi$, this forces $m_j = 0$ and proves (4.15). At an endpoint contact the image is $x_j$, forcing equality with $\Theta_j$. Iteration gives (4.16); replacing the final endpoint by a relative-interior contact gives (4.17). ∎

For the vertex-replacement and return theory we now assume

$$1 \le \kappa < N. \tag{4.18}$$

The identity cyclic side permutation $\kappa = N$ is treated directly in lemma 8.4; it requires no vertex replacement or projective deformation.

## 5 Vertex replacement and interval reduction

Call contact $i$ a *relative-interior contact* if $\beta_i > 0$ and an *endpoint contact* if $\beta_i = 0$.

Write

$$S = \{i \in \mathbb{Z}/N\mathbb{Z} : \beta_i > 0\}$$

for the set of indices whose assigned image points lie in the relative interiors of their sides. When $i \in S$ and $i+1 \notin S$, clipping the corner $x_i$ replaces the index $i$ in this set by $i+\kappa$:

$$S' = (S \setminus \{i\}) \cup \{i+\kappa\}.$$

If $i+\kappa$ already belongs to $S$, this union has one fewer element. This is only a statement about the index set: no geometric image vertices coalesce. The old target image is replaced when $x_i$ is removed, and multiplication by $\lambda \neq 0$ keeps the $N$ images of the $N$ primed vertices distinct. The next proposition proves that this set update is realized by an actual vertex replacement and preserves the strict polygon structure, invariance, the half-open assignment, and every unchanged contact.

**Coinciding-index cases.** The vertex-replacement theorem is used only in the proper-shift range (4.18). Put $j_0 = i+\kappa$ in $\mathbb{Z}/N\mathbb{Z}$. The complete list of possible index coincidences is

$$j_0 = i+1 \iff \kappa = 1, \qquad j_0 - 1 = i+1 \iff \kappa = 2, \qquad j_0 = i-1 \iff \kappa = N-1.$$

The first equality is the only overlap between the changed source index and the changed target-side index. The second is the only case in which the preceding image $A$ below is the unchanged endpoint $x_{i+1}$ of a changed side. In the third case the changed image is assigned to the preceding side index $i-1$, but neither endpoint of that target side is the replaced vertex $x_i$; hence it belongs to the generic incidence row. For $N \geq 4$, the cases $\kappa = 1$, $\kappa = 2$, $\kappa = N-1$, and $3 \leq \kappa \leq N-2$ are disjoint and exhaustive, with the last two sharing the same algebraic formula. When $N = 3$, the cases $\kappa = 2$ and $\kappa = N-1$ coincide, and the $\kappa = 2$ row applies. All equalities and side labels in the proposition below are interpreted modulo $N$, so wrap-around creates no additional case.

**Proposition 5.1** (Exact vertex replacement and its set update)**.** *Let*

$$(P, \lambda, \kappa; (\alpha_j, \beta_j)_{j\in\mathbb{Z}/N\mathbb{Z}})$$

*be contact data satisfying* (A0)–(A4) *with* $1 \leq \kappa < N$*, and put*

$$S = \{j : \beta_j > 0\}.$$

*Suppose* $i \in S$ *and* $i+1 \notin S$*. Thus*

$$\xi_i = \beta_i x_{i-1} + \alpha_i x_i \in \operatorname{relint} E_i, \qquad \xi_{i+1} = x_{i+1}.$$

*Set*

$$x'_i = \xi_i, \qquad x'_j = x_j \quad (j \neq i), \qquad P' = \operatorname{conv}\{x'_0, \ldots, x'_{N-1}\},$$

*and put* $\eta_j = \lambda x'_{j-\kappa}$*. Then* $P'$ *is a strict invariant* $N$*-gon and every* $\eta_j$ *belongs to* $E'^{+}_j$*. Consequently there are unique coefficients* $(\alpha'_j, \beta'_j)$ *satisfying*

$$\eta_j = \beta'_j x'_{j-1} + \alpha'_j x'_j, \qquad \alpha'_j > 0, \quad \beta'_j \geq 0, \quad \alpha'_j + \beta'_j = 1,$$

*and the primed tuple*

$$(P', \lambda, \kappa; (\alpha'_j, \beta'_j)_{j\in\mathbb{Z}/N\mathbb{Z}})$$

*again satisfies* (A0)–(A4)*. The chosen representative $\kappa \in \{1, \dots, N-1\}$ of the cyclic shift is unchanged. More precisely, for some $t \in (0, 1]$ the new coefficients are*

$$\alpha'_i = 1, \quad \beta'_i = 0, \qquad \alpha'_{i+\kappa} = \alpha_i t, \quad \beta'_{i+\kappa} = 1 - \alpha_i t, \tag{5.1}$$

*and $(\alpha'_j, \beta'_j) = (\alpha_j, \beta_j)$ for $j \notin \{i, i+\kappa\}$. The corresponding index set after replacement is*

$$S' = (S \setminus \{i\}) \cup \{i + \kappa\}. \tag{5.2}$$

*In particular, if $i + \kappa \in S$, the number of relative-interior contacts decreases by one.*

*Proof.* All indices are modulo $N$. Write

$$D(a, b, c) = \det(b - a, c - a).$$

For distinct vertices of a positively oriented strict polygon, $D(x_a, x_b, x_c) > 0$ whenever $x_a, x_b, x_c$ occur in positive cyclic order.

Put $\alpha = \alpha_i$ and $\beta = \beta_i$. The relative-interior condition at contact $i$ gives

$$\xi_i = \beta x_{i-1} + \alpha x_i, \qquad 0 < \alpha, \beta < 1, \qquad \alpha + \beta = 1.$$

*Step 1: intersect with the retained half-plane bounded by an image edge.* The global half-open assignment puts no image vertex in the boundary arc from $\xi_i$ to $\xi_{i+1} = x_{i+1}$, apart from its endpoints. Since the cyclic order of the vertices of $Q = \lambda P$ agrees with their order on $\partial P$, the two points $\xi_i$ and $\xi_{i+1}$ are consecutive vertices of $Q$. Hence, for

$$H = \{z : D(\xi_i, x_{i+1}, z) \geq 0\},$$

one has $Q \subseteq H$. The sign of every displayed vertex of $P$ relative to the cutting line is explicit:

$$D(\xi_i, x_{i+1}, x_i) = -\beta D(x_{i-1}, x_i, x_{i+1}) < 0, \tag{5.3}$$

$$D(\xi_i, x_{i+1}, x_k) = \beta D(x_{i-1}, x_{i+1}, x_k) + \alpha D(x_i, x_{i+1}, x_k) > 0 \tag{5.4}$$

for $k \notin \{i, i+1\}$, while $D(\xi_i, x_{i+1}, x_{i+1}) = 0$. In (5.4), convexity makes the first summand nonnegative and the second positive; this also covers $k = i - 1$. Thus the cutting line meets $\partial P$ exactly at $\xi_i$ and $x_{i+1}$, and $x_i$ is the only discarded displayed vertex. Consequently

$$P' = P \cap H = \operatorname{conv}(x_0, \dots, x_{i-1}, \xi_i, x_{i+1}, \dots, x_{N-1}). \tag{5.5}$$

Since $Q \subseteq P' \subseteq P$,

$$\lambda P' \subseteq \lambda P = Q \subseteq P'. \tag{5.6}$$

For completeness, the only changed consecutive-turn determinants in the displayed boundary list are

$$D(x_{i-2}, x_{i-1}, \xi_i) = \alpha D(x_{i-2}, x_{i-1}, x_i) > 0,$$
$$D(x_{i-1}, \xi_i, x_{i+1}) = \alpha D(x_{i-1}, x_i, x_{i+1}) > 0,$$
$$D(\xi_i, x_{i+1}, x_{i+2}) = \beta D(x_{i-1}, x_{i+1}, x_{i+2}) + \alpha D(x_i, x_{i+1}, x_{i+2}) > 0.$$

All other consecutive-turn determinants are unchanged. Together with the convex intersection description (5.5), these inequalities show that the displayed list consists of $N$ distinct extreme points in positive cyclic order. Hence $P'$ is a strict invariant $N$-gon.

Because the multiplier is the same $N$-critical map, theorem 3.2 additionally gives full side and vertex touching for $P'$. Thus the only remaining part of the standing assumptions is the labeled right-half-open incidence.

*Step 2: all unchanged sources and target sides are accounted for.* Put $j_0 = i + \kappa$. Apart from the location of the changed image $\eta_{j_0}$, the complete incidence table is

| side index $j$ | new image | target side | contact type |
|---|---|---|---|
| $i$ | $\eta_i = \xi_i = x'_i$ | $E'_i = [x_{i-1}, \xi_i]$ | endpoint |
| $i+1$, $\kappa \neq 1$ | $\eta_{i+1} = \xi_{i+1} = x_{i+1}$ | $E'_{i+1} = [\xi_i, x_{i+1}]$ | endpoint |
| $j_0$ | $\eta_{j_0} = \lambda \xi_i$ | $E'_{j_0}$ | checked below |
| $j \notin \{i, i+1, j_0\}$ | $\eta_j = \xi_j$ | $E'_j = E_j$ | unchanged |

The rows are disjoint: if $\kappa = 1$, the second row is absent and $j_0 = i+1$; otherwise $i, i+1, j_0$ are distinct. Thus only index $j_0$ remains to be located.

*Step 3: the changed image has exactly one possible side.* Set

$$A = \lambda x_{i-1} = \xi_{j_0-1}, \qquad B = \lambda x_i = \xi_{j_0}.$$

Then

$$\eta_{j_0} = \lambda \xi_i = \beta A + \alpha B \in \operatorname{relint}[A, B]. \tag{5.7}$$

The endpoint incidences after replacement are exhausted by

| $\kappa$ | $A$ | $B$ |
|---|---|---|
| 1 | $A = x'_i \in E'^{+}_i = E'^{+}_{j_0-1}$ | $B = x_{i+1} \in E'^{+}_{i+1} = E'^{+}_{j_0}$ |
| 2 | $A = x_{i+1} \in E'^{+}_{i+1} = E'^{+}_{j_0-1}$ | $B = \xi_{i+2} \in E^{+}_{i+2} = E'^{+}_{j_0}$ |
| $\kappa \notin \{1, 2\}$ | $A = \xi_{j_0-1} \in E^{+}_{j_0-1} = E'^{+}_{j_0-1}$ | $B = \xi_{j_0} \in E^{+}_{j_0} = E'^{+}_{j_0}$ |

Thus, uniformly in $\kappa$,

$$A \in E'^{+}_{j_0-1}, \qquad B \in E'^{+}_{j_0}. \tag{5.8}$$

The point $x'_i$ is an extreme point of $P'$, and multiplication by the nonzero scalar $\lambda$ maps extreme points bijectively to extreme points of $\lambda P'$. Thus $\eta_{j_0} = \lambda x'_i$ is a vertex of $\lambda P'$. Full vertex touching gives $\eta_{j_0} \in \partial P'$. Since $\eta_{j_0} = \beta A + \alpha B$ with $A \neq B$ and $0 < \alpha, \beta < 1$, the hypotheses of lemma 4.5 apply to the primed polygon, using (5.8). Hence

$$A = x'_{j_0-1}, \qquad [A, B] \subseteq E'_{j_0},$$

Notice that when $\kappa \geq 2$, contact $j_0 - 1 = i + \kappa - 1$ was already an endpoint contact before the replacement. For $\kappa = 2$, this is the permitted-replacement hypothesis $i + 1 \notin S$. For $\kappa \geq 3$, the vertex in question is unchanged, and the identities

$$A = \xi_{j_0-1} = x'_{j_0-1} = x_{j_0-1}$$

imply $\beta_{j_0-1} = 0$. Thus the "unchanged" contact-type entry for this index in Step 2 is consistent; the endpoint property is a consequence of the existing hypotheses, not an additional assumption.

Because $B \in E'^{+}_{j_0}$ and $B \neq A$, there is a unique $t \in (0, 1]$ such that

$$B = A + t(x'_{j_0} - A).$$

Substitution in (5.7) now yields

$$\eta_{j_0} = A + \alpha t(x'_{j_0} - A), \qquad 0 < \alpha t < 1.$$

Therefore $\eta_{j_0} \in \operatorname{relint} E'_{j_0}$; equivalently, its new barycentric coefficients are

$$\alpha'_{j_0} = \alpha_i t \in (0, 1), \qquad \beta'_{j_0} = 1 - \alpha_i t \in (0, 1).$$

*Step 4: complete assignment and check every side inequality.* Thus contact $i$ changes from relative-interior to endpoint contact, contact $i+\kappa$ lies in the relative interior after replacement, and every other contact retains its type. This proves the coefficient update (5.1) and the set update (5.2). We now verify all labelled side incidences, not merely the local ones shown in the construction.

For the strict polygon $P'$ put

$$C'_{r,k} = \det(x'_r - x'_{r-1}, x'_k - x'_{r-1})$$

with cyclic indices, and extend this notation by $C'_{r,r-1} = C'_{r,r} = 0$. Since $P'$ is strict and positively oriented,

$$P' = \{z : \det(x'_r - x'_{r-1}, z - x'_{r-1}) \geq 0 \text{ for every } r\},$$

and the half-open sides $E'^{+}_r$ form a disjoint partition of $\partial P'$. The following list exhausts the incidences.

(a) $\eta_i = x'_i \in E'^{+}_i$.
(b) If $\kappa \neq 1$, then $\eta_{i+1} = x'_{i+1} \in E'^{+}_{i+1}$.
(c) If $j \notin \{i, i+1, i+\kappa\}$, then the source and the target side are unchanged, so $\eta_j = \xi_j \in E^{+}_j = E'^{+}_j$.
(d) For $j_0 = i + \kappa$, the preceding step proved $\eta_{j_0} \in \operatorname{relint} E'_{j_0}$.

These alternatives are disjoint and cover all $j \in \mathbb{Z}/N\mathbb{Z}$; when $\kappa = 1$ the second item is intentionally absent and index $i + 1$ is covered by the final item. Hence each image vertex lies on its specified half-open side, and the disjoint half-open partition precludes any conflict between the assignments.

Equivalently, lemma 4.6, applied to the primed vertices and the just-listed points $\eta_j$, gives the complete side matrix. In expanded form: if $\eta_j = x'_j$, then

$$\det(x'_r - x'_{r-1}, \eta_j - x'_{r-1}) = C'_{r,j} \geq 0;$$

if $\eta_j = (1 - t_j)x'_{j-1} + t_j x'_j$ with $0 < t_j < 1$, then for every $r$

$$\det(x'_r - x'_{r-1}, \eta_j - x'_{r-1}) = (1 - t_j)C'_{r,j-1} + t_j C'_{r,j} \geq 0.$$

The formula gives the exact zero on the assigned side and strict positivity away from incident sides. Thus the labeled incidences independently imply $\lambda P' \subseteq P'$; the earlier inclusion (5.6) was not being used to infer the side assignment. Finally, the algebraic identities remain

$$\eta_j = \lambda x'_{j-\kappa} \qquad (j \in \mathbb{Z}/N\mathbb{Z}),$$

with the same chosen representative $\kappa \in \{1, \ldots, N-1\}$ of the cyclic shift. Since the target side index of the source $j - \kappa$ is still exactly $j$, this chosen representative, not only its residue class, is preserved. Together with the conclusion of theorem 3.2 above, this proves that the primed tuple satisfies (A0)–(A4). □

**Corollary 5.2** (Equivariance under the label-preserving bijection between old and new side sets)**.** *Let $(P, T, \chi)$ be the half-open contact bijection obtained in the orientation selected by lemma 4.13, and write its positive indexed form as in lemma 4.13. Thus*

$$E_i = [x_{i-1}, x_i], \qquad \chi(x_{i-\kappa}) = E_i.$$

*Then*

$$\operatorname{succ}(E_i) = E_{i+1}, \qquad \sigma(E_i) = E_{i+\kappa}. \tag{5.9}$$

*Let a permitted vertex replacement at $e = E_i$ produce the primed half-open contact assignment, write $E'_j = [x'_{j-1}, x'_j]$, and let*

$$b = b_e : \mathcal{E}(P) \longrightarrow \mathcal{E}(P'), \qquad b(E_j) = E'_j,$$

*be the label-preserving bijection between the old and new side sets. Denote the primed successor, the set of sides with relative-interior contact, and the induced cyclic permutation of side labels by* $\operatorname{succ}'$, $I'$, *and* $\sigma'$, *respectively. The permitted replacement condition is exactly* $i \in S$, $i+1 \notin S$, *and the vertex-replacement proposition gives the well-typed identities*

$$\begin{aligned} b \circ \operatorname{succ} &= \operatorname{succ}' \circ b, \\ I' &= b((I \setminus \{e\}) \cup \{\sigma(e)\}), \\ \sigma' &= b \circ \sigma \circ b^{-1}. \end{aligned} \tag{5.10}$$

*Under an invertible real-linear conjugacy, with the transported boundary orientation, these identities are carried by the induced old and new side bijections. In particular, the earlier choice between the two ambient orientations introduces no additional sign or index convention into* (5.10).

*Proof.* The successor identity in (5.9) is the definition of the positive side indexing. Since the head of $E_i$ is $x_i$, its image under the induced cyclic permutation is

$$\sigma(E_i) = \chi(\operatorname{head}(E_i)) = \chi(x_i) = E_{i+\kappa},$$

because $x_i = x_{(i+\kappa)-\kappa}$. The sides in $I$ are precisely those whose assigned image points lie in their relative interiors, and the condition $\operatorname{succ}(E_i) \notin I$ is precisely $i+1 \notin S$. The indexed update (5.2), together with $b(E_j) = E'_j$, gives

$$I' = b((I \setminus \{e\}) \cup \{\sigma(e)\}).$$

Moreover,

$$\operatorname{succ}'(b(E_j)) = E'_{j+1} = b(\operatorname{succ}(E_j)),$$

and preservation of the chosen representative $\kappa$ of the cyclic shift in proposition 5.1 gives

$$\sigma'(b(E_j)) = E'_{j+\kappa} = b(\sigma(E_j)).$$

These are exactly the three identities in (5.10). Real-linear covariance follows from (2.3) in proposition 2.3. The chosen orientation is already the positively indexed orientation used above, including when it arose from the conjugate alternative in lemma 4.11; no subsequent orientation reversal occurs in the vertex-replacement module. □

**Corollary 5.3** (Geometric realization of every permitted update sequence)**.** *Starting from the half-open contact assignment, every finite sequence of updates* $S \mapsto (S \setminus \{i\}) \cup \{i+\kappa\}$ *for which* $i \in S$ *and* $i+1 \notin S$ *at that step is realized by successive strict invariant* $N$*-gons satisfying* (A0)–(A4)*, with the same chosen representative* $\kappa$ *of the cyclic shift. The index set* $S$ *after each step is exactly the set produced by* (5.2).

*Proof.* Apply proposition 5.1 inductively. Its conclusion includes preservation of the right-half-open assignment and of $\kappa$, so the hypotheses and the description by the corresponding index set $S$ remain valid at every later step. □

**Corollary 5.4** (Right-to-left set updates are geometrically realizable)**.** *Every right-to-left sequence of set updates used in the proof of lemma 5.5 is a sequence of actual vertex replacements. More precisely, start from the current index set* $S^{(0)}$ *and define*

$$S^{(r)} = (S^{(r-1)} \setminus \{i_r\}) \cup \{i_r + \kappa\} \qquad (1 \le r \le m).$$

*If* $i_r \in S^{(r-1)}$ *and* $i_r + 1 \notin S^{(r-1)}$ *at every step, then there are strict invariant* $N$*-gons*

$$P = P^{(0)}, P^{(1)}, \ldots, P^{(m)}$$

*with the same multiplier and the same chosen representative $\kappa$ of the cyclic shift whose sets of indices with relative-interior contact are exactly the sets produced after the corresponding updates.*

*Proof.* This is a finite induction on the length of the sequence. At each stage the permitted-replacement condition means precisely that the contact at the current index lies in the relative interior and the contact at the next index is an endpoint, which is the hypothesis of proposition 5.1. That proposition returns a strict invariant polygon satisfying (A0)–(A4), with the corresponding index set $S$ updated by (5.2). Since the standing assumptions and the chosen representative $\kappa$ of the cyclic shift are part of the conclusion, the next permitted set update is again realized by a vertex replacement. No separate geometric-reachability assumption enters lemma 5.5. $\square$

### 5.1 Reduction to one cyclic interval

Let $C_N$ be the cycle graph on $\mathbb{Z}/N\mathbb{Z}$, with consecutive residues joined. For $S \subseteq \mathbb{Z}/N\mathbb{Z}$, write $\operatorname{comp}(S)$ for the number of connected components of the induced subgraph $C_N[S]$. When $S$ is proper, each such component is a maximal cyclic interval contained in $S$.

The reduction minimizes the pair $(|S|, \operatorname{comp}(S))$ lexicographically. An update whose target already belongs to the current set decreases the first coordinate, while a completed translation that joins two components decreases the second. Neither can occur from a minimizing set. Because the $\kappa$-orbits are exactly the residue classes modulo $\gcd(N, \kappa)$, every such class must meet $S$; otherwise a full orbit of endpoint equalities would imply $x_j = \lambda^{N/\gcd(N,\kappa)} x_j$. These two facts force a single cyclic interval of length at least $\gcd(N, \kappa)$.

**Lemma 5.5** (Reduction to one cyclic interval and a first-entrance identity)**.** *Put*

$$\delta = \gcd(N, \kappa). \tag{5.11}$$

*Starting from the half-open contact assignment fixed in lemma 4.13, consider only tuples satisfying* (A0)–(A4) *that are obtainable by a finite sequence of permitted vertex replacements. Choose one for which the following pair is lexicographically least:*

$$(|S|,\ \operatorname{comp}(S)). \tag{5.12}$$

*This minimization is only among such tuples reached by finitely many vertex replacements; it is neither a new vertex-count minimization nor an area minimization. Write $\varphi = |S|$. After a cyclic relabelling, its indices with relative-interior contact form one cyclic interval*

$$\{1, 2, \ldots, \varphi\}, \qquad \varphi \geq \delta. \tag{5.13}$$

*If $\varphi < N$, then for some $h > 0$*

$$[h\kappa]_N = N - \varphi, \qquad [m\kappa]_N < N - \varphi \quad (0 \leq m < h), \tag{5.14}$$

*where $[\cdot]_N$ denotes the representative in $\{0, \ldots, N-1\}$. For $\varphi = N$, the corresponding case is the time-zero identity $[0\kappa]_N = 0$.*

*Proof.* Let $\mathcal{R}$ be the family of corresponding index sets of reachable representatives. It is a nonempty subset of the finite set $2^{\mathbb{Z}/N\mathbb{Z}}$, so a lexicographic minimizer $S \in \mathcal{R}$ exists. If $S = \mathbb{Z}/N\mathbb{Z}$, the conclusion holds with $\varphi = N$ and the stated time-zero identity. Assume henceforth that $S \neq \mathbb{Z}/N\mathbb{Z}$; every connected component of $C_N[S]$ is then a proper cyclic interval.

We first record the complete transition table for updating one connected component from right to left. Let $G = \{a, a+1, \ldots, b\}$ be a connected component of $C_N[S]$, let $\ell = |G| < N$, and put $R = S \setminus G$. Starting from $S^{(0)} = S$, consider, in order, the updates at

$$j_r = b - r, \qquad 0 \le r < \ell,$$

and denote the resulting sets by

$$S^{(r+1)} = (S^{(r)} \setminus \{j_r\}) \cup \{j_r + \kappa\}.$$

Until the first step for which $j_r + \kappa \in S^{(r)}$, every source $j_r$ belongs to $S^{(r)}$ when used: it belonged to $G$ initially, has not yet appeared as a source, and earlier updates remove only their distinct sources. For $r = 0$, its right neighbor $b+1$ does not belong to $S$ by maximality of $G$. For $r > 0$, the right neighbor $j_r + 1 = j_{r-1}$ was removed in the preceding update. Before the first target-overlap step it cannot have been reinserted by an earlier target: if $j_s + \kappa = j_{r-1}$ for some $s < r-1$, then $j_{r-1}$ still belonged to the current set when the update at $j_s$ was made, so that earlier target would already have belonged to the current set; the case $s = r-1$ would force $\kappa \equiv 0 \pmod N$, contrary to $1 \le \kappa < N$. Hence every update up to and including the first target-overlap step satisfies the permitted-replacement condition $j_r \in S^{(r)}$, $j_r + 1 \notin S^{(r)}$. Before that step, direct induction gives the disjoint-union invariant

$$S^{(r)} = R \mathbin{\dot\cup} \{a, \ldots, b-r\} \mathbin{\dot\cup} (\{b-r+1, \ldots, b\} + \kappa). \tag{5.15}$$

The pair $(|S^{(r)}|, \mathrm{comp}(S^{(r)}))$ at the first target-overlap step, or at the end when no target overlap occurs, is therefore exactly as follows.

| event | cardinality | component count |
|---|---|---|
| $j_r + \kappa \in S^{(r)}$ for the first time | $\lvert S^{(r+1)}\rvert = \lvert S\rvert - 1$ | irrelevant |
| no target overlap and $G + \kappa$ is not adjacent to $R$ | $\lvert S^{(\ell)}\rvert = \lvert S\rvert$ | same as $\mathrm{comp}(S)$ |
| no target overlap and $G + \kappa$ is adjacent to $R$ | $\lvert S^{(\ell)}\rvert = \lvert S\rvert$ | at most $\mathrm{comp}(S) - 1$ |

Indeed, in the last two rows

$$S^{(\ell)} = R \mathbin{\dot\cup} (G + \kappa). \tag{5.16}$$

This table also makes the minimality consequence and the repeatability precise. During a partial right-to-left update, an adjacency of the translated suffix to a component of $R$ does not obstruct the next permitted replacement: the next source's right neighbor does not belong to the current set, by the right-to-left order. If such a partial adjacency ever lowers the component count, the reachable intermediate state already contradicts lexicographic minimality; otherwise it is ignored, and only the completed translate is used in the repeatability invariant. For later use, after $m$ completed right-to-left update sequences write

$$G_m = G + m\kappa, \qquad T_m = R \mathbin{\dot\cup} G_m.$$

The repeatability invariant is that $T_m$ is reachable with

$$(|T_m|, \mathrm{comp}(T_m)) = (|S|, \mathrm{comp}(S)),$$

while $G_m$ is a proper connected component of $C_N[T_m]$, disjoint from and nonadjacent to every connected component contained in $R$. It holds at $m = 0$: $G$ is a proper connected component of $C_N[S]$ and $R = S \setminus G$.

Suppose it holds at $m$. Apply the same transition table to the connected component $G_m$ in $T_m$. A target already belonging to the current set would produce a reachable set of smaller cardinality, and an overlap-free sequence whose terminal translate is adjacent to $R$ would produce one with the same cardinality and fewer connected components. Both contradict the lexicographic minimality of $S$. Thus no target overlap occurs, the completed translate is nonadjacent to $R$, and

$$T_{m+1} = R \mathbin{\dot\cup} G_{m+1}.$$

Translation preserves $|G_{m+1}| = |G| < N$, so $G_{m+1}$ is proper; its nonadjacency to the unchanged set $R$ makes it a connected component of $C_N[T_{m+1}]$. The minimizing pair is unchanged, and hence the invariant holds at $m+1$. Therefore the right-to-left update sequence may be repeated as many times as required. No assertion about the number of connected components during a partial sequence is being used.

Every residue class modulo $\delta$ meets $S$. If one did not, all $L = N/\delta$ indices in the corresponding $\kappa$-orbit would be endpoint contacts. Iterating their identities $\lambda x_{j-\kappa} = x_j$ gives

$$x_j = \lambda^L x_j,$$

contrary to $x_j \neq 0$ and $|\lambda| < 1$.

Distinct connected components of $C_N[S]$ have disjoint residue sets modulo $\delta$. For if $p \in G$ and $q \in H$, where $G \neq H$ are connected components and $p \equiv q \pmod{\delta}$, there is $t \in \{1, \ldots, L-1\}$ such that

$$p + t\kappa = q \pmod{N}.$$

Translate $G$ through $t$ complete right-to-left update sequences, leaving all other connected components fixed. The preceding transition argument permits each repetition unless it has already produced a set with smaller cardinality or a completed translate with fewer connected components; either outcome contradicts minimality. Otherwise, during the $t$-th sequence, the index that started at $p$ reaches the index $q$, which already belongs to $H$. The corresponding update would reduce the cardinality, again contradicting minimality. This proves residue disjointness.

If some connected component has at least $\delta$ consecutive indices, its residue set is all of $\mathbb{Z}/\delta\mathbb{Z}$, and residue disjointness excludes every other connected component. Suppose instead that every connected component has length less than $\delta$. The residue set of each component is then a proper cyclic interval in $\mathbb{Z}/\delta\mathbb{Z}$. These intervals are pairwise disjoint and, by residue coverage, cover the cyclic group $\mathbb{Z}/\delta\mathbb{Z}$. Two of them are therefore consecutive. Write the corresponding connected components as $G = \{a, \ldots, b\}$ and $H = \{c, \ldots, d\}$, choosing their terminal and initial indices so that

$$\delta \mid (c - b - 1).$$

Because the subgroup generated by $\kappa$ is $\delta\mathbb{Z}/N\mathbb{Z}$, there is $t \in \{0, \ldots, L-1\}$ with

$$t\kappa \equiv c - b - 1 \pmod{N}. \tag{5.17}$$

Here $t \neq 0$, since $t = 0$ would make $b$ and $c$ adjacent indices and would therefore place them in the same connected component. Translate $G$ through $t$ complete right-to-left update sequences. If a target already belongs to the current set, the resulting set has smaller cardinality. Otherwise the terminal index of the final translate is $c-1$, so the final translate is adjacent to $H$. The third row of the transition table then gives the same cardinality and fewer connected components. Both alternatives contradict minimality. Hence $C_N[S]$ has one connected component. Since it meets every residue class, the length $\varphi$ of this cyclic interval is at least $\delta$. A cyclic relabelling gives (5.13).

Assume now that $\varphi < N$. Relabel temporarily so that

$$S = F \mathbin{\dot\cup} \{0\}, \qquad F = \{N - \varphi + 1, \ldots, N-1\}.$$

Put

$$a_m = [m\kappa]_N, \qquad I = \{N-\varphi, \dots, N-1\}.$$

The orbit of 0 consists of the multiples of $\delta$ modulo $N$. Because $|I| = \varphi \geq \delta$, the interval $I$ meets this orbit; because $0 \notin I$, there is a least $h > 0$ such that $a_h \in I$.

The exact single-index induction is

$$S_m = F \,\dot\cup\, \{a_m\} \qquad (0 \leq m \leq h), \tag{5.18}$$

where $S_m$ is reachable from $S$ by successively updating only the displayed index. For $m = 0$ the formula reads $S_0 = F \,\dot\cup\, \{0\} = S$, so it holds. If $m < h$, minimality of $h$ gives $a_m < N - \varphi$, and therefore

$$a_m + 1 \notin F \cup \{a_m\};$$

the update at $a_m$ is permitted. If its target $a_{m+1}$ belonged to $F$, the updated set would have cardinality $\varphi - 1$, contrary to minimality. Thus $a_{m+1}$ does not belong to the current set, and (5.18) follows at $m+1$. At the first entrance time one consequently has

$$a_h \in I \setminus F = \{N - \varphi\},$$

whereas $a_m \notin I$, hence $a_m < N - \varphi$, for every $m < h$. This is exactly (5.14). The computation used displacements from the terminal index, so the final common cyclic relabelling to (5.13) changes neither $\kappa$ nor the residue statement. For $\varphi = N$, the time-zero identity $[0\kappa]_N = 0$ was already handled at the start of the proof. □

# 6 Finite rotation and first-return decomposition from the lattice sail

The cyclic arithmetic is isolated in this section. It depends only on the rotation of a finite cyclic set; no polygon is used until the final tower identities are read through endpoint contacts.

Fix integers

$$N \geq 2, \qquad 1 \leq \kappa < N, \qquad \delta = \gcd(N, \kappa),$$

and write $[a]_N$ for the representative of $a$ in $\{0, \dots, N-1\}$. Whenever an interval $\{1, \dots, \nu\}$ is used as a set of cyclic labels, its entries are read modulo $N$. In the full-length case $\nu = N$, the terminal label $N$ therefore denotes the residue 0; the displayed list records one complete circuit and does not introduce an additional label. Put

$$L(a,b) = a\kappa - bN. \tag{6.1}$$

A time $h \geq 0$ is an *upper-record time* when

$$[h\kappa]_N > [t\kappa]_N \qquad (0 \leq t < h).$$

Its deficit is $\nu = N - [h\kappa]_N$, with the convention that time zero has deficit $N$. Equivalently, with

$$b = \begin{cases} 1, & h = 0, \\ \lceil h\kappa/N \rceil, & h > 0, \end{cases} \qquad V = (h, b),$$

one has

$$L(V) = -\nu, \qquad [t\kappa]_N < N - \nu = [h\kappa]_N \quad (0 \leq t < h). \tag{6.2}$$

The terminal record has deficit $\delta$, because the attainable residues are exactly the multiples of $\delta$ and $N - \delta$ is the largest one. The upper-record vectors are ordered by their time coordinates. Two of them are called consecutive when no upper-record time lies strictly between their time coordinates; the next upper-record vector is the latter member of such a pair.

**Theorem 6.1** (Finite rotation and first-return decomposition)**.** *Let $V = (h, b)$ be an upper-record vector with deficit $\nu > \delta$, and let $V' = (h', b')$ be the next upper-record vector. Define*

$$U = V' - V = (q, p), \qquad \Delta = \nu - \nu'.$$

*Since the records are listed chronologically, $h' > h$ and therefore $q = h' - h > 0$. Then*

$$q\kappa - pN = \Delta > 0, \qquad h\kappa - bN = -\nu, \tag{6.3}$$

$$0 < \Delta < \nu, \qquad \det(U, V) = 1, \tag{6.4}$$

$$q\nu + h\Delta = N, \qquad \gcd(\Delta, \nu) = \delta. \tag{6.5}$$

*The initial record $\nu = N$ has $V = (0, 1)$. Moreover $h = 0$ if and only if $\nu = N$.*

*Let*

$$q\kappa - pN = \Delta, \qquad h\kappa - bN = -\nu, \qquad q\nu + h\Delta = N, \tag{6.6}$$

*and define*

$$H_i = \begin{cases} q, & 1 \le i \le \nu - \Delta, \\ q + h, & \nu - \Delta < i \le \nu. \end{cases}$$

*Then*

$$\mathcal{D} = \{(t, i) : 1 \le i \le \nu,\ 0 \le t < H_i\} \longrightarrow \mathbb{Z}/N\mathbb{Z}, \qquad F(t, i) = [i + t\kappa]_N \tag{6.7}$$

*is a bijection. The first-return map on the base interval $\{1, \ldots, \nu\}$ is addition of $\Delta$ modulo $\nu$ and has the two return times displayed above.*

*If $\Delta = 1$, put*

$$w = \lfloor h/q \rfloor, \qquad E = V - wU = (e, c), \qquad d = -L(E) = \nu + w.$$

*Then $E$ is an upper-record vector, and*

$$q\kappa - pN = 1, \qquad e\kappa - cN = -d, \qquad qd + e = N, \qquad 0 \le e < q. \tag{6.8}$$

*In particular $d = \lfloor N/q \rfloor \ge \nu$, and*

$$E, E + U, \ldots, E + (w + 1)U$$

*are consecutive upper-record vectors.*

*Finally, suppose a polygonal contact assignment satisfying the standing conditions* (A0)–(A4) *has no relative-interior contact index outside the base interval $\{1, \ldots, \nu\}$ and has the record data above. Then every internal tower state has an assigned contact at a side endpoint, and*

$$\lambda^t x_i = x_{F(t,i)} \qquad (0 \le t < H_i). \tag{6.9}$$

*The top and closure relations are*

$$\lambda^q x_i = \xi_{i+\Delta} \qquad (1 \le i \le \nu - \Delta), \tag{6.10}$$

$$\lambda^{q+h} x_i = \xi_{i+\Delta-\nu} \qquad (\nu - \Delta < i \le \nu), \tag{6.11}$$

$$\lambda^h x_\nu = x_0, \tag{6.12}$$

$$\lambda^q x_0 = \xi_\Delta. \tag{6.13}$$

*Proof.* We proceed in four stages.

*1. Consecutive record vectors are unimodular.* List the upper-record times chronologically. The list is finite and ends at deficit $\delta$. Indeed,

$$\gcd(\kappa/\delta, N/\delta) = 1,$$

so multiplication by $\kappa/\delta$ permutes the residue classes modulo $N/\delta$. Hence there is an $h$ such that

$$h\kappa/\delta \equiv -1 \pmod{N/\delta},$$

equivalently $h\kappa \equiv N - \delta \pmod N$. Thus the attainable residue $N-\delta$ occurs, and the upper-record list ends at deficit $\delta$. The time-zero record vector $V_0 = (0,1)$ is primitive. Every other record vector $V = (h,b)$ has $h > 0$ and is primitive as well. Indeed, if $V = gW$ with an integer $g > 1$, then $L(W) = -\nu/g < 0$, while the time coordinate of $W$ is a nonnegative integer strictly smaller than $h$. Its residue is $N - \nu/g > N - \nu$, contradicting the record property.

Let $V = (h,b)$ and $V' = (h',b')$ be consecutive records, with deficits $\nu > \nu' > 0$. Since

$$b = \frac{h\kappa + \nu}{N}, \qquad b' = \frac{h'\kappa + \nu'}{N},$$

we have

$$bh' - hb' = \frac{h'\nu - h\nu'}{N} > 0.$$

Thus $\det(V,V') < 0$. Suppose $D = -\det(V,V') > 1$. By lemma A.6, the half-open fundamental parallelogram generated by $V, V'$ contains a nonzero lattice point $W = \alpha V + \beta V'$ with $0 \le \alpha, \beta < 1$. Primitivity excludes a nonzero point on either radial edge. Replacing $W$ by $V + V' - W$ when $\alpha + \beta > 1$, we obtain a lattice point in the triangle $\operatorname{conv}\{0, V, V'\}$ different from its three vertices. Hence

$$W = \alpha V + \beta V', \qquad \alpha, \beta > 0, \qquad \alpha + \beta \le 1.$$

Write $W = (t,c)$. Its time $t = \alpha h + \beta h'$ is an integer with $0 < t < h'$, while

$$-L(W) = \alpha\nu + \beta\nu' < \nu.$$

Since $0 < -L(W) < \nu \le N$, one has $c = \lceil t\kappa/N \rceil$ and therefore

$$[t\kappa]_N = N + L(W) > N - \nu.$$

This contradicts that $V$ and $V'$ are consecutive upper-record vectors. Therefore $D = 1$ and

$$\det(V,V') = -1.$$

For $U = V' - V$, this is $\det(U,V) = 1$. Linearity of $L$ gives $L(U) = \nu - \nu' = \Delta$, so $0 < \Delta < \nu$. Expanding the determinant,

$$q\nu + h\Delta = q(bN - h\kappa) + h(q\kappa - pN) = N(qb - ph) = N.$$

Because $U, V$ are a unimodular basis, the subgroup generated by $L(U) = \Delta$ and $L(V) = -\nu$ equals $L(\mathbb{Z}^2)$. The latter is $\delta\mathbb{Z}$. Hence $\gcd(\Delta, \nu) = \delta$. The time-zero record has $h = 0$, $V = (0,1)$, and deficit $N$. Conversely, suppose $\nu = N$. Then $[h\kappa]_N = N - \nu = 0$. If $h > 0$, the upper-record inequality compared with time zero would require

$$[h\kappa]_N > [0\kappa]_N = 0,$$

which is impossible. Hence $h = 0$.

*2. The tower map is bijective.* Give $\mathcal{D}$ the successor map

$$\sigma(t,i) = \begin{cases} (t+1, i), & t+1 < H_i, \\ (0, i+\Delta), & t = q-1,\ i \le \nu - \Delta, \\ (0, i+\Delta-\nu), & t = q+h-1,\ i > \nu - \Delta. \end{cases}$$

The congruences in (6.6) give

$$F(\sigma(t,i)) = F(t,i) + \kappa \pmod{N}. \tag{6.14}$$

On the bases, $\sigma$ induces addition of $\Delta$ modulo $\nu$. It has $\delta = \gcd(\Delta, \nu)$ cycles, namely the residue classes modulo $\delta$. Each base cycle contains $\nu/\delta$ bases. The long-base labels form the block of $\Delta$ consecutive integers $\nu - \Delta + 1, \dots, \nu$. Since $\delta$ divides both $\nu$ and $\Delta$, this block contains exactly $\Delta/\delta$ representatives of each residue class modulo $\delta$. Hence each base cycle contains $\Delta/\delta$ long bases, so its total number of states is

$$q\frac{\nu}{\delta} + h\frac{\Delta}{\delta} = \frac{N}{\delta}.$$

Addition of $\kappa$ on $\mathbb{Z}/N\mathbb{Z}$ also has $\delta$ cycles of length $N/\delta$. Since $F(t,i) \equiv i \pmod{\delta}$, distinct base cycles map into distinct $\kappa$-orbits. Equation (6.14) makes the restriction to each cycle an equivariant map between two cycles of the same finite length, hence a bijection. This proves (6.7) and the return-time statement.

*3. When $\Delta = 1$, the preceding record vectors form the stated arithmetic progression.* Assume $\Delta = 1$. By construction $E = V - wU = (e, c)$ has $0 \le e < q$ and $L(E) = -(\nu + w) = -d$. The determinant remains $\det(U, E) = 1$, so

$$qd + e = N.$$

It remains to prove that $E$ and the displayed arithmetic progression are exactly consecutive record vectors.

For a putative earlier improving time $t$ for $E$, put

$$z = \lceil t\kappa/N \rceil, \qquad Z = (t, z).$$

If $[t\kappa]_N > N - d$, then $-d < L(Z) < 0$; a tie $[t\kappa]_N = N - d$ is represented by choosing the integer second coordinate so that $L(Z) = -d$ (with the declared time-zero convention when $d = N$). Thus excluding integer vectors in these ranges excludes every earlier improvement and tie. The same observation applies below with $d$ replaced by $d - j$.

Because $U, E$ are a basis, write any lattice vector $Z = (t, z)$ uniquely as $Z = AE + BU$. If $e = 0$, write $E = (0, c)$. The identity $\det(U, E) = 1$ becomes $qc = 1$. Since $q > 0$, it follows that $q = c = 1$, and hence

$$E = (0, 1), \qquad d = -L(E) = N.$$

Thus $E$ is precisely the declared time-zero record vector.

If $e > 0$, suppose that $0 \le t < e$ and $-d < L(Z) < 0$. The inequalities are

$$-d < -Ad + B < 0.$$

If $A \le 0$, then $B < Ad \le 0$ and $t = Ae + Bq < 0$. If $A = 1$, then $B \ge 1$ and $t \ge e + q$. If $A \ge 2$, then $B > (A-1)d \ge 1$ and again $t = Ae + Bq > e$. All alternatives contradict $0 \le t < e$. Hence $E$ is an upper record unless an earlier time has the same residue. Equality with the residue of $E$ would mean $L(Z) = -d$, so $B = (A-1)d$ and

$$t = Ae + (A-1)dq = e + (A-1)N,$$

where $qd + e = N$. No number of this form lies in $0 \le t < e$. Thus an earlier tie is impossible, and $E$ is an upper record when $e > 0$. Together with the time-zero case, this proves that $E$ is an upper record in all cases.

More generally, fix $j$ with $0 \le j \le w$. Since $d - w = \nu > \delta \ge 1$, one has $d - j \ge \nu > 1$. Suppose $0 \le t < e + (j+1)q$ while

$$-(d-j) < L(Z) < 0.$$

Equivalently,

$$-(d-j) < -Ad + B < 0.$$

If $A \le 0$, then $t < 0$. If $A = 1$, the left inequality gives $B > j$, so $B \ge j+1$ and $t \ge e+(j+1)q$. If $A \ge 2$, it gives $B > (A-1)d + j$, hence $B \ge j+1$ and $t = Ae + Bq \ge e + (j+1)q$. Thus no improved residue occurs before the time of $E + (j+1)U$, whose deficit is one less than that of $E + jU$. Induction shows that the displayed vectors are consecutive records. Since $V = E + wU$, this includes the given pair $V, V + U$. Finally $N = qd + e$ with $0 \le e < q$ gives $d = \lfloor N/q \rfloor$.

*4. Geometric tower identities.* Assume now that every relative-interior contact index belongs to $\{1, \dots, \nu\}$. If an internal state $F(t,i)$ with $1 \le t < H_i$ belonged to the base block, then it and the corresponding level-zero state would have the same image under the bijection $F$, impossible. Hence every internal destination index has its assigned contact at a side endpoint. If $j = F(t,i)$, the endpoint-contact identity at index $j$ is

$$\lambda x_{F(t-1,i)} = x_{F(t,i)}.$$

Induction gives (6.9). One further multiplication at a short or long top gives (6.10) and (6.11). Since $F(h,\nu) \equiv \nu + h\kappa \equiv 0$ and $h < H_\nu$, the internal identity gives (6.12); combining it with the long top at base $\nu$ gives the additional identity (6.13) at base 0. □

**Corollary 6.2** (Upper-record intervals extended by endpoint contacts)**.** *Let $S$ be the set of relative-interior contact indices of a contact assignment satisfying the standing conditions* (A0)–(A4)*. Suppose $U = (q,p)$ and $E = (e,c)$ are integer vectors for which $E$ and $E + U$ are consecutive upper-record vectors and*

$$q\kappa - pN = 1, \qquad e\kappa - cN = -d, \qquad qd + e = N, \qquad 0 \le e < q, \qquad d > 1. \tag{6.15}$$

*Assume*

$$S \subseteq J := \{1, \dots, d\}. \tag{6.16}$$

*Then the first-return decomposition over the base interval $J$ has height $q$ at bases $1, \dots, d-1$, height $q + e$ at base $d$, and first return equal to the successor. Its tower states form a bijection with the $N$ polygon labels. Every internal destination index lies outside $J$ and therefore has its assigned contact at a side endpoint, and*

$$\lambda^q x_{j-1} = \xi_j \qquad\qquad (1 \le j \le d), \tag{6.17}$$
$$\lambda^e x_d = x_0. \tag{6.18}$$

*For every $j \in J \setminus S$, one has $\xi_j = x_j$. Thus extending a consecutive block of relative-interior contact indices to such an upper-record interval adds only indices whose assigned contacts are side endpoints. At each added index one has $\beta_j = 0$ and $\alpha_j = 1$; no further return datum is needed in this topic.*

*Proof.* Since $q\kappa - pN = 1$, every common divisor of $N$ and $\kappa$ divides 1. Hence $\delta = \gcd(N,\kappa) = 1$, and the assumption $d > 1$ gives $d > \delta$. We may therefore apply theorem 6.1 to the consecutive record pair $V = E$, $V' = E + U$. In the notation of that theorem,

$$\nu = d, \qquad \Delta = 1, \qquad h = e.$$

The arithmetic assumptions in (6.15) are precisely (6.6); hence the tower map is bijective, its return map is addition of one modulo $d$, and its heights are the stated two values.

The geometric conclusion of theorem 6.1 requires only that there be no relative-interior contact index outside the chosen base interval. This is exactly (6.16). More explicitly, if an internal tower state had destination label $j \in J$, then it would have the same image under the tower bijection as the level-zero state $(0, j)$, which is impossible. Thus every internal destination lies outside $J$ and, since all relative-interior contact indices belong to $J$, its assigned contact is a side endpoint. The short-return identities for bases $1, \dots, d-1$ give $\lambda^q x_i = \xi_{i+1}$; the additional relation at base 0 gives $\lambda^q x_0 = \xi_1$; and the tower closure gives $\lambda^e x_d = x_0$. These are (6.17)–(6.18). Finally, in a contact assignment satisfying (A0)–(A4), an index outside $S$ has $\beta_j = 0$ in (4.12), so $\xi_j = x_j$ and the asserted endpoint-extension statement follows. □

*Remark* 6.3. Let $C$ be the closed cone bounded by the positive $b$-axis and the ray $L(h, b) = 0$. The Klein sail of $C$ is the boundary visible from the origin of

$$\operatorname{conv}\big((C \cap \mathbb{Z}^2) \setminus \{0\}\big).$$

Theorem 6.1 proves that every record vector is primitive and that every consecutive pair is unimodular. These are the only lattice-geometric properties used below. No identification of the record chain with the sail, or of every record vector with a sail vertex, is asserted or needed. The sail and continued fractions remain useful geometric context for the arithmetic.

**Why the projective proof of $\Delta = 1$ starts at four vertices.** The finite-rotation theorem above remains valid for $N \geq 2$, but the projective proof that the first return reaches the cyclic successor assumes $N \geq 4$. This restriction is essential. For $1/2 < a < 1$, the doubly stochastic matrix

$$B_a = \begin{pmatrix} 0 & 1-a & a \\ a & 0 & 1-a \\ 1-a & a & 0 \end{pmatrix}$$

has eigenvalues, counted with algebraic multiplicity, 1, $\lambda_a = -1/2 + i\sqrt{3}(2a-1)/2$, and $\overline{\lambda_a}$. It preserves the plane $u_0 + u_1 + u_2 = 0$ and restricts there to the elliptic contraction $T_a$ with eigenvalues $\lambda_a$ and $\overline{\lambda_a}$. The centred standard simplex $P$ is independent of $a$ and is an invariant triangle with $(\varphi, \kappa, \Delta) = (3, 2, 2)$. No invariant segment is possible for a nonreal planar map, while any invariant triangle for a radial enlargement $tT_a$, $t > 1$, would yield a nonnegative stochastic $3 \times 3$ realization with trace $1 + 2t \operatorname{Re} \lambda_a = 1 - t < 0$, which is impossible. Consequently $\nu_{\text{poly}}(T_a) = 3$ and $\nu_{\text{poly}}(tT_a) > 3$ for every $t > 1$: in the terminology of definition 1.1, each $T_a$ is 3-critical. Thus the conclusion $\Delta = 1$ is false for $N = 3$. The small orders, including order three, are treated directly in proposition II.9.2.

# 7 Why the first return cannot skip the cyclic successor

Throughout this section assume $N \geq 4$. The positive integer $\Delta = \nu - \nu'$ from theorem 6.1 is the advance of its first-return map on the base interval. Assume $\varphi > \delta$, so the interval of relative-interior contact indices contains more labels than the number $\delta$ of $\kappa$-orbits. The first-return map is the translation $i \mapsto i + \Delta$ on that interval. The proof that it lands at the next base label, equivalently $\Delta = 1$, is begun in this topic and completed in Topic VI. It is separated into a preparatory part and a deformation part. The results through proposition 7.5 establish only the preparation: the return towers produce consecutive boundary vertices while omitting at least one side, account for every nonclosing source–target pair, and provide an affine chart adapted to the resulting composition of perspectivities. They do not yet rule out $\Delta > 1$. The subsequent projective deformation opens the single unconstrained closing contact and completes the contradiction. The contradiction is with the literal image-vertex boundary conclusion of theorem 3.2: every vertex of $TR$ must lie on $\partial R$ for every invariant polygon $R$ with at most $N$ vertices.

Under this standing assumption $\varphi > \delta$, lemma 5.5 gives

$$[h\kappa]_N = N - \varphi, \qquad [t\kappa]_N < N - \varphi \quad (0 \le t < h)$$

when $\varphi < N$. Hence $h > 0$ is an upper-record time and its deficit is $\varphi$. When $\varphi = N$, the same role is played by the time-zero record $h = 0$, whose deficit is $N = \varphi$ by convention. Thus in both cases theorem 6.1 applies with $\nu = \varphi$ and gives integers $q, h, p, b, \Delta$ such that

$$q\kappa - pN = \Delta, \qquad h\kappa - bN = -\varphi, \qquad q\varphi + h\Delta = N, \qquad 0 < \Delta < \varphi. \tag{7.1}$$

We shall prove that the alternative $\Delta > 1$ is impossible. For $1 \le i \le \varphi$, define the return height $H_i$ and return target index $r(i)$ by

$$H_i = \begin{cases} q, & 1 \le i \le \varphi - \Delta, \\ q + h, & \varphi - \Delta < i \le \varphi, \end{cases} \qquad r(i) = \begin{cases} i + \Delta, & 1 \le i \le \varphi - \Delta, \\ i + \Delta - \varphi, & \varphi - \Delta < i \le \varphi. \end{cases} \tag{7.2}$$

Thus the two top relations (6.10)–(6.11) are the single identity

$$\lambda^{H_i} x_i = \xi_{r(i)} \in \operatorname{relint} E_{r(i)} \qquad (1 \le i \le \varphi). \tag{7.3}$$

The additional relation at base index 0 is

$$\lambda^q x_0 = \xi_\Delta. \tag{7.4}$$

For the remainder of the finite-return construction, through proposition 7.3, assume toward a contradiction that $\Delta > 1$. The projective statements beginning with definition 7.4 do not retain this arithmetic assumption. Because $\Delta > 1$, the base $\varphi - 1$ belongs to the long-return range, so

$$H_{\varphi-1} = q + h, \qquad h < H_{\varphi-1},$$

where the strict inequality uses $q > 0$. We may therefore apply the internal-tower identity at level $h$ to this base. Together with (6.12), this records the transported terminal side:

$$\lambda^h x_\varphi = x_0, \qquad \lambda^h x_{\varphi-1} = x_{-1}, \qquad \lambda^h \operatorname{relint} E_\varphi = \operatorname{relint} E_0. \tag{7.5}$$

If $h = 0$, then theorem 6.1 gives $\varphi = N$, and the same display is just the cyclic identity $E_\varphi = E_0$.

For a base $i$, put

$$\ell_i := \lambda^{-H_i} \operatorname{aff} E_{r(i)}. \tag{7.6}$$

Since $\lambda^{H_i} P \subseteq P$ and (7.3) places $\lambda^{H_i} x_i$ in the relative interior of the side $E_{r(i)}$, the line $\ell_i$ supports $P$ at $x_i$.

**Lemma 7.1** (Pulled-back supporting lines expose one vertex)**.** *Assume $\Delta > 1$, and set*

$$m_+ = \Delta, \qquad m_- = \varphi - \Delta + 1, \qquad m = \min\{m_+, m_-\}. \tag{7.7}$$

*If $m = m_+$, then $\ell_i$ supports $P$ and exposes only $x_i$ for $2 \le i \le \Delta$. If $m = m_-$, then $\ell_i$ supports $P$ and exposes only $x_i$ for $\Delta - 1 \le i \le \varphi - 2$. When $2\Delta = \varphi + 1$, one has $m_+ = m_-$, so both conclusions are valid; the orientation table below assigns this equality case to the forward orientation.*

*Proof.* By the support-face test lemma 2.9, it is enough to show that neither neighboring vertex of $x_i$ lies on $\ell_i$. Since multiplication by $\lambda^{H_i}$ is invertible, this is equivalent to showing that $\lambda^{H_i} x_{i-1}$ and $\lambda^{H_i} x_{i+1}$ avoid aff $E_{r(i)}$.

If a neighboring base index has the same height as $i$, its image is the relative-interior contact point on the adjacent target side:

$$\lambda^{H_i} x_{i-1} = \xi_{r(i)-1}, \qquad \lambda^{H_i} x_{i+1} = \xi_{r(i)+1},$$

with cyclic side labels. A relative-interior point of an adjacent side of a strict polygon is not on the line of $E_{r(i)}$. Thus only the interface after the last base index of height $q$,

$$i_0 = \varphi - \Delta$$

and before the first base index of height $q + h$, namely $i_0 + 1$, needs separate checking.

For the base index $i_0$, the target side is $E_\varphi$. Its outgoing neighbor has height $q + h$. If $h > 0$, then level $q$ in that tower is internal, and

$$\lambda^q x_{i_0+1} = x_{\varphi+1} \notin \operatorname{aff} E_\varphi.$$

If $h = 0$, then $\varphi = N$ and the long top relation gives $\lambda^q x_{i_0+1} = \xi_1 \in \operatorname{relint} E_1$, again not on the line of $E_\varphi$.

For the base index $i_0 + 1$, the target side is $E_1$. Its incoming neighbor is $i_0$, and (7.5) gives

$$\lambda^{q+h} x_{i_0} = \lambda^h \xi_\varphi \in \operatorname{relint} E_0.$$

The outgoing neighbor returns to $\xi_2 \in \operatorname{relint} E_2$. Neither point is on the line of $E_1$.

The two index ranges in the statement meet the short–long interface at most once and contain no other height transition. Hence both adjacent vertices avoid every displayed pulled-back line, so each such supporting line exposes only the indicated vertex. $\square$

The shorter of the two cyclic arcs is now read as a single boundary chain. The following table gives both orientation cases; after this point the projective argument is orientation-free.

| | forward orientation | reverse orientation |
|---|---|---|
| condition | $2\Delta \le \varphi + 1$ | $2\Delta > \varphi + 1$ |
| length | $m = \Delta$ | $m = \varphi - \Delta + 1$ |
| chain vertices | $X_i = x_i$ | $X_i = x_{\varphi - i}$ |
| chain contacts | $C_i = \xi_i$ | $C_i = \xi_{\varphi - i + 1}$ |
| transport exponent | $a = q$ | $a = q + h$ |
| base subset $M$ | $\{1, \dots, \Delta\}$ | $\{\Delta - 1, \dots, \varphi - 1\}$ |
| target $c$ | $\Delta + 1$ | $\Delta - 1$ |

Here $0 \le i \le m + 1$ for the chain vertices, while the contact formula is used only for $2 \le i \le m + 1$. In either column,

$$C_i \in \operatorname{relint}[X_{i-1}, X_i] \qquad (2 \le i \le m + 1), \tag{7.8}$$

and the initial side satisfies

$$\lambda^a X_1 = C_{m+1}, \qquad \lambda^a X_0 = C_m. \tag{7.9}$$

In the forward orientation the first identity is the short-top relation for base 1, and the second is (7.4). In the reverse orientation they are the long-top relations for bases $\varphi - 1$ and $\varphi$.

For $2 \le i \le m$, define

$$\mathcal{L}_i = \ell_{b_i}, \qquad b_i = \begin{cases} i, & \text{in the forward orientation,} \\ \varphi - i, & \text{in the reverse orientation.} \end{cases} \tag{7.10}$$

By lemma 7.1, $\mathcal{L}_i$ supports $P$ and exposes only $X_i$.

**Lemma 7.2** (The selected boundary arc omits at least one side)**.** *Assume $N \geq 4$, $1 < \Delta < \varphi \leq N$, and choose the branch in the preceding table. Then $m \geq 2$ and*

$$m + 1 < N. \tag{7.11}$$

*Consequently the displayed vertices $X_0, \dots, X_{m+1}$ are pairwise distinct consecutive vertices in the chosen cyclic orientation and the chain omits at least one side of $P$.*

*This conclusion includes all boundary values used later: if $\Delta = 2$, the forward branch has $m = 2$; if $2\Delta = \varphi + 1$, the equality belongs to the forward branch and does not identify two chain labels; and if $h = 0$, then $\varphi = N$ but the same strict inequality (7.11) still holds.*

*Proof.* In the forward branch, $m = \Delta$. If $\varphi < N$, then

$$m + 1 = \Delta + 1 \leq \varphi < N$$

because $\Delta < \varphi$. If $\varphi = N$ and equality $m + 1 = N$ were to hold, then $\Delta = N - 1$; the branch inequality $2\Delta \leq \varphi + 1 = N + 1$ would give $2N - 2 \leq N + 1$, hence $N \leq 3$, a contradiction. Thus (7.11) holds. The labels $0, 1, \dots, \Delta + 1$ therefore represent distinct vertices modulo $N$. The special equality $2\Delta = \varphi + 1$ changes the return-edge wrap but not this label interval. When $\Delta = 2$, the same calculation gives $m = 2$ and $m + 1 = 3 < N$.

In the reverse branch, $m = \varphi - \Delta + 1 \geq 2$. If $\varphi < N$, then $\Delta \geq 2$ gives

$$m + 1 = \varphi - \Delta + 2 \leq \varphi < N.$$

If $\varphi = N$, the strict branch inequality $2\Delta > N + 1$ forces $\Delta \geq 3$ because $N \geq 4$. Hence

$$m + 1 = N - \Delta + 2 < N.$$

The reverse labels run through the integer interval $\varphi, \varphi - 1, \dots, \Delta - 2$, whose length is $m + 2$ and whose total index variation is $m + 1 < N$; they are therefore distinct modulo $N$. When $\varphi = N$, the first label $x_\varphi = x_0$ is not repeated because $\Delta - 2 \geq 1$. Finally, theorem 6.1 gives $h = 0 \iff \varphi = N$, so the preceding two $\varphi = N$ arguments also cover the zero-height case. In either branch, $m + 1 < N$ means that the consecutive chain uses fewer than all $N$ sides and is proper. □

**Proposition 7.3** (Partition of the return source–target pairs)**.** *Let*

$$\mathcal{B} = \{1, \dots, \varphi\}, \qquad s = r^{-1},$$

*where $r$ is addition of $\Delta$ modulo $\varphi$ in the representative set $\mathcal{B}$. Assume $\Delta > 1$ and choose the forward or reverse orientation from the preceding table. Define the subset $M$ of base indices, the distinguished base index $b_*$, the three target-index sets $D, R, A$, and the distinguished target index $c$ by the following exact formulas. We classify the pairs $(i, r(i))$ consisting of a base index and its return target.*

*An integer interval $\{u, \dots, v\}$ is understood to be empty when $u > v$. In the forward branch $2\Delta \leq \varphi + 1$, put*

$$\begin{aligned} M &= \{1, \dots, \Delta\}, & b_* &= 1, \\ M \setminus \{b_*\} &= \{2, \dots, \Delta\}, & & \qquad (7.12) \\ D &= \{2, \dots, \Delta\}, & c &= \Delta + 1, \\ R &= r(M \setminus \{b_*\}), & & \\ A &= \mathcal{B} \setminus (D \cup R \cup \{c\}). & & \qquad (7.13) \end{aligned}$$

*More explicitly,*

$$R = \begin{cases} \{\Delta+2,\dots,2\Delta\}, & 2\Delta \le \varphi, \\ \{\Delta+2,\dots,\varphi\} \cup \{1\}, & 2\Delta = \varphi+1. \end{cases} \tag{7.14}$$

*In the reverse branch $2\Delta > \varphi + 1$, equivalently $2\Delta \ge \varphi + 2$, put*

$$\begin{aligned} M &= \{\Delta-1,\dots,\varphi-1\}, & b_* &= \varphi-1, \\ M \setminus \{b_*\} &= \{\Delta-1,\dots,\varphi-2\}, & & \qquad (7.15) \\ D &= \{\Delta,\dots,\varphi-1\}, & c &= \Delta-1, \\ R &= r(M \setminus \{b_*\}), \\ A &= \mathcal{B} \setminus (D \cup R \cup \{c\}). & & \qquad (7.16) \end{aligned}$$

*In this branch*

$$R = \{2\Delta - \varphi - 1, \dots, \Delta - 2\}. \tag{7.17}$$

*For either branch the four sets*

$$\mathcal{B} = D \,\dot\cup\, R \,\dot\cup\, \{c\} \,\dot\cup\, A \tag{7.18}$$

*are pairwise disjoint, and*

$$r(M) = R \,\dot\cup\, \{c\}, \qquad r(b_*) = c, \qquad r(M \setminus \{b_*\}) = R. \tag{7.19}$$

*Consequently*

$$s(D) \cap M = \varnothing, \qquad s(R) = M \setminus \{b_*\}, \qquad s(c) = b_*, \qquad s(A) \cap M = \varnothing. \tag{7.20}$$

*Thus every source–target pair $(i, r(i))$ is classified exactly once by its target in $D$, $R$, $\{c\}$, or $A$, while (7.20) records whether its source lies outside $M$, in $M \setminus \{b_*\}$, or at the distinguished base index $b_*$. No deformation is being assumed in this proposition.*

*Proof.* Because $r$ is a translation of the finite cyclic set $\mathcal{B}$, it is a bijection. It is therefore enough to compute the images of the displayed integer intervals and to verify the stated disjointness.

*Forward branch.* For $2 \le i \le \Delta - 1$ one has

$$i + \Delta \le 2\Delta - 1 \le \varphi,$$

so $r(i) = i + \Delta$. For $i = \Delta$, either $2\Delta \le \varphi$ and $r(\Delta) = 2\Delta$, or $2\Delta = \varphi + 1$ and $r(\Delta) = 1$. This proves (7.14). In particular,

$$R \subseteq \{\Delta+2,\dots,\varphi\} \cup \{1\},$$

which is disjoint from $D = \{2,\dots,\Delta\}$ and from $c = \Delta + 1$. Since $r(1) = \Delta + 1 = c$ and $r$ is injective, (7.18) and (7.19) follow.

For $j \in D$, the inverse formula is

$$s(j) = \varphi - \Delta + j.$$

The branch inequality gives

$$s(j) \ge \varphi - \Delta + 2 \ge \Delta + 1,$$

so $s(D) \cap M = \varnothing$. The assertions for $R$, $c$, and $A$ now follow from bijectivity and (7.19).

*Reverse branch.* For every $i \in M \setminus \{b_*\}$,

$$i + \Delta \ge 2\Delta - 1 \ge \varphi + 1, \qquad i + \Delta \le \varphi + \Delta - 2 < 2\varphi,$$

so there is exactly one wrap and

$$r(i) = i + \Delta - \varphi.$$

As $i$ runs from $\Delta - 1$ to $\varphi - 2$, this gives precisely (7.17). Its lower endpoint is at least 1 because $2\Delta \geq \varphi + 2$. Hence

$$R \subseteq \{1, \dots, \Delta - 2\},$$

which is disjoint from $c = \Delta - 1$ and from $D = \{\Delta, \dots, \varphi - 1\}$. Also $r(\varphi - 1) = \Delta - 1 = c$. Injectivity of $r$ again proves (7.18) and (7.19).

For $j = \Delta$ one has $s(j) = \varphi$. For $\Delta < j \leq \varphi - 1$ one has $s(j) = j - \Delta$, and

$$1 \leq j - \Delta \leq \varphi - \Delta - 1 \leq \Delta - 3.$$

Thus $s(D) \cap M = \varnothing$, and the remaining assertions in (7.20) again follow from bijectivity. This completes the reverse branch and the proof. □

## 7.1 From the boundary chain to a projectivity

The next theorem is purely projective. It contains no stochastic matrices, no Euclidean algorithm, and no modular labels.

**Definition 7.4** (Projectivity associated with a boundary-contact chain)**.** Let $P$ be a strict polygon. Consider consecutive boundary vertices

$$X_0, X_1, \dots, X_{m+1}, \qquad m \geq 2,$$

read in either cyclic orientation, which are pairwise distinct and whose displayed sides do not exhaust the sides of $P$; points

$$C_i \in \operatorname{relint}[X_{i-1}, X_i] \qquad (2 \leq i \leq m+1),$$

and supporting lines $\mathcal{L}_i$ of $P$ that expose only $X_i$ for $2 \leq i \leq m$. Regard every affine line below as its projective completion. For distinct projective points $A, B$, write $\langle A, B\rangle$ for their join. Put

$$\Lambda_1 = \operatorname{aff}(X_0, X_1), \qquad \Lambda_i = \mathcal{L}_i \quad (2 \leq i \leq m), \qquad K = \operatorname{aff}(C_m, C_{m+1}).$$

Thus $\Lambda_i$ and $K$ denote the projective completions of the displayed affine lines. Starting with $Y_1 \in \Lambda_1$, define successively

$$Y_i = \Lambda_i \cap \langle C_i, Y_{i-1}\rangle \quad (2 \leq i \leq m), \qquad Y_{m+1} = K \cap \langle X_{m+1}, Y_m\rangle.$$

Here every projection is well defined. For the first step, $C_2 \notin \Lambda_1 = \operatorname{aff}(X_0, X_1)$ because the three consecutive vertices $X_0, X_1, X_2$ are not collinear, while $C_2 \notin \Lambda_2$ because $\Lambda_2$ meets $P$ only at $X_2$. The lines $\Lambda_1$ and $\Lambda_2$ are distinct for the same noncollinearity reason. For $3 \leq i \leq m$, the center $C_i \in \operatorname{relint}[X_{i-1}, X_i]$ is different from both endpoints. Since the source line $\Lambda_{i-1}$ and target line $\Lambda_i$ expose only $X_{i-1}$ and $X_i$, respectively, $C_i$ lies on neither line; those two lines are also distinct, since a common line would meet $P$ at both named vertices.

For the closing step, $X_{m+1} \notin \Lambda_m$ because $\Lambda_m$ exposes only $X_m$. Also $X_{m+1} \notin K$. Indeed, if $X_{m+1}$ lay on $K$, then $K$, which contains $C_{m+1} \in \operatorname{relint}[X_m, X_{m+1}]$, would be the last side line $\operatorname{aff}(X_m, X_{m+1})$. It would then also contain $C_m \in \operatorname{relint}[X_{m-1}, X_m]$, forcing the two adjacent side lines to coincide, which would make three consecutive vertices collinear. Finally, $\Lambda_m \neq K$ because $K$ contains $C_m \in P \setminus \{X_m\}$ whereas $\Lambda_m \cap P = \{X_m\}$. Thus each projection center lies on neither its source nor its target line, and each source line is distinct from its target line. Hence

$$\Pi : \Lambda_1 \longrightarrow K, \qquad \Pi(Y_1) = Y_{m+1},$$

is a projectivity, being a composition of perspectivities. It is a map between two generally different projective lines. Only after an explicit projective identification $\Theta : K \to \Lambda_1$ has been specified does $\Theta \circ \Pi : \Lambda_1 \to \Lambda_1$ become a self-map of one projective line.

**Proposition 7.5** (Affine chart with parallel endpoint supporting lines)**.** *For every boundary-contact chain in definition 7.4, there exist supporting lines $M_0, M_1$ of $P$ such that*

$$P \cap M_0 = \{X_0\}, \qquad P \cap M_1 = \{X_{m+1}\},$$

*and there is an affine chart of the projective plane in which the image of $P$ is a bounded convex polygon, the images of the same labelled vertices and sides have the same incidences and cyclic boundary order up to a simultaneous reversal, $M_0$ and $M_1$ are parallel, no side of the displayed chain and no line $\mathcal{L}_i$ is parallel to them, and the displayed chain is the graph of a convex piecewise-linear function whose consecutive edge slopes are strictly increasing over a strictly increasing coordinate.*

*Proof.* At a vertex of a strict polygon, the supporting lines that meet the polygon only at that vertex form a nonempty open interval in the pencil of all lines through the vertex. Let $\mathcal{F}$ be the finite family consisting of all side lines of $P$ and all lines $\mathcal{L}_i$. Choose a supporting line $M_0$ that meets $P$ only at $X_0$. It is distinct from every line in $\mathcal{F}$: a side line in $\mathcal{F}$ meets $P$ along an entire side, while a line in $\mathcal{F}$ exposing another displayed vertex cannot contain $X_0$.

Now vary a supporting line $M_1$ that meets $P$ only at $X_{m+1}$. Exclude first the single direction parallel to $M_0$, so that $O = M_0 \cap M_1$ is a finite point. For a fixed $F \in \mathcal{F}$, the condition $O \in F$ excludes at most one further member of the pencil. Indeed, if $M_0$ is not parallel to $F$, put $K_F = M_0 \cap F$; then $O \in F$ is equivalent to $M_1 = \operatorname{aff}(X_{m+1}, K_F)$. If $M_0$ is parallel to $F$, no finite point of $M_0$ lies on $F$. Removing these finitely many exceptional lines from the open interval of such supporting lines leaves a valid choice of $M_1$. For this choice,

$$O = M_0 \cap M_1 \notin F \qquad (F \in \mathcal{F}). \tag{7.21}$$

The point $O$ lies outside $P$. Otherwise the defining contact property of $M_0$ would give $O = X_0$, while that of $M_1$ would give $O = X_{m+1}$, contradicting that $X_0$ and $X_{m+1}$ are distinct displayed vertices. Since $P$ is compact and convex, strict separation of $O$ from $P$ gives an affine functional $\ell$ such that

$$\ell(z) < \ell(O) \qquad (z \in P).$$

Hence the line $J = \{z : \ell(z) = \ell(O)\}$ passes through $O$ and is disjoint from $P$. Apply a projective automorphism $\Phi$ sending $J$ to the line at infinity. In affine coordinates it has the form

$$\Phi(z) = \frac{Az + b}{\omega(z)},$$

where $\omega$ is affine and vanishes precisely on $J$. It has constant sign on the connected set $P$; after multiplying the homogeneous matrix by $-1$ if necessary, $\omega > 0$ there. Compactness therefore gives a constant $\varepsilon_0 > 0$ such that $\omega(z) \geq \varepsilon_0$ for every $z \in P$. For $x, y \in P$ and $0 \leq t \leq 1$, direct substitution gives

$$\Phi((1-t)x + ty) = \frac{(1-t)\omega(x)}{(1-t)\omega(x) + t\omega(y)}\Phi(x) + \frac{t\omega(y)}{(1-t)\omega(x) + t\omega(y)}\Phi(y). \tag{7.22}$$

The two coefficients are nonnegative and sum to one, and they are both strictly positive when $0 < t < 1$. Thus $\Phi$ maps every segment in $P$ onto the segment joining the endpoint images and maps relative interiors to relative interiors. Applying the same formula to $\Phi^{-1}$ in the image chart shows that no additional points enter those segments. Consequently $\Phi(P)$

is a compact, hence bounded, convex polygon, and the images of the labelled vertices and sides have the same incidences and cyclic boundary order, up to a simultaneous reversal of orientation. Supporting lines that expose one vertex are also preserved.

The images of $M_0$ and $M_1$ are distinct parallel supporting lines, because their projective closures meet the new line at infinity at the common point which is the image of $O$. By (7.21), no side line of $P$ and no $\mathcal{L}_i$ has that point at infinity; equivalently, none is parallel to the two endpoint supporting lines.

Choose an affine coordinate $t$ constant on lines parallel to the two endpoint supporting lines, with $t(X_0) < t(X_{m+1})$. These are the two opposite supporting lines of $P$ in this direction and, by construction, meet $P$ only at $X_0$ and $X_{m+1}$. For every intermediate value of $t$, the corresponding transverse line cuts the convex polygon in a nondegenerate segment; its two endpoints lie on the two boundary arcs joining $X_0$ to $X_{m+1}$. Thus each arc is a graph over $t$. Since no side of the displayed arc is parallel to the endpoint supporting lines, no graph segment is vertical and $t$ is strictly increasing along the displayed chain. After reflecting the second affine coordinate if necessary, this chain is the lower graph of the polygon. Convexity makes its consecutive slopes nondecreasing. Equality of two consecutive slopes would make the corresponding three consecutive displayed vertices collinear, contrary to the definition of a strict polygon. Hence the slopes are strictly increasing. □

**Lemma 7.6** (Location of the final intersection when $Z_1 = X_0$)**.** *Regard every affine line below as its projective completion, and write $\overline{AB}$ for the projective line through two distinct points $A, B$. Set $Z_1 = X_0$ and recursively define the projective points*

$$Z_i = \mathcal{L}_i \cap \overline{Z_{i-1}C_i} \qquad (2 \le i \le m). \tag{7.23}$$

*There exists an admissible affine chart as in proposition 7.5 in which all the points $Z_i$ are finite. The two projective lines in the final intersection are distinct, and the projectively defined point*

$$W_* := \overline{Z_m X_{m+1}} \cap \overline{C_m C_{m+1}} \in \operatorname{relint}[C_{m+1}, C_m] \tag{7.24}$$

*in the original affine chart. The same assertion holds when the original boundary chain is read in the opposite cyclic orientation.*

*Proof.* Apply a projective transformation supplied by proposition 7.5, and in its target affine chart use the same symbols for the transformed points and lines. The recursion (7.23) uses only joins and intersections, so projective transformations preserve all its incidence relations. Write

$$X_i = (t_i, f_i), \qquad t_0 < t_1 < \cdots < t_{m+1},$$

and let $m_i$ be the slope of the edge $[X_{i-1}, X_i]$. Since the chain is the lower graph of a strict polygon,

$$m_1 < m_2 < \cdots < m_{m+1}. \tag{7.25}$$

If $\ell_i$ is the slope of the strict support $\mathcal{L}_i$ at $X_i$, the supporting-slope interval at that vertex is $[m_i, m_{i+1}]$, and strictness of the support excludes both endpoint slopes. Hence

$$m_i < \ell_i < m_{i+1} \qquad (2 \le i \le m). \tag{7.26}$$

Write $C_i = (c_i, \zeta_i)$. Because $C_i \in \operatorname{relint}[X_{i-1}, X_i]$,

$$t_{i-1} < c_i < t_i, \qquad \zeta_i = f_i + m_i(c_i - t_i). \tag{7.27}$$

We prove inductively that every projectively defined point $Z_i$ is affine, say $Z_i = (z_i, g_i)$, and that, for $2 \le i \le m$,

$$c_i < z_i < t_i, \qquad \rho_i < m_i, \tag{7.28}$$

where $\rho_i$ is the slope of $\Gamma_i = \overline{Z_{i-1}C_i}$. When $i < m$, we also prove that $Z_i$ lies strictly above the backward extension of the next edge $[X_i, X_{i+1}]$.

For $i = 2$, the point $Z_1 = X_0$, and the slope of $\Gamma_2$ is

$$\rho_2 = \frac{(t_1 - t_0)m_1 + (c_2 - t_1)m_2}{c_2 - t_0}. \tag{7.29}$$

Both coefficients $t_1 - t_0$ and $c_2 - t_1$ are positive, and their sum is the denominator $c_2 - t_0$. Thus $\rho_2$ is a strict convex combination of $m_1$ and $m_2$, so $m_1 < \rho_2 < m_2$, in particular $\rho_2 < m_2$.

Assume now that $i \geq 3$ and the assertion has been proved at $i - 1$. The previous "above the next edge" statement says

$$g_{i-1} > f_{i-1} + m_i(z_{i-1} - t_{i-1}).$$

Since $C_i$ lies on that same edge line, $\zeta_i = f_{i-1} + m_i(c_i - t_{i-1})$. Also $z_{i-1} < t_{i-1} < c_i$. Subtracting the two displayed relations gives

$$\zeta_i - g_{i-1} < m_i(c_i - z_{i-1}),$$

and division by the positive denominator $c_i - z_{i-1}$ yields

$$\rho_i = \frac{\zeta_i - g_{i-1}}{c_i - z_{i-1}} < m_i. \tag{7.30}$$

Regard $\Gamma_i$ and $\mathcal{L}_i$ as affine functions of the coordinate $t$. At $t = c_i$, using (7.27),

$$\begin{aligned} \Gamma_i(c_i) - \mathcal{L}_i(c_i) &= \zeta_i - (f_i + \ell_i(c_i - t_i)) \\ &= (\ell_i - m_i)(t_i - c_i) > 0. \end{aligned} \tag{7.31}$$

At $t = t_i$,

$$\Gamma_i(t_i) - \mathcal{L}_i(t_i) = (\rho_i - m_i)(t_i - c_i) < 0. \tag{7.32}$$

Since $\rho_i < \ell_i$, the two lines have a unique intersection, and the opposite signs at $c_i$ and $t_i$ locate it at $c_i < z_i < t_i$. Because $Z_i \in \mathcal{L}_i$, its vertical difference from the backward extension of $[X_i, X_{i+1}]$ equals

$$g_i - (f_i + m_{i+1}(z_i - t_i)) = (\ell_i - m_{i+1})(z_i - t_i) > 0. \tag{7.33}$$

Here both factors on the right are negative. This completes the induction.

Let $L = \mathrm{aff}(Z_m, X_{m+1})$, and denote its slope by $\eta$. Since $Z_m \in \mathcal{L}_m$, direct subtraction of the endpoint ordinates gives

$$\eta = \frac{(t_m - z_m)\ell_m + (t_{m+1} - t_m)m_{m+1}}{t_{m+1} - z_m}. \tag{7.34}$$

The two coefficients in the numerator are positive and sum to the denominator. Therefore

$$\ell_m < \eta < m_{m+1}. \tag{7.35}$$

The line $\Gamma_m = \mathrm{aff}(C_m, Z_m)$ has slope $\rho_m < m_m < \ell_m < \eta$. Since $c_m < z_m$,

$$L(c_m) - \zeta_m = (\eta - \rho_m)(c_m - z_m) < 0. \tag{7.36}$$

On the other hand, $L$ passes through $X_{m+1}$, while $C_{m+1}$ lies on the last chain edge of slope $m_{m+1}$; hence

$$L(c_{m+1}) - \zeta_{m+1} = (\eta - m_{m+1})(c_{m+1} - t_{m+1}) > 0. \tag{7.37}$$

Let $K = \operatorname{aff}(C_m, C_{m+1})$. The affine function $L - K$ is negative at $c_m$ and positive at $c_{m+1}$. Because $c_m < t_m < c_{m+1}$, it has a zero at an abscissa strictly between $c_m$ and $c_{m+1}$. This zero is exactly the transformed point corresponding to $W_*$, and it lies in $\operatorname{relint}[C_m, C_{m+1}]$ in the admissible chart. The projective transformation used above and its inverse map every segment contained in the polygon onto the corresponding segment and preserve relative interiors, as proved in proposition 7.5. Applying the inverse transformation therefore gives (7.24) in the original affine chart. Distinctness of the two final projective lines is likewise preserved.

If the boundary chain is read in the opposite cyclic orientation, apply proposition 7.5 to that oriented chain and repeat the same calculation; no sign or index change is required inside the argument. □

**Lemma 7.7** (Sign of $t - u(t)$ near a fixed point of a real projectivity)**.** *Let $u$ be a real projective automorphism, written in an affine coordinate near* 0*, such that*

$$u(0) = 0, \qquad 0 < u(1) < 1.$$

*Then there are arbitrarily small nonzero real $\tau$ for which*

$$\tau - u(\tau) > 0. \tag{7.38}$$

*Proof.* A real projective automorphism fixing 0 has, on an open neighborhood of 0, the normal form

$$u(\tau) = \frac{a\tau}{1 + c\tau}, \qquad a \neq 0. \tag{7.39}$$

Consequently

$$\tau - u(\tau) = \frac{\tau(1 - a + c\tau)}{1 + c\tau}. \tag{7.40}$$

If $a \neq 1$, choose the sign of $\tau$ so that $\tau(1 - a) > 0$, and then choose $|\tau|$ small enough that $1 + c\tau > 0$ and $1 - a + c\tau$ has the sign of $1 - a$. The right side of (7.40) is then positive. If $a = 1$, the condition $0 < u(1) < 1$ gives $0 < 1/(1 + c) < 1$, hence $c > 0$; for every sufficiently small nonzero $\tau$,

$$\tau - u(\tau) = \frac{c\tau^2}{1 + c\tau} > 0.$$

In either case the admissible $\tau$'s can be chosen arbitrarily close to 0. □

**Theorem 7.8** (A small deformation moving the final image into the interior half-plane)**.** *Let the boundary-contact data of definition 7.4 be given, and define the moving chain by*

$$X_1(\tau) = (1 - \tau)X_1 + \tau X_0, \qquad X_i(\tau) = \mathcal{L}_i \cap \operatorname{aff}(X_{i-1}(\tau), C_i) \quad (2 \leq i \leq m). \tag{7.41}$$

*Put*

$$Y(\tau) = (1 - \tau)C_{m+1} + \tau C_m.$$

*The recursive intersections are finite and continuous on some open interval about* 0*. Every sufficiently small neighborhood of* 0 *contains a nonzero real parameter $\tau$ for which $Y(\tau)$ lies in the open half-plane bounded by* $\operatorname{aff}(X_m(\tau), X_{m+1})$ *that contains $X_{m-1}(\tau)$. For all parameters in a sufficiently small such interval, the displayed vertices remain in strict convex position with their original cyclic order.*

*Proof.* Put

$$\Lambda_1 = \overline{\operatorname{aff}(X_0, X_1)}, \qquad \Lambda_i = \overline{\mathcal{L}_i} \quad (2 \leq i \leq m), \qquad K = \overline{\operatorname{aff}(C_m, C_{m+1})},$$

where the bars denote projective completion. We first verify every perspectivity in the construction.

For $i = 2$, the center $C_2 \in \operatorname{relint}[X_1, X_2]$ is not on $\Lambda_1$, and strictness of $\mathcal{L}_2$ gives $C_2 \notin \Lambda_2$. For $3 \leq i \leq m$, the point $C_i \in \operatorname{relint}[X_{i-1}, X_i]$ lies on neither the strict support $\Lambda_{i-1}$ at $X_{i-1}$ nor the strict support $\Lambda_i$ at $X_i$. Hence projection from center $C_i$ defines a projective isomorphism $\Lambda_{i-1} \to \Lambda_i$. For the closing projection, strictness gives $X_{m+1} \notin \Lambda_m$. Also $X_{m+1} \notin K$: if it were, then $K$, which already contains $C_{m+1} \in \operatorname{relint}[X_m, X_{m+1}]$, would equal the side line $\operatorname{aff}(X_m, X_{m+1})$; this would put $C_m \in \operatorname{relint}[X_{m-1}, X_m]$ on that adjacent side line, contradicting strictness. Thus projection from $X_{m+1}$ is a projective isomorphism $\Lambda_m \to K$.

The affine parametrization of the distinguished starting point

$$\tau \longmapsto X_1(\tau) = (1-\tau)X_1 + \tau X_0$$

extends to a projective isomorphism from the compactified parameter line to $\Lambda_1$. Composing it with the verified perspectivities gives the following projective isomorphism from the compactified parameter line to the final contact line:

$$\mathcal{H} : \mathbb{P}^1(\mathbb{R}) \longrightarrow K. \tag{7.42}$$

Let

$$\upsilon : K \longrightarrow \mathbb{P}^1(\mathbb{R})$$

be the projective coordinate induced by the original affine line $\operatorname{aff}(C_m, C_{m+1})$, normalized by $\upsilon(C_{m+1}) = 0$ and $\upsilon(C_m) = 1$, and define the real projective automorphism

$$u = \upsilon \circ \mathcal{H}.$$

Both $\mathcal{H}$ and $\upsilon$ are projective isomorphisms; hence $u$ is a nonconstant real projectivity, as required by lemma 7.7. At $\tau = 0$, the projective recursion gives $\mathcal{H}(0) = C_{m+1}$, so $u(0) = 0$. At $\tau = 1$, the distinguished starting point has moved to $X_0$; preservation of joins and intersections under projective transformations gives $\mathcal{H}(1) = W_*$, where $W_*$ is the point of lemma 7.6. Since $W_* \in \operatorname{relint}[C_{m+1}, C_m]$,

$$0 < u(1) < 1. \tag{7.43}$$

At $\tau = 0$, every recursive intersection is the original vertex $X_i$, and the closing intersection is $C_{m+1}$. Therefore none of the intermediate affine coordinate maps has a pole at 0. Since each has at most one pole, there is an open interval $J$ about 0 on which all points $X_i(\tau)$ and the closing intersection

$$W(\tau) = \operatorname{aff}(X_m(\tau), X_{m+1}) \cap \operatorname{aff}(C_m, C_{m+1}) \tag{7.44}$$

are finite and continuous. In particular, the two lines in (7.44) are distinct throughout $J$. On this interval the local affine recursion agrees with the global line-to-line projectivity: $W(\tau) = \mathcal{H}(\tau)$, and $u(\tau)$ is precisely the normalized affine coordinate of $W(\tau)$.

By lemma 7.7, the set of nonzero parameters satisfying $\tau - u(\tau) > 0$ accumulates at 0. We postpone the choice of such a parameter until all remaining open conditions have been fixed.

Set

$$z(t) = (1-t)C_{m+1} + tC_m.$$

Choose $\varepsilon \in \{1, -1\}$ so that

$$\varepsilon \det(X_{m+1} - X_m, X_{m-1} - X_m) > 0,$$

and define the signed-side functional $\mathcal{S}$ together with its restriction $d$ to the parametrized final contact line:

$$\begin{aligned} \mathcal{S}(x, \tau) &= \varepsilon \det\big(X_{m+1} - X_m(\tau), x - X_m(\tau)\big) \quad (x \in V), \\ d(t, \tau) &= \mathcal{S}(z(t), \tau). \end{aligned} \tag{7.45}$$

Thus $\mathcal{S}(x,\tau) > 0$ selects one of the two open half-planes bounded by the moving line $\mathrm{aff}(X_m(\tau), X_{m+1})$, and the choice of sign places the preceding chain vertex in that half-plane at $\tau = 0$. For $\tau \in J$, the point $z(u(\tau)) = W(\tau)$ is the unique intersection of that closing line with $\mathrm{aff}(C_m, C_{m+1})$. Since $d$ is affine in $t$,

$$d(t,\tau) = \gamma(\tau)(t - u(\tau)), \qquad \gamma(\tau) = \varepsilon \det\bigl(X_{m+1} - X_m(\tau), C_m - C_{m+1}\bigr). \tag{7.46}$$

The coefficient $\gamma$ is continuous. Write $C_m = (1-\alpha)X_{m-1} + \alpha X_m$ with $0 < \alpha < 1$. The defining choice of $\varepsilon$ gives

$$\gamma(0) = d(1,0) = (1-\alpha)\varepsilon \det(X_{m+1} - X_m, X_{m-1} - X_m) > 0. \tag{7.47}$$

After shrinking $J$, if necessary, $\gamma(\tau) > 0$ throughout $J$.

Fix the orientation of the original displayed chain. By lemma 2.6, every cyclically ordered triple of its displayed vertices has a nonzero determinant with one common sign at $\tau = 0$. There are finitely many such determinants, and all vertices depend continuously on $\tau$ on $J$. Shrink $J$ once more so that every one of these determinants keeps its sign throughout $J$. The triple-sign criterion, lemma 2.7, then shows that the displayed vertices remain distinct, in the same cyclic order, and in strict convex position for every $\tau \in J$. In particular,

$$\varepsilon \det\bigl(X_{m+1} - X_m(\tau), X_{m-1}(\tau) - X_m(\tau)\bigr) > 0$$

throughout $J$. Hence the half-plane described by $\mathcal{S}(x,\tau) > 0$ is exactly the open half-plane bounded by the moving closing line that contains the preceding chain vertex.

Now use the accumulation statement from lemma 7.7 to choose a nonzero $\tau \in J$ with $\tau - u(\tau) > 0$. Evaluating (7.46) at $t = \tau$ gives

$$d(\tau,\tau) = \gamma(\tau)(\tau - u(\tau)) > 0.$$

Since $Y(\tau) = z(\tau)$, we have $\mathcal{S}(Y(\tau),\tau) = d(\tau,\tau) > 0$. Thus the final image lies in the open half-plane of the moving closing line that contains $X_{m-1}(\tau)$. Because such parameters accumulate at 0, the chosen parameter can be made arbitrarily small. $\square$

### 7.2 Global realization of the boundary-chain motion

The local projective motion is useful only after it has been extended to all $N$ polygon vertices and all return incidences. The next lemma makes every required assignment and incidence explicit.

**Lemma 7.9** (Extension of the local deformation to all polygon vertices)**.** *Assume $\varphi > \delta$ and $\Delta > 1$, choose the forward or reverse boundary-contact chain from the orientation table, and suppose that its displayed boundary chain is proper. Let $M, b_*, D, R, c, A$ be the data of proposition 7.3; the moved bases other than the distinguished base $b_*$ are $M \setminus \{b_*\}$. For $0 \le i \le m+1$, define the chain label*

$$\iota_i = \begin{cases} i, & \text{in the forward branch,} \\ \varphi - i, & \text{in the reverse branch,} \end{cases}$$

*so that $X_i = x_{\iota_i}$. For $2 \le i \le m+1$, define the side index*

$$k_i = \begin{cases} i, & \text{in the forward branch,} \\ \varphi - i + 1, & \text{in the reverse branch,} \end{cases}$$

*so that $C_i = \xi_{k_i}$. Then the map $i \mapsto \iota_i$ is a bijection $\{1,\dots,m\} \to M$, the map $i \mapsto k_i$ is a bijection $\{2,\dots,m\} \to D$, and*

$$b_* = \iota_1, \qquad c = k_{m+1} = r(b_*), \qquad a = H_{b_*}.$$

*In particular, the chain and contact labels have no repetitions. Moreover, $a = q$ in the forward branch while $a = q + h$ in the reverse branch.*

*Define the moving chain by*

$$X_1(\tau) = (1-\tau)X_1 + \tau X_0, \qquad X_i(\tau) = \mathcal{L}_i \cap \operatorname{aff}\bigl(X_{i-1}(\tau), C_i\bigr) \quad (2 \le i \le m). \tag{7.48}$$

*There is an open interval $U_0$ about $0$ on which all these points are finite and continuous. On $U_0$, define one base function for each $j \in \mathcal{B} = \{1,\dots,\varphi\}$ by*

$$B_j(\tau) = \begin{cases} X_i(\tau), & j = \iota_i \text{ for some } 1 \le i \le m, \\ x_j, & j \notin M, \end{cases} \tag{7.49}$$

*and define all $N$ polygon-vertex functions by the tower formula*

$$\widehat{x}_{F(t,j)}(\tau) = \lambda^t B_j(\tau) \qquad (1 \le j \le \varphi,\ 0 \le t < H_j). \tag{7.50}$$

*For $k \in \mathbb{Z}/N\mathbb{Z}$, put*

$$\widehat{E}_k(\tau) = [\widehat{x}_{k-1}(\tau), \widehat{x}_k(\tau)], \qquad P_\tau = \operatorname{conv}\{\widehat{x}_k(\tau) : k \in \mathbb{Z}/N\mathbb{Z}\}.$$

*Then, after shrinking $U_0$ to an open interval $U$ about $0$, the following statements hold for every $\tau \in U$.*

(i) Unique global assignment and cyclic order. *Equation (7.50) assigns exactly one function to each polygon label, every function is continuous, $\widehat{x}_k(0) = x_k$, and the $N$ points $\widehat{x}_k(\tau)$ are distinct extreme points in their original positive cyclic order. Hence $P_\tau$ is a strict $N$-gon with sides $\widehat{E}_k(\tau)$.*

(ii) Exact internal propagation. *For every tower state with $t + 1 < H_j$,*

$$\lambda \widehat{x}_{F(t,j)}(\tau) = \widehat{x}_{F(t+1,j)}(\tau). \tag{7.51}$$

*Thus every internal tower image is exactly a polygon vertex, with no side or endpoint convention involved.*

(iii) Final images assigned to every side. *For each strict side index $k \in \mathcal{B}$, let*

$$\widehat{V}_k(\tau) = \lambda^{H_{s(k)}} B_{s(k)}(\tau). \tag{7.52}$$

*For every $k \in \mathcal{B} \setminus \{c\}$,*

$$\widehat{V}_k(\tau) \in \operatorname{relint}\bigl(\widehat{E}_k(\tau)\bigr). \tag{7.53}$$

*The only final-image incidence not constrained to its assigned side is the return from the distinguished base $b_*$ to the side labelled $c$:*

$$Y(\tau) := \widehat{V}_c(\tau) = \lambda^a X_1(\tau) = (1-\tau)C_{m+1} + \tau C_m. \tag{7.54}$$

*Its assigned side is exactly*

$$\widehat{E}_c(\tau) = [X_m(\tau), X_{m+1}], \tag{7.55}$$

*where the endpoints may appear in the reverse order relative to the positive boundary orientation in the reverse branch.*

(iv) All nonclosing side inequalities for the closing image. *For every polygon side label $k$, define*

$$G_k(\tau) = \det\big(\widehat{x}_k(\tau) - \widehat{x}_{k-1}(\tau), Y(\tau) - \widehat{x}_{k-1}(\tau)\big). \tag{7.56}$$

*Then*

$$G_k(\tau) > 0 \qquad (k \neq c,\ \tau \in U). \tag{7.57}$$

*Thus the closing side is the only side inequality not fixed by openness.*

(v) Invariant-polygon criterion and the only unchecked side inequality. *For every $\tau \in U$, all image vertices other than $Y(\tau)$ lie on $\partial P_\tau$, while*

$$Y(\tau) \in \operatorname{Ext}(\lambda P_\tau).$$

*If, in addition, $G_c(\tau) > 0$, then*

$$Y(\tau) \in \operatorname{int} P_\tau, \qquad \lambda P_\tau \subseteq P_\tau, \qquad \operatorname{Ext}(\lambda P_\tau) \cap \operatorname{int} P_\tau = \{Y(\tau)\}. \tag{7.58}$$

*Proof.* We prove the five conclusions in six explicit stages.

*1. The local chain and the fixed chain endpoints.* By lemma 7.1, every line $\mathcal{L}_i$ in (7.48) is a strict support of $P$ at $X_i$. Together with (7.8), these are the boundary-contact data of definition 7.4. The first assertion of theorem 7.8 therefore supplies an open interval $U_0$ on which every relevant denominator is nonzero and the recursion is finite and continuous. At $\tau = 0$, induction gives $X_i(0) = X_i$: for $i = 1$ this is immediate, and for $i \geq 2$ both lines in the defining intersection pass through $X_i$ and are distinct because $\mathcal{L}_i$ is a strict support while $C_i \neq X_i$ lies on the incident side.

The endpoints $X_0$ and $X_{m+1}$ are fixed global tower vertices. In the forward branch, $X_{m+1} = x_{\Delta+1}$ is an unmoved base. Also $F(h, \varphi) = 0$ and $h < H_\varphi = q + h$; when $h = 0$, this statement is the same identity with $F(0, \varphi) = 0$ and $\varphi = N$. Hence

$$\widehat{x}_0(\tau) = \lambda^h B_\varphi(\tau) = \lambda^h x_\varphi = x_0 = X_0.$$

In the reverse branch, $X_0 = x_\varphi$ and $X_{m+1} = x_{\Delta-2}$ are unmoved bases. The reverse inequality forces $\Delta \geq 3$, so the latter label belongs to $\mathcal{B}$. Thus in both branches the constants appearing in the local boundary-contact chain are exactly the corresponding vertices in the global tower assignment.

*2. The tower formula is a unique continuous labeling.* The branch formulas show directly that $i \mapsto \iota_i$ is a bijection from $\{1, \ldots, m\}$ onto $M$. Consequently (7.49) assigns exactly one formula to every base label: a moved formula for $j \in M$ and the fixed formula $x_j$ for $j \notin M$.

The map $F$ in (6.7) is a bijection from the finite state set

$$\mathcal{D} = \{(t, j) : 1 \leq j \leq \varphi,\ 0 \leq t < H_j\}$$

onto $\mathbb{Z}/N\mathbb{Z}$. Therefore every polygon label has exactly one preimage $(t, j)$, so (7.50) gives one and only one definition for every $\widehat{x}_k(\tau)$. In particular, no moved formula and no fixed formula can assign different points to the same polygon label. Each base function is continuous and multiplication by the fixed linear map $\lambda^t$ is continuous, hence all global vertex functions are continuous.

At $\tau = 0$, the geometric tower identity (6.9) gives

$$\widehat{x}_{F(t,j)}(0) = \lambda^t B_j(0) = \lambda^t x_j = x_{F(t,j)}.$$

Since $F$ is onto, $\widehat{x}_k(0) = x_k$ for every polygon label $k$.

Apply the simultaneous convex-admissibility lemma lemma 2.8 first with no auxiliary test points. It gives an open interval $U_1 \subseteq U_0$ on which the global vertex list is distinct, retains its positive cyclic order, and is in strict convex position. This proves part (i) on $U_1$.

Equation (7.51) is now an identity:

$$\lambda\widehat{x}_{F(t,j)}(\tau) = \lambda^{t+1}B_j(\tau) = \widehat{x}_{F(t+1,j)}(\tau).$$

This proves part (ii).

*3. Supporting lines for the strict side indices.* Write

$$L_k(\tau) = \operatorname{aff}\big(\widehat{x}_{k-1}(\tau), \widehat{x}_k(\tau)\big) \qquad (k \in \mathcal{B}).$$

We identify every one of these lines.

The branch formulas also show that $i \mapsto k_i$ maps $\{2, \dots, m\}$ bijectively onto $D$ and sends $m+1$ to $c$.

If $k \in D$, there is therefore a unique chain index

$$\psi(k) = \begin{cases} k, & \text{forward branch,} \\ \varphi - k + 1, & \text{reverse branch,} \end{cases} \qquad 2 \le \psi(k) \le m,$$

for which $C_{\psi(k)} = \xi_k$. The two endpoints of $\widehat{E}_k(\tau)$ are $X_{\psi(k)-1}(\tau)$ and $X_{\psi(k)}(\tau)$, possibly in reverse order. By the recursion,

$$L_k(\tau) = \operatorname{aff}\big(X_{\psi(k)-1}(\tau), C_{\psi(k)}\big). \tag{7.59}$$

We now verify directly, using the vertex functions already defined in (7.49)–(7.50), that every line with label in $R \cup A$ is fixed. In the forward branch,

$$R \cup A = \{1\} \cup \{\Delta + 2, \dots, \varphi\}.$$

The tower based at $\varphi \notin M$ is fixed, and the tower identity $F(h, \varphi) = 0$ gives $\widehat{x}_0(\tau) = \lambda^h B_\varphi(\tau) = x_0$. The distinguished starting point $X_1(\tau)$ moves on the original line $\operatorname{aff} E_1$, so $L_1(\tau) = L_1(0)$. If $\Delta + 2 \le k \le \varphi$, both endpoints of $E_k$ have base labels outside $M$, except that when $k = \varphi = N$ the second endpoint is the already fixed vertex $x_0$. Hence $L_k(\tau) = L_k(0)$ throughout this range.

In the reverse branch,

$$R \cup A = \{1, \dots, \Delta - 2\} \cup \{\varphi\}.$$

Here $B_\varphi(\tau) = x_\varphi$, while the distinguished starting point $X_1(\tau) = \widehat{x}_{\varphi-1}(\tau)$ moves on the original line $\operatorname{aff} E_\varphi$; thus $L_\varphi(\tau) = L_\varphi(0)$. For $2 \le k \le \Delta - 2$, both endpoints have base labels outside $M$, so $L_k(\tau) = L_k(0)$. Finally, $\widehat{x}_1(\tau) = x_1$ and $\widehat{x}_0(\tau) = \lambda^h B_\varphi(\tau) = x_0$ by (7.5), so $L_1(\tau) = L_1(0)$ as well.

The one remaining strict side has label $c$, and direct reading of the branch table gives (7.55). Thus the partition

$$\mathcal{B} = D \mathbin{\dot\cup} R \mathbin{\dot\cup} \{c\} \mathbin{\dot\cup} A$$

accounts for every strict side exactly once.

*4. Four-case partition and line membership.* First partition the index pairs in the return-time tower into those whose next iterate remains in the same orbit segment and those that index its final point:

$$\begin{aligned} \mathcal{D}_{\text{int}} &= \{(t, j) \in \mathcal{D} : t + 1 < H_j\}, \\ \mathcal{D}_{\text{top}} &= \{(H_j - 1, j) : j \in \mathcal{B}\}. \end{aligned} \tag{7.60}$$

These sets are disjoint, their union is $\mathcal{D}$, and

$$|\mathcal{D}_{\text{int}}| = N - \varphi, \qquad |\mathcal{D}_{\text{top}}| = \varphi.$$

Because $F$ is bijective, they index disjoint sets of polygon vertices whose union is the complete set of $N$ source vertices. Part (ii) has already accounted for every source in $\mathcal{D}_{\text{int}}$. The image of the final point $(H_j - 1, j)$ in the orbit segment based at $j$ is $\lambda^{H_j} B_j(\tau) = \widehat{V}_{r(j)}(\tau)$. Since $r$ is a bijection of $\mathcal{B}$, every such image is therefore indexed exactly once by its assigned side label $k$, with unique source base $s(k)$.

We now apply the inverse-source partition (7.20) to those $\varphi$ final images according to their assigned sides. The four cases are

| side class | source base | assigned side line | reason for line membership |
|---|---|---|---|
| $D$ | fixed, since $s(D) \cap M = \varnothing$ | controlled by the projection recursion | the fixed final image $\xi_k = C_{\psi(k)}$ is built into that line |
| $R$ | the supported moved base $s(k) \in M \setminus \{b_*\}$ | fixed | the base remains on the pulled-back support $\ell_{s(k)}$ |
| $A$ | fixed | fixed | the original incidence is unchanged |
| $\{c\}$ | the distinguished base $b_*$ | the closing line | the only final incidence not imposed in advance, namely $Y(\tau)$ |

The row sets are disjoint and exhaustive by (7.18); the source assertions are exactly (7.20).

If $k \in D$, then $s(k) \notin M$, so its source base is fixed. The original top relation gives

$$\widehat{V}_k(\tau) = \lambda^{H_{s(k)}} x_{s(k)} = \xi_k = C_{\psi(k)}.$$

Equation (7.59) puts this fixed point on the moving line $L_k(\tau)$.

If $k \in R$, then $s(k) \in M \setminus \{b_*\}$. Let $i$ be the unique chain index with $\iota_i = s(k)$; necessarily $2 \le i \le m$. The recursion puts $B_{s(k)}(\tau) = X_i(\tau)$ on

$$\mathcal{L}_i = \ell_{s(k)} = \lambda^{-H_{s(k)}} L_k(0).$$

The target line is fixed, so

$$\widehat{V}_k(\tau) = \lambda^{H_{s(k)}} B_{s(k)}(\tau) \in L_k(0) = L_k(\tau).$$

If $k \in A$, both its source base and its target side line are fixed. Hence

$$\widehat{V}_k(\tau) = \xi_k \in L_k(0) = L_k(\tau).$$

The only remaining side label is $c$. Here $s(c) = b_*$, so

$$\widehat{V}_c(\tau) = \lambda^{H_{b_*}} B_{b_*}(\tau) = \lambda^a X_1(\tau).$$

Using the interpolation of the distinguished starting point and (7.9),

$$\lambda^a X_1(\tau) = (1-\tau)\lambda^a X_1 + \tau\lambda^a X_0 = (1-\tau)C_{m+1} + \tau C_m.$$

This proves (7.54). The partition from proposition 7.3 proves simultaneously that every final image has been treated and that no final image belongs to two cases.

For every nonclosing side label $k$, exact collinearity has now been proved, and at $\tau = 0$ the point $\widehat{V}_k(0) = \xi_k$ belongs to relint $E_k$. Apply lemma 2.8 to the finite family

$$w_k(\tau) = \widehat{V}_k(\tau), \qquad k \in \mathcal{B} \setminus \{c\}.$$

After shrinking $U_1$ to an interval $U_2$, every one of these points lies in the relative interior of its assigned moving side. This proves (7.53) and part (iii).

*5. All nonclosing inequalities for the image of the distinguished base.* At $\tau = 0$,

$$Y(0) = C_{m+1} = \xi_c \in \operatorname{relint} E_c.$$

By (2.11) in lemma 2.8, this point satisfies the strict side inequality for every $k \neq c$. Apply the same lemma to the finite family of test pairs

$$(Y(\tau), k), \qquad k \in \mathbb{Z}/N\mathbb{Z} \setminus \{c\}.$$

After shrinking $U_2$ to an interval $U \ni 0$, the functions (7.56) satisfy (7.57). This is a global statement: it includes sides whose endpoints are internal tower vertices, not merely the displayed boundary-chain sides. This proves part (iv).

*6. Invariance and the only remaining unchecked side inequality.* Let $k \in \mathbb{Z}/N\mathbb{Z}$ and let $(t, j)$ be its unique tower preimage under $F$. If $t + 1 < H_j$, then (7.51) gives $\lambda \widehat{x}_k(\tau)$ as another polygon vertex. If $t = H_j - 1$ and $r(j) \neq c$, then

$$\lambda \widehat{x}_k(\tau) = \lambda^{H_j} B_j(\tau) = \widehat{V}_{r(j)}(\tau) \in \operatorname{relint}\big(\widehat{E}_{r(j)}(\tau)\big).$$

Thus every image vertex other than the closing image lies on $\partial P_\tau$ for every $\tau \in U$.

Consecutive record times satisfy $q > 0$ and $h \geq 0$, so $a = H_{b_*} \geq 1$. Put

$$k_* = F(a - 1, b_*).$$

Then

$$\widehat{x}_{k_*}(\tau) = \lambda^{a-1} B_{b_*}(\tau), \qquad \lambda \widehat{x}_{k_*}(\tau) = Y(\tau).$$

The point $\widehat{x}_{k_*}(\tau)$ is an extreme point of the strict polygon $P_\tau$. Since multiplication by $\lambda \neq 0$ is invertible, it maps extreme points bijectively to extreme points; hence $Y(\tau) \in \operatorname{Ext}(\lambda P_\tau)$ for every $\tau \in U$.

Now fix any $\tau \in U$ for which $G_c(\tau) > 0$. Together with (7.57), this gives all $N$ strict side inequalities for $Y(\tau)$. The final assertion of lemma 2.8 therefore yields $Y(\tau) \in \operatorname{int} P_\tau$. Every vertex image belongs to $P_\tau$: internal images are polygon vertices, nonclosing top images lie in relative side interiors, and the closing top image is $Y(\tau)$. Convexity and linearity give

$$\lambda P_\tau = \operatorname{conv}\{\lambda \widehat{x}_k(\tau) : k \in \mathbb{Z}/N\mathbb{Z}\} \subseteq P_\tau.$$

All image vertices except $Y(\tau)$ lie on the boundary, while $Y(\tau)$ is interior. This proves (7.58) and part (v). $\square$

**Theorem 7.10** (Existence of an invariant deformation with one interior image vertex)**.** *Under the hypotheses and notation of lemma 7.9, every neighborhood of* 0 *contains a nonzero parameter* $\tau \in U$ *for which*

$$Y(\tau) \in \operatorname{int} P_\tau, \qquad \lambda P_\tau \subseteq P_\tau, \qquad \operatorname{Ext}(\lambda P_\tau) \cap \operatorname{int} P_\tau = \{Y(\tau)\}. \tag{7.61}$$

*In particular,* $Y(\tau) \in \operatorname{Ext}(\lambda P_\tau) \cap \operatorname{int} P_\tau$*, while every other vertex of* $\lambda P_\tau$ *lies on* $\partial P_\tau$*.*

*Proof.* The projective-deformation theorem theorem 7.8 supplies, arbitrarily close to 0, a nonzero parameter for which $Y(\tau)$ lies in the open half-plane of the moving closing line that contains $X_{m-1}(\tau)$. Choose that parameter inside the common admissibility interval $U$ from lemma 7.9. By part (i) of that lemma, $P_\tau$ is a strict polygon with the original cyclic order. The nonincident vertex $X_{m-1}(\tau)$ therefore lies strictly in the interior half-plane of the closing side. This half-plane condition is exactly $G_c(\tau) > 0$. Part (v) of lemma 7.9 now gives (7.61). $\square$

### 7.3 The first-return step

**Theorem 7.11** (The first-return step satisfies $\Delta = 1$)**.** *Assume $N \geq 4$. If $\varphi > \delta$, then $\Delta = 1$.*

*Proof.* Assume $\Delta > 1$. By lemma 7.2, the selected branch has $m \geq 2$, its displayed vertices are pairwise distinct consecutive boundary vertices, and it omits at least one polygon side. Thus it gives a proper boundary-contact chain. The same lemma verifies this directly at $\Delta = 2$, $2\Delta = \varphi + 1$, $h = 0$, and $\varphi = N$. Together with lemma 7.1 and proposition 7.3, all hypotheses of lemma 7.9 and theorem 7.10 are satisfied.

Choose the nonzero parameter supplied by that theorem. Then $P_\tau$ is a strict invariant $N$-gon for the same $N$-critical map, while the vertex $Y(\tau)$ of $\lambda P_\tau$ lies in int $P_\tau$. This contradicts the literal conclusion of theorem 3.2 that every extreme point of $\lambda P_\tau$ lies on $\partial P_\tau$. Hence $\Delta > 1$ is impossible. Since $\Delta$ is a positive integer, $\Delta = 1$. □

*Remark* 7.12 (Boundary cases in the proof that $\Delta = 1$)*.* No exceptional parameter is suppressed in the preceding proof.

(a) If $\Delta = 2$, the reverse inequality is impossible because $\Delta < \varphi$; the forward boundary-contact chain has $m = 2$, one exposing support $\mathcal{L}_2$, one controlled side index, and the same exhaustive partition of the return targets.

(b) The equality $2\Delta = \varphi + 1$ belongs to the forward branch. It is exactly the reduction modulo $\varphi$ expressed by $r(\Delta) = 1$ in (7.14); side label 1 remains on the fixed supporting line.

(c) The equality $h = 0$ is equivalent to $\varphi = N$ by theorem 6.1. Then $\lambda^h E_\varphi = E_0$ is the cyclic identity, and every formula in the boundary-contact chain and global tower construction remains literal.

(d) The orbit number $\delta$, including $\delta = 1$, enters this argument only through the hypothesis $\varphi > \delta$ and the existence of the first-return decomposition. It creates no additional deformation case.

*Remark* 7.13 (Local and global data used in the deformation argument)*.* The local choice of the deformation parameter depends only on the behavior near 0 of the normalized projective map $u$. The preceding extension lemma supplies the separate global assertions needed to use that local choice: every polygon index is assigned exactly once, every nonclosing image lies on the line containing its assigned side and remains in the relative interior of that side, and every side inequality except $G_c(\tau) > 0$ persists for sufficiently small $\tau$.

*Proof of theorem 1.3.* Choose an adapted complex coordinate by proposition 2.1. The boundary conclusion for every invariant polygon with at most $N$ vertices is theorem 3.2. The least-area selection and cyclic interlacing in lemma 4.13 give a half-open contact assignment after choosing the appropriate orientation. Write the resulting positive indexing as in lemma 4.13, so that $E_i = [x_{i-1}, x_i]$ and $\chi(x_{i-\kappa}) = E_i$. If $\kappa = N$, then

$$\sigma(E_i) = \chi(\mathrm{head}(E_i)) = \chi(x_i) = E_{i+\kappa} = E_i,$$

so $\sigma$ is the identity. The final assertion of lemma 4.13 says that every contact lies in the relative interior of its assigned side, and hence $I = \mathcal{E}(P)$. There are therefore no permitted vertex replacements, so assertion (ii) is vacuous; the only relative-interior contact set obtainable by such replacements is $I$ itself, which is a cyclic interval with $\varphi = N = \delta$. Assertions (iii) and (iv) follow immediately. This completes the identity case.

Henceforth assume $1 \leq \kappa < N$. The permitted vertex replacement is proposition 5.1, and corollary 5.2 translates its set update $i \mapsto i + \kappa$ exactly into $e \mapsto \sigma(e)$. Under the induced side bijection $b$, the new contact permutation is $\sigma' = b \circ \sigma \circ b^{-1}$, and the construction is covariant under real-linear conjugacy. Apply lemma 5.5 to a lexicographically minimal relative-interior contact set obtainable by permitted vertex replacements. It is one interval of length $\varphi \geq \delta$, where $\delta$ is the number of contact-permutation orbits. If $\varphi = \delta$, the interval meets every orbit and has exactly one point in each; hence it is a complete orbit transversal, and the common return time is the orbit length $N/\delta$.

Suppose $\varphi > \delta$. The last assertion of lemma 5.5 makes $N - \varphi$ an upper-record residue. Apply theorem 6.1 with $\nu = \varphi$. It gives a two-height tower partition and a first-return translation by $\Delta$. Theorem theorem 7.11 gives $\Delta = 1$. Thus the first return is the cyclic successor on the relative-interior contact interval. All parts of the theorem have now been proved. □

# 8 Complex return monodromy and Farey arithmetic

Let $F_n$ denote the Farey sequence of order $n$ on $[0, 1]$. Here *return monodromy* means the product obtained by multiplying the recurrence factors once around a closed return-recurrence chain and using its closing relation to cancel the initial and terminal vertices. No additional geometric structure is hidden in this term.

**Lemma 8.1** (Farey adjacency criterion)**.** *Two reduced fractions $a/b < c/d$ in $F_n$ are consecutive if and only if*

$$bc - ad = 1, \qquad b + d > n. \tag{8.1}$$

*Proof.* Assume first that $bc - ad = 1$. If a reduced fraction $h/k$ lies strictly between $a/b$ and $c/d$, define

$$m = ck - dh, \qquad \ell = bh - ak.$$

Cross multiplication shows that $m$ and $\ell$ are positive integers. The determinant-one identity gives the exact decomposition

$$(k, h) = m(b, a) + \ell(d, c).$$

Looking at the first coordinate yields $k = mb + \ell d \geq b + d$. Hence, if $b + d > n$, no fraction of denominator at most $n$ can lie between the two endpoints, so they are consecutive in $F_n$.

Conversely, suppose the two fractions are consecutive in $F_n$ and put $D = bc - ad$. The determinant is a positive integer. If $D > 1$, the half-open parallelogram spanned by the primitive integer vectors $u = (b, a)$ and $v = (d, c)$ contains a nonzero integer point

$$w = \alpha u + \beta v, \qquad 0 < \alpha, \beta < 1.$$

Indeed, lemma A.6 says that the half-open fundamental parallelogram contains exactly $D = |\det(u, v)|$ representatives of the quotient group $\mathbb{Z}^2/(\mathbb{Z}u + \mathbb{Z}v)$. When $D > 1$, one of these representatives is different from the origin. It cannot lie on either open coordinate edge of the parallelogram: such a point would be a nonintegral proper multiple of the primitive vector $u$ or $v$. The strict inequalities follow because a primitive vector has no integer point in the relative interior of its first unit segment. Both $w$ and $u + v - w$ are positive combinations of $u, v$, so their slopes lie strictly between $a/b$ and $c/d$. Their first coordinates are positive integers $k$ and $b + d - k$; one is at most $(b + d)/2 \leq n$ because $b, d \leq n$. After reducing that intermediate slope, its denominator can only decrease. This produces an element of $F_n$ strictly between the alleged consecutive fractions, a contradiction. Therefore $D = 1$. Finally, if $b + d \leq n$, the reduced mediant $(a + c)/(b + d)$ has denominator at most $n$ and lies strictly between the endpoints. Thus consecutiveness also forces $b + d > n$. □

**Lemma 8.2** (Reflection of a Farey cell)**.** *Let $a/b < x < c/d$ be the open cell between two consecutive fractions of $F_N$. Then*

$$\frac{d-c}{d} < 1 - x < \frac{b-a}{b} \tag{8.2}$$

*is also an open cell between consecutive fractions of $F_N$. The reflection exchanges the endpoint denominators: the left and right denominators in (8.2) are $d$ and $b$, respectively. In particular, if $d \le b$, the reflected ordered cell has the smaller denominator first.*

*Proof.* Reducedness is preserved because

$$\gcd(d-c, d) = \gcd(c, d) = 1, \qquad \gcd(b-a, b) = \gcd(a, b) = 1.$$

The inequalities follow by subtracting the original inequalities from 1. Moreover

$$(b-a)d - (d-c)b = bc - ad = 1,$$

and the sum of the reflected denominators is the unchanged number $b + d > N$. The Farey adjacency criterion, lemma 8.1, therefore proves consecutiveness. The denominator statement is immediate from the displayed fractions. □

**Lemma 8.3** (Reflection of a closed backward return-recurrence chain)**.** *Let $\lambda \in \mathbb{C} \setminus \mathbb{R}$ satisfy $0 < |\lambda| < 1$, let $\theta = \arg_+(\lambda)$, and let $q, d \ge 1$ and $0 \le h < dq$. Suppose nonzero points $z_0, \dots, z_d$, numbers*

$$0 \le b_i < 1, \qquad a_i > 0 \qquad (1 \le i \le d),$$

*and lifted arguments $\Theta_i$ satisfy*

$$(\lambda^q - b_i)z_i = a_i z_{i-1} \qquad (1 \le i \le d), \tag{8.3}$$

$$\lambda^h z_d = z_0, \tag{8.4}$$

$$0 < \Theta_i - \Theta_{i-1} < \pi \qquad (1 \le i \le d), \tag{8.5}$$

$$\Theta_d + h\theta = \Theta_0 + 2\pi m \tag{8.6}$$

*for an integer $m$.*

*Put*

$$\mu = \overline{\lambda}, \qquad \vartheta = \arg_+(\mu) = 2\pi - \theta, \qquad e = -h, \qquad s = dq - h,$$

*and, for $1 \le j \le d$, define*

$$w_j = \overline{z}_{d-j}, \qquad w_0 = \overline{z}_d, \qquad \beta'_j = b_{d-j+1}, \qquad \alpha'_j = a_{d-j+1}.$$

*Then*

$$(\mu^q - \beta'_j)w_{j-1} = \alpha'_j w_j \qquad (1 \le j \le d), \tag{8.7}$$

$$\mu^e w_d = w_0. \tag{8.8}$$

*Consequently*

$$\mu^e \prod_{j=1}^{d} (\mu^q - \beta'_j) = \prod_{j=1}^{d} \alpha'_j, \tag{8.9}$$

$$\mu^s \prod_{j=1}^{d} (\mu^q - \beta'_j) = \mu^{dq} \prod_{j=1}^{d} \alpha'_j. \tag{8.10}$$

*If*

$$u_j = \Theta_{d-j+1} - \Theta_{d-j} \in (0,\pi),$$

*then $u_j$ is the argument selected by* (8.7)*, and*

$$e\vartheta + \sum_{j=1}^{d} u_j = 2\pi(m-h). \tag{8.11}$$

*Proof.* Conjugate (8.3) with $i = d-j+1$. Since

$$\overline{z}_{d-j+1} = w_{j-1}, \qquad \overline{z}_{d-j} = w_j,$$

the conjugated equation is exactly (8.7). Conjugating (8.4) gives

$$\mu^h \overline{z}_d = \overline{z}_0.$$

Because $w_0 = \overline{z}_d$ and $w_d = \overline{z}_0$, this is equivalent, using $\mu \neq 0$, to (8.8) with $e = -h$. Multiplying the forward cell equations and cancelling the nonzero points $w_0, w_d$ by the signed closure proves (8.9). Multiplication by $\mu^{dq}$ gives (8.10); the exponent on its left is $dq - h = s > 0$ by hypothesis.

Choose lifted arguments for the reflected points by

$$\Phi_j = -\Theta_{d-j} + C,$$

where the constant $C$ is arbitrary. Then

$$\Phi_j - \Phi_{j-1} = \Theta_{d-j+1} - \Theta_{d-j} = u_j \in (0,\pi).$$

Since $\alpha'_j > 0$, equation (8.7) shows that $u_j$ is congruent modulo $2\pi$ to the argument of $\mu^q - \beta'_j$. Its membership in $(0,\pi)$ selects the oriented branch uniquely. Finally,

$$\sum_{j=1}^{d} u_j = \Theta_d - \Theta_0 = 2\pi m - h\theta$$

by (8.6); therefore

$$e\vartheta + \sum_{j=1}^{d} u_j = -h(2\pi - \theta) + (2\pi m - h\theta) = 2\pi(m-h),$$

which is (8.11). □

**Lemma 8.4** (The identity contact permutation closes after reflection)**.** *Assume the integer lift in lemma 4.13 is $\kappa = N$. For the closed-return monodromy data, take the eigenvalue in the opposite orientation:*

$$\mu = \overline{\lambda}, \qquad y = 1 - x.$$

*For this eigenvalue the ordered Farey cell is*

$$\frac{0}{1} < y < \frac{1}{N},$$

*so $p = 0$, $q = 1$, $r = 1$, $s = N$, $d = N$, and $e = 0$. There are parameters satisfying* (1.5) *for which* (1.6)*–*(1.9) *hold.*

*Proof.* Every contact lies in the relative interior of its assigned side by lemma 4.13; hence $0 < \alpha_i, \beta_i < 1$. Extend the indices periodically and set $z_i = x_i$ for $0 \le i \le N$, with $z_N = z_0$ as a point and with lifted arguments $\Theta_N = \Theta_0 + 2\pi$. The contact equation for $\kappa = N$ is

$$\lambda x_i = \beta_i x_{i-1} + \alpha_i x_i.$$

After rearrangement it is the backward cell relation

$$(\lambda - \alpha_i) z_i = \beta_i z_{i-1} \qquad (1 \le i \le N),$$

where the coefficients at $i = N$ are the periodic coefficients at index 0. The inequalities required in (8.5) follow from lemma 2.10. Apply lemma 8.3 with

$$q = 1, \quad d = N, \quad h = 0, \quad m = 1, \quad b_i = \alpha_i, \quad a_i = \beta_i.$$

It produces the forward reflected cells with

$$\beta'_j = \alpha_{N-j+1}, \qquad \alpha'_j = \beta_{N-j+1}.$$

Because $\alpha_i + \beta_i = 1$, these coefficients satisfy $\alpha'_j = 1 - \beta'_j$ and hence (1.5). The reflection lemma also gives the polynomial and Laurent identities with $s = N$ and $e = 0$, and the phase identity

$$\sum_{j=1}^{N} u_j = 2\pi.$$

This is (1.9) because $r - dp = 1$.

It remains to identify the ordered Farey cell. The angle estimate (4.13) gives

$$\frac{N-1}{N} < x < 1.$$

The endpoints are consecutive in $F_N$, and lemma 8.2 gives

$$0 < 1 - x < \frac{1}{N}.$$

Thus the reflected orientation has exactly the Farey labels stated in the lemma, with the smaller denominator on the left. □

## 8.1 Closed-return monodromy products

**Proposition 8.5** (More than one relative-interior contact in some orbit: $\varphi > \delta$)**.** *Assume $N \ge 4$ and $\varphi > \delta$. Then $\delta = 1$, and the closed-return monodromy data use the contact eigenvalue*

$$\mu = \lambda.$$

*There are integers $p, q, d, e$ with*

$$q\kappa - pN = 1, \qquad N = qd + e, \qquad 0 \le e < q, \qquad d = \lfloor N/q \rfloor, \tag{8.12}$$

*and parameters satisfying (1.5) such that*

$$(\lambda^q - \beta_j) x_{j-1} = \alpha_j x_j \qquad (1 \le j \le d), \tag{8.13}$$

$$\lambda^e x_d = x_0. \tag{8.14}$$

*The ordered Farey cell is*

$$\frac{p}{q} < x < \frac{\kappa}{N}. \tag{8.15}$$

*Thus $r = \kappa$, $s = N$, and $q < s$. For these data (1.6)–(1.9) hold.*

*Proof.* By theorem 7.11 and (6.5), $\Delta = 1$ and $\delta = \gcd(\Delta, \varphi) = 1$.

Suppose first that $\varphi = N$. The initial record pair is $U = (1, 0)$, $V = (0, 1)$, so $\Delta = \kappa$. Since $\Delta = 1$, this gives $\kappa = 1$. Put

$$q = 1, \quad p = 0, \quad d = N, \quad e = 0.$$

Every contact index is a relative-interior contact index and $\lambda x_{j-1} = \xi_j$, hence (8.13); the identity $x_N = x_0$ is (8.14).

Now assume $\varphi < N$. Use the determinant-one record extension in theorem 6.1: let $E = (e, c)$ be the earliest record in the maximal backward arithmetic run with increment $U = (q, p)$, and put $d = -L(E)$. Equation (6.8) is exactly (8.12); in particular $d \geq \varphi$, and the proof of theorem 6.1 shows that $E$ and $E + U$ are consecutive upper records. Since the actual relative-interior contact interval $\{1, \ldots, \varphi\}$ is contained in $J = \{1, \ldots, d\}$, the endpoint-padding corollary corollary 6.2 applies. It gives

$$\lambda^q x_{j-1} = \xi_j \qquad (1 \leq j \leq d), \tag{8.16}$$

and its closure is (8.14). For $j \leq \varphi$, substitute the half-open contact equation for $\xi_j$. For $j > \varphi$, the same corollary says that index $j$ is an endpoint-contact index, so put $\beta_j = 0$, $\alpha_j = 1$ and again obtain (8.13).

Multiplying (8.13) gives

$$\left(\prod_{j=1}^{d} (\lambda^q - \beta_j)\right) x_0 = \left(\prod_{j=1}^{d} \alpha_j\right) x_d.$$

No vertex is zero because $0 \in \operatorname{int} P$. Using $x_0 = \lambda^e x_d$ proves (1.7). Multiplying by $\lambda^{dq}$ and using $N = qd + e = s$ proves (1.6).

The return identity $q\varphi + h = N$ from (6.5) gives $q \leq N$. Equality is impossible, because $q = N$ would make $q\kappa - pN$ divisible by $N$, whereas (8.12) makes it equal to 1. Hence $q < N = s$. The determinant identity $q\kappa - pN = 1$ and the denominator sum $q + N > N$ now show that $p/q$ and $\kappa/N$ are consecutive in $F_N$. The upper inequality $x < \kappa/N$ follows from (4.13). To locate the lower endpoint, follow the $q$ successive $\kappa$-steps beginning at the lifted vertex $x_0$. Their total index advance is $q\kappa = pN + 1$, and (8.16) ends strictly on the lifted side $E_{pN+1}$. By lemma 4.14,

$$\Theta_0 + 2\pi p < \Theta_0 + q\theta < \Theta_1 + 2\pi p,$$

or equivalently

$$0 < q\theta - 2\pi p < \Theta_1 - \Theta_0.$$

Thus $p/q < x$.

Finally, in each cell relation choose the factor argument $u_j$ so that

$$u_j = \Theta_j - \Theta_{j-1}.$$

This is the continuous rooted branch selected by the return path. The arithmetic identity

$$d + e\kappa = (\kappa - dp)N \tag{8.17}$$

follows from $q\kappa - pN = 1$ and $N = qd + e$. The closure path consists entirely of endpoint contacts and ends at lifted index $d + e\kappa = (\kappa - dp)N$; therefore lemma 4.14 gives

$$\Theta_d + e\theta = \Theta_0 + 2\pi(\kappa - dp).$$

Telescoping the $u_j$ gives

$$e\theta + \sum_{j=1}^{d} u_j = 2\pi(\kappa - dp),$$

which is (1.9) with $(r, s) = (\kappa, N)$. □

**Proposition 8.6** (Exactly one relative-interior contact in each orbit: $\varphi = \delta$)**.** *Assume $\varphi = \delta$. Put*

$$L = \frac{N}{\delta}, \qquad K = \frac{\kappa}{\delta},$$

*choose the unique $h \in \{1, \dots, L-1\}$ with $Kh \equiv -1 \pmod{L}$, and define*

$$b = \frac{Kh+1}{L}, \qquad S = N - h, \qquad R = \kappa - b.$$

*Then*

$$KS - RL = 1, \qquad L + S > N, \qquad \frac{R}{S} < x < \frac{K}{L}. \tag{8.18}$$

*The closed-return monodromy data are as follows.*

(a) *If $\delta \geq 2$, use the opposite eigenvalue $\mu = \overline{\lambda}$. Then $y = 1 - x$ lies in the reflected Farey cell*

$$\frac{L-K}{L} < y < \frac{S-R}{S}, \tag{8.19}$$

*whose denominators satisfy $q = L < s = S$. With*

$$d = \delta, \qquad e = s - dq = -h,$$

*there are parameters satisfying (1.5) for which (1.6)–(1.9) hold.*

(b) *If $\delta = 1$, retain the contact eigenvalue $\mu = \lambda$. Then the ordered cell in (8.18) has*

$$\frac{p}{q} = \frac{R}{S}, \qquad \frac{r}{s} = \frac{K}{L}, \qquad q = S < s = L = N.$$

*With $d = \lfloor N/q \rfloor$ and $e = s - dq$, there are parameters satisfying (1.5) for which the same identities hold.*

*Proof.* The $\kappa$-orbits are the $\delta$ residue classes modulo $\delta$, each of length $L$. The relative-interior contact interval $\{1, \dots, \delta\}$ contains exactly one relative-interior contact index in each orbit; in standard orbit terminology, it is an orbit transversal. Starting at $x_i$, the first $L-1$ destination indices are therefore endpoint contacts and the last step reaches the relative-interior contact index $i$. Consequently

$$\lambda^L x_i = \xi_i = \beta_i x_{i-1} + \alpha_i x_i \qquad (1 \leq i \leq \delta). \tag{8.20}$$

Set

$$\widehat{\beta}_i = \alpha_i, \qquad \widehat{\alpha}_i = \beta_i.$$

Every contact in the interval lies in the relative interior of its side, so both numbers belong to $(0, 1)$, and

$$(\lambda^L - \widehat{\beta}_i) x_i = \widehat{\alpha}_i x_{i-1}. \tag{8.21}$$

The congruence $h\kappa \equiv -\delta \pmod{N}$ moves $x_\delta$ to $x_0$. No intermediate destination index is a relative-interior contact: the orbit remains in residue class 0 modulo $\delta$ and meets the relative-interior contact interval again only after $L$ steps. Hence

$$\lambda^h x_\delta = x_0. \tag{8.22}$$

Multiplying (8.21), cancelling the nonzero vertices, and using (8.22) gives

$$\lambda^{-h} \prod_{i=1}^{\delta} (\lambda^{L} - \widehat{\beta}_i) = \prod_{i=1}^{\delta} \widehat{\alpha}_i. \tag{8.23}$$

We next prove the Farey assertions and the lifted closure needed for both cases. The determinant computation is

$$KS - RL = K(N-h) - (\kappa - b)L = bL - Kh = 1,$$

and $L + S > N$ because $h < L$. Let $\gamma_i = \Theta_i - \Theta_{i-1}$ and $\Gamma = \sum_{i=1}^{\delta} \gamma_i$. The strict $L$-return in (8.20) lies in relint $E_i$. Since $L\kappa = KN$, lemma 4.14 places its lift strictly between $\Theta_{i-1} + 2\pi K$ and $\Theta_i + 2\pi K$. Thus

$$\varepsilon = 2\pi K - L\theta \quad \text{satisfies} \quad 0 < \varepsilon < \gamma_i.$$

Likewise $h\kappa = bN - \delta$. The endpoint closure (8.22) therefore has the exact lifted form

$$\Theta_\delta + h\theta = \Theta_0 + 2\pi b, \qquad \Gamma = 2\pi b - h\theta. \tag{8.24}$$

Using $N = \delta L$ and $\kappa = \delta K$ gives

$$S\theta - 2\pi R = (N-h)\theta - 2\pi(\kappa - b) = \Gamma - \delta\varepsilon > 0,$$

whereas $L\theta < 2\pi K$. Hence $R/S < x < K/L$. Together with determinant one and $L + S > N$, the Farey adjacency criterion proves (8.18).

Assume first that $\delta \geq 2$. Since $1 \leq h < L$,

$$S = \delta L - h > L.$$

Apply lemma 8.2 to (8.18); this gives (8.19), with the smaller denominator $L$ on the left. Set

$$\frac{p}{q} = \frac{L-K}{L}, \qquad \frac{r}{s} = \frac{S-R}{S}, \qquad d = \delta, \qquad e = -h.$$

Then $d = \lfloor N/q \rfloor$ because $N = \delta L$, and

$$s - dq = S - \delta L = -h = e.$$

The inequalities $0 < \gamma_i < \pi$ required in (8.5) follow from lemma 2.10. Apply lemma 8.3 to (8.21)–(8.24) with

$$q = L, \quad d = \delta, \quad b_i = \widehat{\beta}_i, \quad a_i = \widehat{\alpha}_i, \quad m = b.$$

The hypothesis $h < dq$ is $h < N$, which is automatic. Because $\widehat{\alpha}_i + \widehat{\beta}_i = 1$, the reflected coefficients satisfy (1.5). The lemma gives the forward recurrences for $\mu = \overline{\lambda}$, the Laurent identity with $e = -h$, and its polynomial form (1.6). It also gives

$$e \arg_+(\mu) + \sum_{j=1}^{d} u_j = 2\pi(b-h).$$

Finally,

$$r - dp = (S-R) - \delta(L-K) = b - h,$$

so this is exactly (1.9). This proves part (a).

Now assume $\delta = 1$. Then $L = N$ and $S = N - h < L$, so the original ordered cell already has the smaller denominator on the left. Put

$$q = S = N - h, \qquad s = N, \qquad p = R, \qquad r = K = \kappa.$$

Equation (8.23) becomes

$$\lambda^{-h}(\lambda^N - \widehat{\beta}_1) = \widehat{\alpha}_1,$$

or, after moving the second term,

$$\lambda^h(\lambda^{N-h} - \widehat{\alpha}_1) = \widehat{\beta}_1. \tag{8.25}$$

Let $d = \lfloor N/q \rfloor$ and $e = N - dq$. Define

$$\beta_1' = \widehat{\alpha}_1, \quad \alpha_1' = \widehat{\beta}_1, \qquad \beta_j' = 0, \quad \alpha_j' = 1 \quad (2 \le j \le d).$$

Since $e + (d-1)q = h$, equation (8.25) is exactly

$$\lambda^e \prod_{j=1}^{d} (\lambda^q - \beta_j') = \prod_{j=1}^{d} \alpha_j'.$$

Multiplying by $\lambda^{dq}$ gives (1.6).

The first factor has a genuine consecutive-vertex relation. Indeed, $\lambda^h x_1 = x_0$ and $\lambda^N x_1 = \widehat{\alpha}_1 x_0 + \widehat{\beta}_1 x_1$ imply

$$(\lambda^q - \beta_1')x_0 = \alpha_1' x_1. \tag{8.26}$$

Let $u_1 = \Theta_1 - \Theta_0 \in (0, \pi)$ be its geometric factor argument. The lifted closure (8.24) is

$$h\theta + u_1 = 2\pi b. \tag{8.27}$$

For $2 \le j \le d$, assign the padded zero factor the argument $u_j = q\theta - 2\pi p$. The Farey inequalities and $q\kappa = pN + 1$ show that this is the unique argument in $(0, \pi)$ of $\lambda^q$. Because $e + (d-1)q = h$,

$$e\theta + \sum_{j=1}^{d} u_j = 2\pi(b - (d-1)p).$$

Here $p = R = \kappa - b$ and $r = K = \kappa$, so $b - (d-1)p = r - dp$. This proves (1.9) and part (b). ∎

The product identity alone determines factor arguments only modulo $2\pi$. The vertex recurrences remove that ambiguity: consecutive lifted vertex angles select the upper-half-plane argument of each nonzero factor, and every padded zero factor is assigned the left endpoint argument $A$. The following lemma records this common continuous argument interval before convexity is applied.

**Lemma 8.7** (The return factors lie in one common continuous argument interval)**.** *Assume $N \ge 4$. For every set of closed-return monodromy data constructed in lemma 8.4 and propositions 8.5 and 8.6, let $\vartheta = \arg_+(\mu)$ and put*

$$A = q\vartheta - 2\pi p.$$

*Then $0 < A < \pi$, and every recurrence argument $u_j$ in (1.9) is the unique argument in $(0, \pi)$ of $\mu^q - \beta_j$. Equivalently, if $M = \arg_+(\mu^q - 1) \in (A, \pi)$, then*

$$u_j \in [A, M) \qquad (1 \le j \le d). \tag{8.28}$$

*Proof.* Because this is an order-$N$ Farey cell, $q + s > N \ge 4$; with $q \le s$ this gives $s \ge 3$. The determinant identity and $p/q < y < r/s$ therefore give

$$0 < A < \frac{2\pi}{s} < \pi.$$

Thus

$$\operatorname{Im}(\mu^q-\beta)=|\mu|^q\sin A>0 \qquad (0\le\beta<1),$$

so each factor has a unique argument in $(0,\pi)$. As $\beta$ increases from 0 to 1, that argument increases continuously and strictly from $A$ to $M$.

It remains only to identify the geometric lifts. In the case $\varphi>\delta$, (8.13) has the form

$$(\mu^q-\beta_j)x_{j-1}=\alpha_j x_j, \qquad \alpha_j>0,$$

where $x_{j-1},x_j$ are consecutive vertices. Their lifted angular gap lies in $(0,\pi)$ and is congruent modulo $2\pi$ to the factor argument; it is therefore the unique upper-half-plane argument. In both reflected cases—$\kappa=N$ and the one-contact-per-orbit case $\delta\ge 2$—this conclusion is exactly the final assertion of lemma 8.3. In the one-contact-per-orbit case $\delta=1$ it applies to (8.26), while every padded zero factor was explicitly assigned the argument $A$. These cases exhaust the constructions listed in the statement, and the range assertion follows from the preceding monotonicity in $\beta$. □

*Proof of theorem 1.4.* Apply lemma 4.13; throughout the contact-return analysis its selected eigenvalue has been denoted by $\lambda$. Determine the output eigenvalue only after the return case is known. The four mutually exclusive outputs are recorded before they are verified:

| return case | output | denominator order | signed $e$ | source |
|---|---|---|---|---|
| $\kappa=N$ | $\mu=\overline{\lambda}$ | $1<N$ | $0$ | lemma 8.4 |
| $\kappa<N,\ \varphi>\delta$ | $\mu=\lambda$ | $q<N$ | $0\le e<q$ | proposition 8.5 |
| $\kappa<N,\ \varphi=\delta\ge 2$ | $\mu=\overline{\lambda}$ | $L<S$ | $e=-h<0$ | proposition 8.6, case (a) |
| $\kappa<N,\ \varphi=\delta=1$ | $\mu=\lambda$ | $S<N$ | $0\le e<S$ | proposition 8.6, case (b) |

If $\kappa=N$, choose $\mu=\overline{\lambda}$ and apply lemma 8.4. Its Farey cell has denominators $1\le N$, and it provides the polynomial identity, the equivalent Laurent identity, and the phase formula.

Assume $\kappa<N$ and pass to the reduced relative-interior contact interval of lemma 5.5. If $\varphi>\delta$, choose $\mu=\lambda$ and apply proposition 8.5, using the equality $\Delta=1$ from theorem 7.11. Its right Farey endpoint is $\kappa/N$, so the ordered denominators satisfy $q\le N=s$, and its Euclidean remainder $e$ is nonnegative.

If $\varphi=\delta$, apply proposition 8.6. That proposition makes the orientation choice explicitly: it chooses $\mu=\overline{\lambda}$ when $\delta\ge 2$ and $\mu=\lambda$ when $\delta=1$. In the first subcase $e=-h<0$ and the homogeneous polynomial identity is primary; in the second subcase $e\ge 0$.

Thus in every case a specific eigenvalue $\mu\in\{\lambda,\overline{\lambda}\}$ has been selected, the ordered Farey cell has $q\le s$, and (1.6)–(1.9) hold for that same $\mu$. Lemma 8.7 verifies that the factor arguments appearing in the case constructions have precisely the normalized-factor description and common-argument-interval range asserted in the theorem. No symbol is restored and no orientation is changed after this selection. □

## 9 Stochastic-matrix corollary

For $n\ge 1$, let

$$\Theta_n=\bigcup\Big\{\operatorname{spec}A : A\in\mathbb{R}_{\ge 0}^{n\times n},\ A\mathbf{1}=\mathbf{1}\Big\}; \tag{9.1}$$

thus $\Theta_n$ is the union of the spectra of all real row-stochastic $n\times n$ matrices.

**Proposition 9.1** (Compactness, conjugation, and disk bound)**.** *For every $n$, the set $\Theta_n$ is compact, invariant under complex conjugation, and contained in the closed unit disk.*

*Proof.* Let $\mathcal{S}_n$ be the compact set of row-stochastic matrices. First note that every eigenvalue lies in the closed unit disk: if $Av = \lambda v$, choose an index with $|v_i| = \|v\|_\infty$; then

$$|\lambda|\|v\|_\infty = |(Av)_i| \leq \|v\|_\infty.$$

Consequently the eigenpair set

$$\mathcal{E}_n = \{(A, \lambda) \in \mathcal{S}_n \times \overline{\mathbb{D}} : \det(\lambda I - A) = 0\}$$

is closed in a compact space and is therefore compact. Its projection to the $\lambda$-coordinate is exactly $\Theta_n$, so $\Theta_n$ is compact. Real characteristic polynomials give conjugation symmetry. □

**Theorem 9.2** (Invariant-polygon criterion)**.** *Let $n \geq 2$. For $\lambda \in \mathbb{C}$, the following are equivalent.*

(i) $\lambda \in \Theta_n$.

(ii) *There is a non-singleton convex polygon $P \subset \mathbb{C}$ with at most $n$ vertices such that*

$$\lambda P \subseteq P. \tag{9.2}$$

*Proof.* If $Av = \lambda v$, put $P = \operatorname{conv}\{v_1, \ldots, v_n\}$. Every coordinate satisfies

$$\lambda v_i = \sum_j a_{ij} v_j \in P,$$

so (9.2) holds. If the hull is a singleton, all coordinates of the nonzero eigenvector are equal to the same nonzero number, so $\lambda = 1$; in that case any fixed segment may be used.

Conversely, let $x_1, \ldots, x_m$ be the vertices of an invariant polygon, $m \leq n$. Choose barycentric coordinates

$$\lambda x_i = \sum_{j=1}^m a_{ij} x_j, \qquad a_{ij} \geq 0, \qquad \sum_j a_{ij} = 1.$$

Then $A = (a_{ij})$ is row-stochastic and $Ax = \lambda x$. If $m < n$, append absorbing states and extend the eigenvector by zero coordinates. □

**Corollary 9.3** (Star-shapedness with respect to the origin)**.** *Let $n \geq 2$. If $\lambda \in \Theta_n$, then $t\lambda \in \Theta_n$ for every $0 \leq t \leq 1$.*

*Proof.* If $|\lambda| < 1$, take an invariant polygon $P$. Iteration gives $\lambda^k P \subseteq P$ and $\lambda^k x \to 0$, hence $0 \in P$; convexity then gives $t\lambda P \subseteq P$.

Suppose $|\lambda| = 1$. If $\lambda = 1$, the segment $[0, 1]$ satisfies $t[0, 1] \subseteq [0, 1]$. Otherwise take an invariant polygon with the least number of vertices. If it has nonzero area, area equality forces $\lambda P = P$, so multiplication by $\lambda$ permutes its vertices and $\lambda$ has finite order $k \leq n$. The regular orbit polygon

$$Q = \operatorname{conv}\{1, \lambda, \ldots, \lambda^{k-1}\}$$

contains 0, satisfies $\lambda Q = Q$, and therefore satisfies $t\lambda Q = tQ \subseteq Q$. If the minimal polygon is a segment, then $\lambda = -1$ and the same conclusion follows from $Q = [-1, 1]$. The invariant-polygon criterion completes the proof. □

**Proposition 9.4** (Unit-circle points)**.** *Let $n \geq 2$. If $\lambda \in \Theta_n$ and $|\lambda| = 1$, then $\lambda$ is a root of unity of order at most $n$. Conversely every such root belongs to $\Theta_n$.*

*Proof.* Take an invariant polygon with the least number of vertices. If it has nonzero area, then $\lambda P \subseteq P$ and equality of areas force $\lambda P = P$ by lemma A.5. Multiplication by $\lambda$ permutes the vertices, so a nonzero vertex has an orbit of length at most $n$ and $\lambda^k = 1$. If the polygon is a segment, $\lambda = \pm 1$. Conversely, a cyclic permutation matrix realizes every root of unity of order at most $n$, after padding. □

For $0 < \theta < \pi$, define

$$R_n(\theta) = \max\left\{\rho \geq 0 : \rho e^{i\theta} \in \Theta_n\right\}. \tag{9.3}$$

The maximum exists by compactness and star-shapedness with respect to the origin.

**Corollary 9.5** (Farey product data for a non-inherited stochastic radial maximum)**.** *Let $N \geq 4$ and let*

$$\lambda = R_N(\theta)e^{i\theta} \in \Theta_N \setminus \Theta_{N-1}, \qquad 0 < |\lambda| < 1.$$

*Then one of $\lambda, \overline{\lambda}$ satisfies the complete conclusion of theorem 1.4; equivalently, it has the Farey product data, the polynomial identity (1.6), its equivalent Laurent form (1.7), normalized factor arguments (1.8) on one common continuous argument interval $0 < A < M < \pi$ and $u_j \in [A, M)$, and the phase identity (1.9).*

*Proof.* Let $T_\lambda z = \lambda z$. Since $0 < \theta < \pi$, we have $\Im\lambda > 0$, and hence

$$(\operatorname{tr} T_\lambda)^2 - 4\det T_\lambda = -4(\Im\lambda)^2 < 0.$$

Together with $0 < |\lambda| < 1$, this shows that $T_\lambda$ is an elliptic contraction. By the invariant-polygon criterion, membership in $\Theta_n$ is equivalent to the existence of a non-singleton invariant polygon with at most $n$ vertices. By lemma 2.5, every such polygon for $T_\lambda$ is nondegenerate, so its least possible number of vertices is exactly $\nu_{\text{poly}}(T_\lambda)$. The assumption $\lambda \in \Theta_N \setminus \Theta_{N-1}$ therefore gives $\nu_{\text{poly}}(T_\lambda) = N$. Radial maximality says that $t\lambda \notin \Theta_N$ for every $t > 1$; applying the same criterion to $tT_\lambda$ gives $\nu_{\text{poly}}(tT_\lambda) > N$. Thus $T_\lambda$ is $N$-critical. Apply theorem 1.4. Reversing the orientation labels the conjugate eigenvalue, which explains the choice between $\lambda$ and $\overline{\lambda}$. □

# 10 Conclusion

For $N \geq 4$, the parent theorem is a theorem about critical planar dynamics, not about stochastic matrices. Radial criticality forces every side to meet the image polygon and every image vertex to lie on the outer boundary; permitted vertex replacements preserve the contact hypotheses, and the finite set reduction makes the relative-interior contact set one cyclic interval. The lattice sail supplies the resulting unimodular first-return data, and the projectivity associated with the boundary-contact chain rules out a nonadjacent return. Farey adjacency and the heterogeneous Ito product are scalar shadows of that adjacent return normal form. Orders at most three are treated directly. The stochastic theorem uses none of the internal deformation or index bookkeeping: it follows from the invariant-polygon criterion in one paragraph.

# A Foundational finite-dimensional tools

This appendix proves the finite-dimensional ingredients invoked in the body of the paper. They are stated in the exact strength used there.

**Lemma A.1** (Weighted spectral-radius bound and nonnegative Perron vectors)**.** *Let $B = (b_{ij}) \in \mathbb{R}^{m\times m}$ be entrywise nonnegative.*

(i) *If $x > 0$ coordinatewise and $Bx \leq cx$, then* $\operatorname{spr}(B) \leq c$.

(ii) *There are nonzero vectors $v, w \geq 0$ such that*

$$Bv = \operatorname{spr}(B)v, \qquad B^T w = \operatorname{spr}(B)w.$$

*Proof.* For $x > 0$, define the weighted supremum norm

$$\|z\|_x = \max_i \frac{|z_i|}{x_i} \qquad (z \in \mathbb{C}^m).$$

If $Bx \le cx$, then entrywise nonnegativity gives

$$|Bz| \le B|z| \le \|z\|_x Bx \le c\|z\|_x x.$$

Hence the operator norm of $B$ in $\|\cdot\|_x$ is at most $c$. Every eigenvalue has modulus bounded by every operator norm, proving (i).

We first prove (ii) for a strictly positive matrix $A$. On the open probability simplex

$$\Sigma^\circ = \{x \in \mathbb{R}^m : x_i > 0,\ \sum_i x_i = 1\}$$

consider the function

$$\Phi(x) = \max_i \frac{(Ax)_i}{x_i}.$$

It satisfies $\Phi(tx) = \Phi(x)$ for every $t > 0$. Let $a_0 = \min_{i,j} a_{ij} > 0$. If a sequence in $\Sigma^\circ$ approaches the boundary, some coordinate $x_i$ tends to zero while $(Ax)_i \ge a_0 \sum_j x_j = a_0$; therefore $\Phi(x) \to \infty$. Fix one $x^{(0)} \in \Sigma^\circ$. The set $\{x \in \Sigma^\circ : \Phi(x) \le \Phi(x^{(0)})\}$ stays a positive distance from the boundary and is closed in the compact probability simplex. It is therefore compact and nonempty, and continuity of $\Phi$ gives an interior minimizer $x$.

Put $M = \Phi(x)$. Then $Ax \le Mx$. If the inequality were strict in at least one coordinate, the nonzero vector $Mx - Ax \ge 0$ would satisfy $A(Mx - Ax) > 0$ coordinatewise because every entry of $A$ is positive. Consequently

$$A^2 x < MAx$$

coordinatewise. With $y = Ax > 0$, this says $\Phi(y) < M$, contradicting the minimality of $x$ after normalizing $y$ back to the simplex. Thus $Ax = Mx$. Part (i), applied with equality, shows $\operatorname{spr}(A) \le M$, while $M$ is itself an eigenvalue; hence $M = \operatorname{spr}(A)$. Applying the same argument to $A^T$ gives a strictly positive left Perron vector.

Now let $B \ge 0$ and put

$$A_\varepsilon = B + \varepsilon \mathbf{1}\mathbf{1}^T \qquad (\varepsilon > 0).$$

The preceding paragraph gives positive right and left Perron vectors $v_\varepsilon, w_\varepsilon$, normalized by $\sum_i (v_\varepsilon)_i = \sum_i (w_\varepsilon)_i = 1$, and a positive spectral radius $r_\varepsilon = \operatorname{spr}(A_\varepsilon)$. We show that $r_\varepsilon \to r := \operatorname{spr}(B)$.

First, $r \le r_\varepsilon$. Choose an eigenvalue $\zeta$ of $B$ with $|\zeta| = r$ and a corresponding nonzero complex eigenvector $z$. Since

$$r|z| = |Bz| \le B|z| \le A_\varepsilon |z|,$$

let $C = \max_i |z_i|/(v_\varepsilon)_i$ and choose an index attaining the maximum. At that index,

$$rC(v_\varepsilon)_i \le (A_\varepsilon |z|)_i \le C(A_\varepsilon v_\varepsilon)_i = C r_\varepsilon (v_\varepsilon)_i,$$

so $r \le r_\varepsilon$.

For the converse bound, fix $c > r$. Choose $r_0$ with $r < r_0 < c$. In Jordan normal form, the $k$th power of a block $J_\zeta = \zeta I + N$ of size $m_\zeta$ is

$$J_\zeta^k = \sum_{j=0}^{m_\zeta - 1} \binom{k}{j} \zeta^{k-j} N^j.$$

Because $|\zeta| \le r < r_0$, this formula gives constants $C, M$ such that $\|B^k\| \le Ck^M r_0^k$ for all $k$. Therefore the matrix series $\sum_{k\ge 0} c^{-k-1}B^k$ converges absolutely. Multiplying its partial sums by $cI - B$ and passing to the limit gives

$$(cI - B)^{-1} = \sum_{k=0}^{\infty} c^{-k-1}B^k.$$

Consequently

$$x_c = (cI - B)^{-1}\mathbf{1} = \sum_{k=0}^{\infty} c^{-k-1}B^k\mathbf{1} > 0, \qquad Bx_c = cx_c - \mathbf{1}.$$

Therefore

$$A_\varepsilon x_c = cx_c - \mathbf{1} + \varepsilon(\mathbf{1}^T x_c)\mathbf{1} \le (c + \eta_\varepsilon)x_c,$$

where

$$\eta_\varepsilon = \varepsilon(\mathbf{1}^T x_c)\max_i (x_c)_i^{-1} \longrightarrow 0.$$

Part (i) gives $r_\varepsilon \le c + \eta_\varepsilon$. Given $\delta > 0$, choose $c$ with $r < c < r + \delta/2$. Since $\eta_\varepsilon \to 0$, for all sufficiently small $\varepsilon$ we have $r \le r_\varepsilon < c + \delta/2 < r + \delta$. Hence $r_\varepsilon \to r$.

The closed probability simplex is compact. Along a sequence $\varepsilon_k \downarrow 0$, the normalized vectors converge to nonzero vectors $v, w \ge 0$. Passing to the limit in

$$A_{\varepsilon_k} v_{\varepsilon_k} = r_{\varepsilon_k} v_{\varepsilon_k}, \qquad A_{\varepsilon_k}^T w_{\varepsilon_k} = r_{\varepsilon_k} w_{\varepsilon_k}$$

gives $Bv = rv$ and $B^T w = rw$. □

**Lemma A.2** (Strict separation from a compact convex set)**.** *Let $K$ be a nonempty compact convex subset of a Euclidean space and let $o \notin K$. There is a linear functional $\ell$ such that*

$$\ell(x) < \ell(o) \qquad (x \in K).$$

*Proof.* Choose $y \in K$ minimizing $|o - y|$ and put $v = o - y \ne 0$. For every $x \in K$, the segment $y + t(x - y)$, $0 \le t \le 1$, belongs to $K$. The function

$$f(t) = |o - y - t(x - y)|^2$$

has a minimum at $t = 0$, so $f'(0) \ge 0$. Thus $\langle v, x - y\rangle \le 0$. With $\ell(z) = \langle v, z\rangle$ we obtain

$$\ell(x) \le \ell(y) < \ell(o),$$

the last inequality being $\ell(o) - \ell(y) = |v|^2 > 0$. □

**Lemma A.3** (Polarity for planar polygons)**.** *Let $K \subset \mathbb{R}^2$ be a compact convex set with $0 \in \operatorname{int} K$, and define*

$$K^\circ = \{y : \langle y, x\rangle \le 1 \text{ for every } x \in K\}.$$

*Then $K^\circ$ is compact, convex, and contains $0$ in its interior. If $A$ is linear and $AK \subseteq K$, then*

$$A^* K^\circ \subseteq K^\circ.$$

*If $K$ is a strict polygon with $N$ vertices, then $K^\circ$ is a strict polygon with $N$ vertices. For every vertex $x$ of $K$, the set*

$$F_x = \{y \in K^\circ : \langle y, x\rangle = 1\}$$

*is a side of $K^\circ$, and every side arises uniquely in this way. Finally, if $x \in K$, $y \in K^\circ$, and $\langle y, x\rangle = 1$, then $x \in \partial K$ and $y \in \partial K^\circ$.*

*Proof.* Choose radii $0 < r < R$ with $r\overline{\mathbb{D}} \subseteq K \subseteq R\overline{\mathbb{D}}$. The defining inequalities show

$$R^{-1}\overline{\mathbb{D}} \subseteq K^\circ \subseteq r^{-1}\overline{\mathbb{D}},$$

so $K^\circ$ has the asserted compactness and interior properties. If $y \in K^\circ$ and $x \in K$, then $Ax \in K$ and

$$\langle A^* y, x\rangle = \langle y, Ax\rangle \leq 1;$$

therefore $A^* y \in K^\circ$.

Let the vertices of the strict polygon be $x_1, \ldots, x_N$. Since a linear functional reaches its maximum over a convex hull at a vertex,

$$K^\circ = \bigcap_{i=1}^{N} \{y : \langle y, x_i\rangle \leq 1\}.$$

Every displayed inequality is irredundant. Indeed, at the vertex $x_i$ choose a strict supporting functional $u_i$ of $K$. Its support value $h_K(u_i)$ is positive because 0 is interior. After replacing $u_i$ by $u_i/h_K(u_i)$, one has

$$\langle u_i, x_i\rangle = 1, \qquad \langle u_i, x_j\rangle < 1 \quad (j \neq i).$$

On the line $\langle y, x_i\rangle = 1$, all other inequalities remain strict in a small interval about $u_i$. Hence $F_{x_i}$ is a nondegenerate side of $K^\circ$. The intersection description has only the $N$ displayed half-planes, so it has at most $N$ sides, while the preceding argument has produced $N$ distinct sides. It therefore has exactly $N$ sides. Traversing the boundary of a bounded full-dimensional polygon alternates sides and vertices, so it also has exactly $N$ vertices. Since no displayed side is redundant or degenerate, the polar polygon is strict. The same argument also proves the asserted bijection between vertices of $K$ and sides of $K^\circ$.

If $x$ were interior to $K$ and $y \neq 0$, then $x + \varepsilon y \in K$ for all sufficiently small $\varepsilon > 0$, giving

$$1 \geq \langle y, x + \varepsilon y\rangle = 1 + \varepsilon |y|^2,$$

a contradiction. Thus equality forces $x \in \partial K$. The defining inequality $\langle \cdot, x\rangle \leq 1$ for $K^\circ$ is attained at $y$, so $y \in \partial K^\circ$. □

**Lemma A.4** (Hausdorff limits of finitely generated polygons)**.** *Fix $N$ and let*

$$P_k = \operatorname{conv}\{z_{k,1}, \ldots, z_{k,N}\}$$

*with all points in one compact subset of $\mathbb{R}^2$. If $z_{k,j} \to z_j$ for every $j$, then*

$$P_k \longrightarrow P := \operatorname{conv}\{z_1, \ldots, z_N\}$$

*in the Hausdorff metric. If $P$ has nonempty interior, then* $\operatorname{area}(P_k) \to \operatorname{area}(P)$*. Moreover, for every fixed linear map $A$, the conditions $AP_k \subseteq P_k$ pass to the limit, and*

$$\max_{z \in P_k} |z| \longrightarrow \max_{z \in P} |z|.$$

*Proof.* For nonempty compact sets $K, L \subseteq \mathbb{R}^2$, their Hausdorff distance is

$$d_H(K, L) = \max\left\{\sup_{x \in K} \operatorname{dist}(x, L), \sup_{y \in L} \operatorname{dist}(y, K)\right\},$$

and we write $\overline{\mathbb{D}} = \{z \in \mathbb{R}^2 : |z| \leq 1\}$ for the closed unit disk.

Put $\varepsilon_k = \max_j |z_{k,j} - z_j|$. If $z = \sum_j t_j z_{k,j} \in P_k$, then $z' = \sum_j t_j z_j \in P$ and $|z - z'| \le \varepsilon_k$; the converse estimate is identical. Hence the Hausdorff distance is at most $\varepsilon_k$.

All polygons lie in one large disk. We prove convergence of their indicator functions away from $\partial P$. If $z \notin P$, its positive distance from $P$ and Hausdorff convergence imply $z \notin P_k$ for large $k$. If $z \in \operatorname{int} P$, choose $r > 0$ with $z + r\overline{\mathbb{D}} \subseteq P$. Hausdorff convergence implies uniform convergence of support functions:

$$|h_{P_k}(u) - h_P(u)| \le d_H(P_k, P) \qquad (|u| = 1).$$

The contained disk gives the uniform support margin

$$h_P(u) \ge \langle u, z\rangle + r \qquad (|u| = 1).$$

For large $k$, when $d_H(P_k, P) < r$, these two estimates give

$$h_{P_k}(u) > \langle u, z\rangle \qquad (|u| = 1).$$

If $z \notin P_k$, lemma A.2 would produce a unit vector $u$ with $\langle u, z\rangle > h_{P_k}(u)$, a contradiction. Thus $z \in P_k$ eventually. All generating points lie in one compact set, so one fixed disk $R\overline{\mathbb{D}}$ contains every $P_k$ and $P$. The indicators of the polygons are therefore dominated by the integrable function $\mathbf{1}_{R\overline{\mathbb{D}}}$. Since the boundary of a polygon is a finite union of line segments and has planar measure zero, dominated convergence of the indicator functions gives area convergence.

If $AP_k \subseteq P_k$ for every $k$ and $x \in P$, choose $x_k \in P_k$ with $x_k \to x$. Then $Ax_k \in P_k$, and Hausdorff convergence gives $Ax \in P$. Finally, by convexity of the norm,

$$\max_{z \in P_k} |z| = \max_j |z_{k,j}|, \qquad \max_{z \in P} |z| = \max_j |z_j|,$$

which proves convergence of the maxima. □

**Lemma A.5** (Strict area monotonicity)**.** *If $K \subseteq L$ are compact convex subsets of $\mathbb{R}^2$, both with nonempty interior, and $K \ne L$, then*

$$\operatorname{area}(K) < \operatorname{area}(L).$$

*Proof.* Choose $y \in L \setminus K$. By lemma A.2, there is a linear functional $\ell$ with

$$\ell(y) > a := \max_{x \in K} \ell(x).$$

Choose a closed disk $D \subset \operatorname{int} K$. The convex hull $C = \operatorname{conv}(D \cup \{y\})$ satisfies $C \subseteq L$. Because $D \subset \operatorname{int} K$, compactness gives $\max_D \ell < a < \ell(y)$. Choose $t \in (0, 1)$ sufficiently close to 1 that

$$(1 - t)\min_D \ell + t\ell(y) > a.$$

Then

$$D_t = (1 - t)D + ty$$

is a disk of positive area contained in $C$, and every $x \in D_t$ satisfies $\ell(x) > a$. Hence $D_t \subseteq L \setminus K$, so $L \setminus K$ has positive area. This proves the strict inequality. □

**Lemma A.6** (Lattice parallelogram count)**.** *Let $u, v \in \mathbb{Z}^2$ be linearly independent and put $D = |\det(u, v)|$. The half-open parallelogram*

$$\mathcal{P} = \{\alpha u + \beta v : 0 \le \alpha, \beta < 1\}$$

*contains exactly $D$ integer points, one in each coset of $\mathbb{Z}^2/(\mathbb{Z}u + \mathbb{Z}v)$.*

*Proof.* Every $z \in \mathbb{R}^2$, and hence every $z \in \mathbb{Z}^2$, has a unique expression $z = au+bv$ with $a, b \in \mathbb{R}$. Subtracting $\lfloor a \rfloor u + \lfloor b \rfloor v$ gives a representative in $\mathcal{P}$. Because $z, u, v$ are integer vectors, this representative is still an integer vector. Two integer points of $\mathcal{P}$ cannot represent the same coset: their difference is $mu + nv$ with integers $m, n$, while both coefficient differences lie in $(-1, 1)$, forcing $m = n = 0$. Thus the integer points of $\mathcal{P}$ are in bijection with the quotient group.

It remains to compute the order of that quotient. Integer elementary row and column operations, with determinant $\pm 1$, do not change either the index of the generated sublattice or the absolute determinant. Applying the Euclidean algorithm to the $2 \times 2$ matrix with columns $u, v$ reduces it to Smith normal form

$$\begin{pmatrix} d_1 & 0 \\ 0 & d_2 \end{pmatrix}, \qquad d_1, d_2 > 0.$$

For this diagonal lattice the quotient has exactly $d_1 d_2$ elements, and $d_1 d_2$ is the absolute determinant. Reversing the unimodular operations gives the result for $u, v$. $\square$

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# Part II: Invariant Polygons and the Farey–Ito Boundary of Stochastic Spectra

**Interface with Part I.** Part II applies the structural results proved in Part I, in particular theorem 1.4. The signed closing exponent $e = s - dq$ follows the convention stated there: it may be negative, and in that case the homogeneous polynomial identity is the primary formulation.

## II.1 Introduction and proof architecture

For $n \geq 1$, define

$$\Theta_n = \bigcup \left\{ \operatorname{spec} A : A \in \mathbb{R}^{n\times n}_{\geq 0},\ A\mathbf{1} = \mathbf{1} \right\}. \tag{II.1.1}$$

Throughout, we identify $\mathbb{C}$ with the oriented plane $\mathbb{R}^2$ and use

$$\langle z, w \rangle = \operatorname{Re}(\overline{z}w), \qquad \det(z, w) = \operatorname{Im}(\overline{z}w).$$

Positive cyclic order means counterclockwise order; oriented boundary intervals inherit that convention.

The problem is to determine the possible location of one eigenvalue, not an entire spectrum. The answer is a compact radial region: its unit-circle points are roots of unity of order at most $n$, and its noncircular boundary arcs are indexed by consecutive fractions in the Farey sequence of order $n$.

The stochastic eigenvalue-region problem was developed by Dmitriev and Dynkin [1] and solved in full by Karpelevič [4]. Ito's reformulation [2] expresses the boundary through one-parameter polynomial families indexed by Farey neighbors. Later matrix realizations and structural treatments include [3, 5, 6]. The present argument derives the Farey–Ito boundary directly from invariant polygons.

The proof passes from stochastic spectra to critical planar dynamics and then to a sharp one-dimensional inequality. Its logical spine is

$$\begin{gathered}
\text{stochastic eigenpair} \iff \text{invariant polygon} \implies \text{non-inherited radial maximum} \\
\implies N\text{-criticality} \xRightarrow[N\geq 4]{\text{critical-polygon monodromy}} \\
\implies \text{heterogeneous Ito product and common argument interval} \\
\implies \text{log-sine equalization} \implies \text{cellwise radial equality} \\
\implies \text{explicit attainment and Farey nesting} \implies \text{Farey–Ito boundary.}
\end{gathered} \tag{II.1.2}$$

The invariant-polygon criterion identifies a non-inherited stochastic radial maximum with an intrinsic $N$-critical elliptic contraction. For $N \geq 4$, the monodromy theorem of Part I converts that criticality into the finite Farey product, phase, and common-argument-interval data needed here. Orders at most three are handled directly and do not pass through the projective cyclic-successor argument. The remaining steps are the local scalar comparison, explicit realization, and global boundary ordering.

**Dependency order.** For non-inherited radial maxima of orders $N \geq 4$, the upper bound uses, in order, the invariant-polygon criterion theorem II.4.2, the implication to criticality proposition II.4.7, the Part I monodromy theorem theorem II.5.1, and the varying-parameter inequality theorem II.6.1. The reverse inclusion is independent of the Part I input: it comes from the sparse stochastic realization theorem II.7.3. Candidate nesting theorem II.8.4 is used only after both bounds are known, to pass from non-inherited radial maxima to all orders.

Thus neither realization nor nesting is used to establish criticality or monodromy. The direct arguments for orders at most three precede this chain.

The Part I assertion is existential: it selects one of the two adapted complex orientations and supplies monodromy data in that orientation. No uniqueness of invariant polygons, contact topology, or stochastic realizers is required.

**Background scope.** No prior Karpelevič boundary reduction or parametrization is used. We use elementary finite-dimensional convexity and compactness for the invariant-polygon criterion and radial structure. The intrinsic critical-polygon geometry, including the boundary conclusion for every invariant polygon within the vertex bound and contact reduction, is proved in Part I; for $N \geq 4$ its return towers and projective cyclic-successor argument culminate in theorem 1.4. Part II uses only the complex monodromy and common-argument-interval conclusions through theorem II.5.1; every other problem-specific implication used in Part II is proved below. The cases $n \leq 3$ are treated directly.

## II.2 Farey data, candidate arcs, and direct extraction

### II.2.1 Farey cells

Let $F_n$ denote the Farey sequence of order $n$ on $[0, 1]$. We use

$$F_n^+ = F_n \cap \left[0, \frac{1}{2}\right]$$

for the upper half-plane.

**Lemma II.2.1** (Farey adjacency criterion)**.** *Two reduced fractions $a/b < c/d$ belonging to $F_n$ are consecutive in $F_n$ if and only if*

$$bc - ad = 1, \qquad b + d > n. \tag{II.2.1}$$

*Proof.* Assume first that $bc - ad = 1$. If a reduced fraction $h/k$ lies strictly between $a/b$ and $c/d$, define

$$m = ck - dh, \qquad \ell = bh - ak.$$

Cross multiplication shows that $m$ and $\ell$ are positive integers. The determinant-one identity gives the exact decomposition

$$(k, h) = m(b, a) + \ell(d, c).$$

Looking at the first coordinate yields $k = mb + \ell d \geq b + d$. Hence, if $b + d > n$, no fraction of denominator at most $n$ can lie between the two endpoints, so they are consecutive in $F_n$.

Conversely, suppose the two fractions are consecutive in $F_n$ and put $D = bc - ad$. The determinant is a positive integer. If $D > 1$, the half-open parallelogram spanned by the primitive integer vectors $u = (b, a)$ and $v = (d, c)$ contains a nonzero integer point

$$w = \alpha u + \beta v, \qquad 0 < \alpha, \beta < 1.$$

Indeed, the half-open fundamental parallelogram contains exactly $D = |\det(u, v)|$ representatives of the quotient group $\mathbb{Z}^2/(\mathbb{Z}u + \mathbb{Z}v)$. When $D > 1$, one of these representatives is different from the origin. It cannot lie on either open coordinate edge of the parallelogram: such a point would be a nonintegral proper multiple of the primitive vector $u$ or $v$. The strict inequalities follow because a primitive vector has no integer point in the relative interior of its first unit segment. Both $w$ and $u + v - w$ are positive combinations of $u, v$, so their slopes lie strictly between $a/b$ and $c/d$. Their first coordinates are positive integers $k$ and $b+d-k$; one is at most $(b+d)/2 \le n$ because $b, d \le n$. After reducing that intermediate slope, its denominator can only decrease. This produces an element of $F_n$ strictly between the alleged consecutive fractions, a contradiction. Therefore $D = 1$. Finally, if $b+d \le n$, the reduced mediant $(a+c)/(b+d)$ has denominator at most $n$ and lies strictly between the endpoints. Thus consecutiveness also forces $b+d > n$. □

Let $f < g$ be consecutive elements of $F_n^+$. Label the two endpoints, independently of their left-right order, as

$$\frac{p}{q}, \qquad \frac{r}{s}, \qquad q \le s, \tag{II.2.2}$$

so that $|rq - ps| = 1$. Put

$$d = \left\lfloor \frac{n}{q} \right\rfloor, \qquad e = s - dq. \tag{II.2.3}$$

The integer $d$ is the number of complete $q$-returns that fit into the vertex budget $n$.

**Definition II.2.2** (Ito polynomial family)**.** For $0 \le \alpha \le 1$, put $\beta = 1 - \alpha$. The Ito equation associated with the cell $[f, g]$ is

$$\lambda^s(\lambda^q - \beta)^d = \alpha^d \lambda^{dq}. \tag{II.2.4}$$

For $\lambda \neq 0$, equivalently, after cancelling the common power of $\lambda$ and clearing a possible negative exponent,

$$\begin{cases} \lambda^e(\lambda^q - \beta)^d = \alpha^d, & e \ge 0, \\ (\lambda^q - \beta)^d = \alpha^d \lambda^{-e}, & e < 0. \end{cases} \tag{II.2.5}$$

We define the cellwise candidate curve by its radial graph, rather than by an unproved choice of an $\alpha$-parametrized algebraic root branch. The next proposition constructs exactly one algebraic point on every open ray.

### II.2.2 The scalar radius equation

Let $x \in (f, g)$ and write $\lambda = \rho e^{2\pi i x}$. Define

$$A = 2\pi \left| qx - p \right|, \qquad B = \frac{2\pi}{d} \left| sx - r \right|. \tag{II.2.6}$$

To verify the range explicitly, put

$$t = \frac{|x - p/q|}{|r/s - p/q|} \in (0, 1).$$

The determinant-one identity gives

$$|qx - p| = \frac{t}{s}, \qquad |sx - r| = \frac{1-t}{q},$$

regardless of which labeled endpoint is on the left. Hence

$$\frac{A+B}{2\pi} = \frac{t}{s} + \frac{1-t}{dq}.$$

For an order-$n \ge 3$ Farey cell labeled by $q \le s$, one has $s \ge 3$ and $dq \ge 2$. If $dq = 2$, the coefficient $1/(dq) = 1/2$ is multiplied by the strict weight $1 - t < 1$, while the other coefficient is smaller than $1/2$. Thus the open-cell convex combination is strictly smaller than $1/2$, and

$$A > 0, \qquad B > 0, \qquad A + B < \pi. \tag{II.2.7}$$

**Proposition II.2.3** (One scalar equation per ray)**.** *There is a unique* $\rho^{(n)}_{f,g}(x) \in (0,1)$ *satisfying*

$$\rho^{s/d}\sin A + \rho^{q}\sin B = \sin(A+B). \tag{II.2.8}$$

*For this radius,*

$$\alpha = \frac{\rho^{s/d}\sin A}{\sin(A+B)}, \qquad \beta = \frac{\rho^{q}\sin B}{\sin(A+B)} \tag{II.2.9}$$

*are nonnegative and sum to one, and* $\lambda = \rho e^{2\pi i x}$ *satisfies* (II.2.4)*. The rooted Ito identity* (II.2.10)*, with the fractional-power branch specified in the proof, varies continuously with* $x$ *throughout the open cell.*

*Proof.* The left side of (II.2.8) is strictly increasing in $\rho$. At $\rho = 0$ it is zero, while at $\rho = 1$ it exceeds the right side because

$$\sin A + \sin B - \sin(A+B) = 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{A+B}{2} > 0.$$

This proves existence and uniqueness. Equation (II.2.9) and (II.2.8) give $\alpha + \beta = 1$.

Set

$$z = \overline{\lambda}^{-1} = \rho^{-1}e^{2\pi i x}.$$

Choose the $d$th-root branch anchored at the endpoint $r/s$ by defining

$$\omega = e^{-2\pi i r/d}, \qquad \omega z^{s/d} := \rho^{-s/d}\exp\left(\frac{2\pi i(sx-r)}{d}\right).$$

The two quantities $qx - p$ and $sx - r$ have opposite signs. Hence, with a common choice of sign,

$$z^{q} = \rho^{-q}e^{\pm iA}, \qquad \omega z^{s/d} = \rho^{-s/d}e^{\mp iB}.$$

Moreover (II.2.9) gives

$$\beta\rho^{-q} = \frac{\sin B}{\sin(A+B)}, \qquad \alpha\rho^{-s/d} = \frac{\sin A}{\sin(A+B)}.$$

The cancellation in the vector identity is completely explicit. The imaginary part of the proposed right side is

$$\frac{\sin B\sin A - \sin A\sin B}{\sin(A+B)} = 0,$$

and its real part is

$$\frac{\sin B\cos A + \sin A\cos B}{\sin(A+B)} = 1.$$

Thus resolving the two vectors into real and imaginary parts gives

$$1 = \beta z^{q} + \alpha\omega z^{s/d}. \tag{II.2.10}$$

Raising to the $d$th power after moving the first term gives

$$(1 - \beta\overline{\lambda}^{-q})^{d} = \alpha^{d}\overline{\lambda}^{-s},$$

and therefore

$$(\overline{\lambda}^{q} - \beta)^{d} = \alpha^{d}\overline{\lambda}^{dq-s}.$$

Conjugation is exactly (II.2.4). All quantities in (II.2.10) depend continuously on $x$ on the open cell. □

**Proposition II.2.4** (Endpoint limits, including the order-three exception). *On every open Farey cell, $x \mapsto \rho^{(n)}_{f,g}(x)$ is continuous. If $n \geq 4$, it extends continuously to the closed cell with value 1 at both endpoints, and the corresponding points tend to the two endpoint roots of unity. The same conclusion holds for $n = 3$ except on the terminal cell $[1/3, 1/2]$. On that cell,*

$$\lim_{x \downarrow 1/3} \rho^{(3)}_{1/3,1/2}(x) = 1, \qquad \lim_{x \uparrow 1/2} \rho^{(3)}_{1/3,1/2}(x) = \frac{1}{2}, \tag{II.2.11}$$

*so the nonreal radial graph tends to $-1/2$, not to the endpoint root $-1$.*

*Proof.* Open-cell continuity follows from strict monotonicity in $\rho$ in (II.2.8), equivalently from the implicit-function theorem. Let $x_k$ tend to an endpoint and pass to a subsequence for which $\rho(x_k) \to r \in [0, 1]$.

At the endpoint $p/q$, where $A \to 0$, determinant one gives

$$B_0 = \frac{2\pi}{dq}.$$

Thus $0 < B_0 \leq \pi$. Equality can occur only when $dq = 2$; under $n \geq 3$, $q \leq s$, and the upper-half-plane Farey restrictions, this is exactly the order-three terminal cell, with $q = 2$, $d = 1$, endpoint $p/q = 1/2$, and opposite endpoint $r/s = 1/3$. Outside that single case, $B_0 \in (0, \pi)$ and the limiting equation

$$r^q \sin B_0 = \sin B_0$$

forces $r = 1$.

At the other endpoint $r/s$, where $B \to 0$, one has

$$A_0 = \frac{2\pi}{s} \in (0, \pi),$$

because $s \geq 3$. Hence $r^{s/d} \sin A_0 = \sin A_0$ and again $r = 1$.

It remains to evaluate the exceptional limit. Put $x = 1/2 - \varepsilon$ in the cell $[1/3, 1/2]$. Then

$$A = 4\pi\varepsilon, \qquad B = \pi - 6\pi\varepsilon, \qquad A + B = \pi - 2\pi\varepsilon,$$

and (II.2.8) is

$$\rho^3 \sin(4\pi\varepsilon) + \rho^2 \sin(6\pi\varepsilon) = \sin(2\pi\varepsilon).$$

After division by $2\pi\varepsilon$ and passage to the limit, every subsequential limit satisfies

$$2r^3 + 3r^2 = 1, \qquad (2r - 1)(r + 1)^2 = 0.$$

The unique solution in $[0, 1]$ is $r = 1/2$. Since every subsequence has the same limit, all asserted endpoint limits follow. In the exceptional asymptotics, (II.2.9) also gives $\alpha \to (1/2)^3 \cdot 4/2 = 1/4$, so the nonreal arc meets the added real branch at the same algebraic parameter. At every nonexceptional endpoint, (II.2.10) tends to the corresponding root of unity. □

**Definition II.2.5** (Cellwise Farey–Ito candidate curve). Let

$$\Gamma^{(n),\circ}_{f,g} = \left\{ \rho^{(n)}_{f,g}(x) e^{2\pi i x} : f < x < g \right\}. \tag{II.2.12}$$

For every cell other than the order-three terminal cell, define

$$\Gamma^{(n)}_{f,g} = \overline{\Gamma^{(n),\circ}_{f,g}}.$$

For the exceptional cell, define

$$\Gamma^{(3)}_{1/3,1/2} = \overline{\Gamma^{(3),\circ}_{1/3,1/2}} \cup [-1, -1/2]. \tag{II.2.13}$$

Every point of the open radial graph satisfies the Ito equation by proposition II.2.3. On the added real segment, writing $\alpha = -\lambda(\lambda+1) \in [0, 1/4]$ gives

$$\lambda^3 - (1-\alpha)\lambda - \alpha = (\lambda - 1)(\lambda^2 + \lambda + \alpha) = 0,$$

so the exceptional segment belongs to the same algebraic family.

**Algorithm II.2.6** (Candidate-curve extraction)**.** For a fixed order $n \geq 3$:

1. Generate $F_n^+$ in increasing order.
2. For each consecutive pair $f < g$, label the endpoint with smaller denominator $p/q$ and the other $r/s$, compute $d = \lfloor n/q \rfloor$, and record the reduced Ito polynomial (II.2.5).
3. For a desired ray $x \in (f, g)$, compute $A, B$ from (II.2.6) and solve the strictly monotone equation (II.2.8) by bisection or Newton iteration.
4. Recover $\alpha, \beta$ from (II.2.9). At this stage the corresponding point is the unique *scalar candidate* on that ray. Appendix II.7 constructs a stochastic matrix realizing it, and appendices II.8 and II.9 prove the nesting and outer-boundary statements needed to identify it as the boundary point and fill the radial segment below it.

## II.3 Main theorem

**Theorem II.3.1** (Karpelevič–Ito, standalone form)**.** *Let $n \geq 3$. For every consecutive pair $f < g$ in $F_n^+$, let $\Gamma_{f,g}^{(n)}$ be the candidate curve of definition II.2.5. Then these curves are the Karpelevič boundary arcs, and their ordered concatenation is the upper boundary of $\Theta_n$, from $1$ to $-1$. The lower boundary is its complex conjugate, and*

$$\Theta_n = \{t\zeta : 0 \leq t \leq 1,\ \zeta \in \partial\Theta_n\}\,. \tag{II.3.1}$$

*The only points of $\Theta_n$ on the unit circle are roots of unity of order at most $n$.*

*For $n = 3$, the terminal candidate curve contains the boundary segment $[-1, -1/2]$ before leaving the real axis. For $n = 1$, one has $\Theta_1 = \{1\}$; for $n = 2$, one has $\Theta_2 = [-1, 1]$.*

The proof occupies appendix II.4–appendix II.9. The principal upper-bound input is theorem 1.4 from Part I, restated for non-inherited radial maxima in appendix II.5. Everything after it is one-dimensional convexity, explicit realization, and Farey nesting.

## II.4 Invariant polygons and elementary structure

**Proposition II.4.1** (Compactness, conjugation, and disk bound)**.** *For every $n$, the set $\Theta_n$ is compact, invariant under complex conjugation, and contained in the closed unit disk.*

*Proof.* Let $\mathcal{S}_n$ be the compact set of row-stochastic matrices. First note that every eigenvalue lies in the closed unit disk: if $Av = \lambda v$, choose an index with $|v_i| = \|v\|_\infty$; then

$$|\lambda|\|v\|_\infty = |(Av)_i| \leq \|v\|_\infty.$$

Consequently the eigenpair set

$$\mathcal{E}_n = \{(A, \lambda) \in \mathcal{S}_n \times \overline{\mathbb{D}} : \det(\lambda I - A) = 0\}$$

is closed in a compact space and is therefore compact. Its projection to the $\lambda$-coordinate is exactly $\Theta_n$, so $\Theta_n$ is compact. Real characteristic polynomials give conjugation symmetry. $\square$

**Theorem II.4.2** (Invariant-polygon criterion)**.** *Let $n \geq 2$. For $\lambda \in \mathbb{C}$, the following are equivalent.*

(i) $\lambda \in \Theta_n$.
(ii) *There is a non-singleton convex polygon $P \subset \mathbb{C}$ with at most $n$ vertices such that*

$$\lambda P \subseteq P. \tag{II.4.1}$$

*Proof.* If $Av = \lambda v$, put $P = \operatorname{conv}\{v_1, \dots, v_n\}$. Every coordinate satisfies

$$\lambda v_i = \sum_j a_{ij} v_j \in P,$$

so (II.4.1) holds. If the hull is a singleton, all coordinates of the nonzero eigenvector are equal to the same nonzero number, so $\lambda = 1$; in that case any fixed segment may be used.

Conversely, let $x_1, \dots, x_m$ be the vertices of an invariant polygon, $m \le n$. Choose barycentric coordinates

$$\lambda x_i = \sum_{j=1}^{m} a_{ij} x_j, \qquad a_{ij} \ge 0, \qquad \sum_j a_{ij} = 1.$$

Then $A = (a_{ij})$ is row-stochastic and $Ax = \lambda x$. If $m < n$, append absorbing states and extend the eigenvector by zero coordinates. □

**Corollary II.4.3** (Star-shapedness with respect to the origin)**.** *Let $n \ge 2$. If $\lambda \in \Theta_n$, then $t\lambda \in \Theta_n$ for every $0 \le t \le 1$.*

*Proof.* If $|\lambda| < 1$, take an invariant polygon $P$. Iteration gives $\lambda^k P \subseteq P$ and $\lambda^k x \to 0$, hence $0 \in P$; convexity then gives $t\lambda P \subseteq P$.

Suppose $|\lambda| = 1$. If $\lambda = 1$, the segment $[0, 1]$ satisfies $t[0, 1] \subseteq [0, 1]$. Otherwise take an invariant polygon with the least number of vertices. If it has nonzero area, area equality forces $\lambda P = P$, so multiplication by $\lambda$ permutes its vertices and $\lambda$ has finite order $k \le n$. The regular orbit polygon

$$Q = \operatorname{conv}\{1, \lambda, \dots, \lambda^{k-1}\}$$

contains 0, satisfies $\lambda Q = Q$, and therefore satisfies $t\lambda Q = tQ \subseteq Q$. If the minimal polygon is a segment, then $\lambda = -1$ and the same conclusion follows from $Q = [-1, 1]$. The invariant-polygon criterion completes the proof. □

**Proposition II.4.4** (Unit-circle points)**.** *Let $n \ge 2$. If $\lambda \in \Theta_n$ and $|\lambda| = 1$, then $\lambda$ is a root of unity of order at most $n$. Conversely every such root belongs to $\Theta_n$.*

*Proof.* Take an invariant polygon with the least number of vertices. If it has nonzero area, then $\lambda P \subseteq P$ and equality of areas force $\lambda P = P$. Multiplication by $\lambda$ permutes the vertices, so a nonzero vertex has an orbit of length at most $n$ and $\lambda^k = 1$. If the polygon is a segment, $\lambda = \pm 1$. Conversely, a cyclic permutation matrix realizes every root of unity of order at most $n$, after padding. □

**Lemma II.4.5** (Interior origin for a nonreal contraction)**.** *Let $P$ be a non-singleton compact convex polygon and let $\lambda = \rho e^{i\theta}$ with $0 < \rho < 1$ and $\theta \not\equiv 0, \pi \pmod{2\pi}$. If $\lambda P \subseteq P$, then $P$ has nonempty interior and $0 \in \operatorname{int} P$.*

*Proof.* Iteration gives $0 \in P$. If $P$ were a segment, its affine line would pass through 0 and would have to contain the nondegenerate segment $\lambda P$; hence multiplication by $e^{i\theta}$ would preserve a real line, forcing $\theta \equiv 0$ or $\pi$, a contradiction. Thus $P$ has nonempty interior.

Suppose $0 \in \partial P$. Choose a nonzero real linear functional $\ell$ such that $\ell \ge 0$ on $P$ and choose $z \in P$ with $\ell(z) > 0$. Invariance would give

$$\ell(e^{ik\theta} z) \ge 0 \qquad (k \ge 0).$$

If $e^{i\theta}$ has finite order $m$, then $m \geq 3$ and $\sum_{k=0}^{m-1} e^{ik\theta} z = 0$; summing the displayed nonnegative quantities forces them all to vanish, contrary to $\ell(z) > 0$. If $e^{i\theta}$ has infinite order, its powers are dense in the unit circle, so some $e^{ik\theta} z$ lies in the open half-plane where $\ell < 0$, again a contradiction. Hence 0 is interior. $\square$

For $0 < \theta < \pi$, define the radial function

$$R_n(\theta) = \max\left\{\rho \geq 0 : \rho e^{i\theta} \in \Theta_n\right\}. \tag{II.4.2}$$

The maximum exists by compactness and star-shapedness with respect to the origin. On an open Farey cell it is strictly less than one by proposition II.4.4.

**Definition II.4.6** (Polygonal complexity and radial criticality)**.** For a real-linear map $A : \mathbb{R}^2 \to \mathbb{R}^2$, let

$$\nu_{\text{poly}}(A) = \min\Big\{|\operatorname{Ext} P| : P \text{ is a nondegenerate compact convex polygon and } AP \subseteq P\Big\},$$

with value $\infty$ if no such polygon exists. An elliptic contraction $T$ is $N$-*critical* when

$$\nu_{\text{poly}}(T) = N, \qquad \nu_{\text{poly}}(tT) > N \quad (t > 1).$$

Here elliptic means $(\operatorname{tr} T)^2 < 4 \det T$, and contraction means that the spectral radius lies in $(0, 1)$.

For $N \geq 4$ and $0 < \theta < \pi$, a *non-inherited radial maximum* is a point satisfying

$$\lambda = R_N(\theta) e^{i\theta} \in \Theta_N \setminus \Theta_{N-1}, \qquad 0 < |\lambda| < 1. \tag{II.4.3}$$

**Proposition II.4.7** (Non-inherited radial maxima are polygonally critical)**.** *If* (II.4.3) *holds and $T_\lambda : \mathbb{C} \to \mathbb{C}$ is multiplication by $\lambda$, regarded as a real-linear map, then $T_\lambda$ is an $N$-critical elliptic contraction.*

*Proof.* In the real basis $(1, i)$,

$$[T_\lambda] = \begin{pmatrix} \operatorname{Re}\lambda & -\operatorname{Im}\lambda \\ \operatorname{Im}\lambda & \operatorname{Re}\lambda \end{pmatrix}.$$

Since $0 < \theta < \pi$, the multiplier is nonreal, and therefore

$$(\operatorname{tr} T_\lambda)^2 - 4 \det T_\lambda = -4(\operatorname{Im}\lambda)^2 < 0.$$

Its spectral radius is $|\lambda| \in (0, 1)$, so it is an elliptic contraction.

By theorem II.4.2, membership $\lambda \in \Theta_N$ gives a non-singleton invariant polygon with at most $N$ vertices. By lemma II.4.5, every such polygon is nondegenerate, so $\nu_{\text{poly}}(T_\lambda) \leq N$. If this complexity were at most $N-1$, the same invariant-polygon criterion would give $\lambda \in \Theta_{N-1}$, contrary to (II.4.3). Consequently

$$\nu_{\text{poly}}(T_\lambda) = N.$$

Finally, radial maximality in the definition of $R_N(\theta)$ gives $t\lambda \notin \Theta_N$ for every $t > 1$. If $\nu_{\text{poly}}(tT_\lambda) \leq N$, a nondegenerate polygon witnessing that inequality would put $t\lambda$ in $\Theta_N$ by theorem II.4.2, a contradiction. Hence $\nu_{\text{poly}}(tT_\lambda) > N$ for all $t > 1$, which is precisely $N$-criticality. $\square$

## II.5 Critical-polygon input from Part I

For a nonzero complex number $z$ that is not positive real, let

$$\arg_+(z) \in (0, 2\pi)$$

denote its positive lifted argument. The following is the complete interface with Part I used below.

**Theorem II.5.1** (Part I critical-polygon monodromy)**.** *Let $N \geq 4$ and suppose (II.4.3) holds. Retain $\lambda = |\lambda| e^{i\theta}$ and $x = \theta/(2\pi)$ for the original upper-half-plane multiplier. There are a selected multiplier*

$$\mu \in \{\lambda, \overline{\lambda}\}, \qquad \vartheta = \arg_+(\mu), \qquad y = \frac{\vartheta}{2\pi},$$

*and consecutive reduced fractions in $F_N$,*

$$\frac{p}{q} < y < \frac{r}{s}, \qquad rq - ps = 1, \qquad q \leq s.$$

*Thus $y = x$ when $\mu = \lambda$ and $y = 1 - x$ when $\mu = \overline{\lambda}$. Put*

$$d = \lfloor N/q \rfloor, \qquad e = s - dq.$$

*There are parameters*

$$0 \leq \beta_j < 1, \qquad \alpha_j = 1 - \beta_j > 0 \qquad (1 \leq j \leq d) \tag{II.5.1}$$

*such that*

$$\mu^s \prod_{j=1}^{d} (\mu^q - \beta_j) = \mu^{dq} \prod_{j=1}^{d} \alpha_j. \tag{II.5.2}$$

*Equivalently, since $\mu \neq 0$,*

$$\mu^e \prod_{j=1}^{d} (\mu^q - \beta_j) = \prod_{j=1}^{d} \alpha_j. \tag{II.5.3}$$

*Any zero factor introduced only to complete a closed return-recurrence chain is zero-factor padding and is not claimed to be an additional relative-interior contact.*

*Set*

$$A = q\vartheta - 2\pi p, \qquad M = \arg_+(\mu^q - 1).$$

*For the unique arguments $u_j \in (0, \pi)$ determined by*

$$e^{iu_j} = \frac{\mu^q - \beta_j}{|\mu^q - \beta_j|},$$

*one has*

$$0 < A < M < \pi, \qquad u_j \in [A, M) \quad (1 \leq j \leq d), \tag{II.5.4}$$

*and the exact lifted phase identity*

$$e\vartheta + \sum_{j=1}^{d} u_j = 2\pi(r - dp). \tag{II.5.5}$$

*Proof.* By proposition II.4.7, $T_\lambda$ is an $N$-critical elliptic contraction. Apply the complex monodromy and Farey-product-data theorem of Part I, theorem 1.4, including its common continuous argument interval. The determinant identity for the consecutive Farey endpoints follows from lemma II.2.1. The two adapted complex orientations give precisely the choice between $\lambda$ and $\overline{\lambda}$. $\square$

**Lemma II.5.2** (Reflection of the selected Farey data)**.** *In theorem II.5.1, suppose $\mu = \overline{\lambda}$, so that $y = 1 - x$. Then*

$$\frac{s-r}{s} < x < \frac{q-p}{q}$$

*is the reflected order-N Farey cell. With the denominator-based labeling used in (II.2.6),*

$$2\pi\,|qx - (q-p)| = 2\pi(qy - p) = A,$$
$$\frac{2\pi}{d}\,|sx - (s-r)| = \frac{2\pi}{d}(r - sy) = \frac{2\pi r - s\vartheta}{d}.$$

*Thus reflection preserves $q, s, d, e$, the modulus $|\lambda| = |\mu|$, and the absolute scalar equation (II.2.8). Conjugating the constant-parameter Ito identity for $\mu$ gives the Ito identity (II.2.4) for $\lambda$ in this reflected cell.*

*Proof.* The map $t \mapsto 1-t$ reverses the endpoint order and preserves denominators, giving the displayed cell. Since $x = 1 - y$,

$$qx - (q-p) = p - qy < 0, \qquad sx - (s-r) = r - sy > 0,$$

which proves the two identities. All remaining assertions follow directly from the definitions and complex conjugation. □

## II.6 Log-sine equalization and the sharp radial inequality

The Part I monodromy input supplies two scalar constraints. Taking moduli of the product turns the factors into a sum of one potential $F(u_j)$, while the lifted phase identity fixes their arithmetic mean at $A+B$. On the common continuous argument interval proved above, $F'' > 0$, so Jensen's inequality is strict unless all factor arguments, and hence all contact parameters, agree. At a non-inherited radial maximum the sparse stochastic realization supplies the matching equality point; the upper inequality is thus sharp rather than merely qualitative.

We work in the oriented cell

$$\frac{p}{q} < x < \frac{r}{s}, \qquad rq - ps = 1, \qquad q \le s. \tag{II.6.1}$$

Put $\theta = 2\pi x$ and

$$A = q\theta - 2\pi p, \qquad B = \frac{2\pi r - s\theta}{d}. \tag{II.6.2}$$

These are the non-absolute versions of (II.2.6). The same determinant-one computation as in (II.2.7) gives

$$A > 0, \qquad B > 0, \qquad A + B < \pi. \tag{II.6.3}$$

### II.6.1 The factor potential

Fix $\lambda = \rho e^{i\theta}$ with $0 < \rho < 1$ and choose

$$M = \operatorname{Arg}(\lambda^q - 1)$$

so that $A < M < \pi$. Since $\lambda^q = \rho^q e^{iA}$ lies in the open upper half-plane, the horizontal segment $\lambda^q - [0,1)$ never meets the origin. Its argument is a strictly increasing continuous function of $\beta$ with range $[A, M)$. For $0 \le \beta < 1$, let

$$u = \operatorname{Arg}(\lambda^q - \beta) \in [A, M)$$

be this uniquely determined branch. The triangle with vertices $\beta$, 1, and $\lambda^q$ gives

$$F(u) := \log \frac{|\lambda^q - \beta|}{1-\beta} = \log \sin M - \log \sin(M-u), \tag{II.6.4}$$

and hence

$$F''(u) = \csc^2(M-u) > 0. \tag{II.6.5}$$

At $\beta = 0$, the same triangle gives

$$\rho^q = \frac{\sin M}{\sin(M-A)}. \tag{II.6.6}$$

**Theorem II.6.1** (Sharp inequality for varying parameters)**.** *Suppose $\lambda$ and parameters* (II.5.1) *satisfy* (II.5.3) *in the cell* (II.6.1)*. For each factor let $u_j \in [A, M)$ be the branch just defined, and suppose these arguments satisfy* (II.5.5)*. Then*

$$\rho^{s/d} \sin A + \rho^q \sin B \leq \sin(A+B). \tag{II.6.7}$$

*Equality holds if and only if*

$$\beta_1 = \cdots = \beta_d. \tag{II.6.8}$$

*Proof.* The phase identity gives

$$\sum_{j=1}^{d} u_j = 2\pi(r - dp) - (s - dq)\theta = d(A+B).$$

Taking moduli in (II.5.3) gives

$$\sum_{j=1}^{d} F(u_j) = (dq - s) \log \rho.$$

Because every $u_j$ belongs to the interval $[A, M)$, their average $A + B$ belongs to the same interval and $F$ is strictly convex on the entire convex hull of the arguments. Strict Jensen convexity gives

$$dF(A+B) \leq (dq - s) \log \rho. \tag{II.6.9}$$

Equality holds exactly when all $u_j$ coincide. The argument map $\beta \mapsto \operatorname{Arg}(\lambda^q - \beta)$ is strictly increasing, so this is equivalent to all $\beta_j$ coinciding.

Exponentiating (II.6.9) and using (II.6.6) yields

$$\frac{\sin(M-A-B)}{\sin M} \geq \rho^{s/d-q}.$$

Using (II.6.6) and the identity

$$\sin(M-A)\sin(A+B) - \sin M \sin B = \sin A \sin(M-A-B)$$

gives

$$\frac{\sin(M-A-B)}{\sin M} = \frac{\rho^{-q}\sin(A+B) - \sin B}{\sin A}. \tag{II.6.10}$$

Multiplication by $\rho^q \sin A$ gives (II.6.7). □

**Corollary II.6.2** (Constant parameters at an outer radial maximum)**.** *For a non-inherited radial maximum in an open Farey cell, the parameters in the monodromy output selected by theorem II.5.1 satisfy*

$$\beta_1 = \cdots = \beta_d = \beta, \qquad \alpha = 1 - \beta,$$

*and the original point lies on* (II.2.4)*. Its modulus is the unique solution of* (II.2.8)*.*

*Proof.* Let $\mu$, $\vartheta$, $y$, and the Farey data be supplied by theorem II.5.1, and put

$$B_\mu = \frac{2\pi r - s\vartheta}{d}.$$

The same theorem supplies exactly the product, phase, and common-argument-interval hypotheses of theorem II.6.1. Applying that theorem to $\mu$ gives

$$\rho^{s/d} \sin A + \rho^q \sin B_\mu \le \sin(A + B_\mu), \qquad \rho = |\mu|.$$

If $\mu = \lambda$, the oriented quantities are the absolute quantities of the original cell. If $\mu = \overline{\lambda}$, lemma II.5.2 identifies them with those absolute quantities. Hence in both cases, with $\rho = |\lambda|$,

$$\rho^{s/d} \sin A + \rho^q \sin B \le \sin(A + B)$$

for the original ray.

Let $\rho_*$ be the unique equality radius supplied by proposition II.2.3. By theorem II.7.3, it is attained in order $N$, so $\rho_* \le \rho = R_N(\theta)$. Strict monotonicity of the scalar left side and the preceding inequality give the reverse inequality. Therefore $\rho = \rho_*$ and equality holds in theorem II.6.1. Its equality statement gives a constant parameter list for $\mu$. For $\mu = \lambda$ this is (II.2.4) directly; for $\mu = \overline{\lambda}$ it becomes the same Ito equation for the original cell by lemma II.5.2. ☐

## II.7 Explicit stochastic realization and attainment

We now prove the inner inclusion directly, without importing an Ito realization theorem.

Throughout this section we use the tail–row convention: an arrow $u \to v$ of weight $w(u, v)$ is encoded by

$$A_{uv} = w(u, v), \tag{II.7.1}$$

with $A_{uv} = 0$ when the arrow is absent. Thus the sum of the weights of arrows leaving $u$ is the row sum of $A$ at $u$.

**Lemma II.7.1** (Cycle-cover coefficient rule)**.** *Let $A$ be the weighted adjacency matrix of a finite directed graph. In $\det(tI - A)$, a collection of $k$ pairwise vertex-disjoint directed cycles contributes*

$$(-1)^k t^{N-L}$$

*times the product of their edge weights, where $L$ is the total number of vertices on the cycles. Vertices not on a selected cycle contribute diagonal factors $t$.*

*Proof.* With the convention (II.7.1), a directed cycle $u_1 \to u_2 \to \cdots \to u_\ell \to u_1$ is selected in the Leibniz expansion by the permutation cycle $u_i \mapsto u_{i+1}$ and contributes the edge product $A_{u_1u_2} \cdots A_{u_\ell u_1}$. Its permutation sign is $(-1)^{\ell-1}$, while selecting the $-A$ entry at its $\ell$ vertices contributes $(-1)^\ell$, so the net sign is $-1$. This also covers $\ell = 1$: one selects $-A_{uu}$ from the diagonal factor $t - A_{uu}$. Disjoint selected cycles multiply, and at every remaining fixed vertex one selects $t$. Hence a collection of $k$ cycles has the stated sign and power of $t$. ☐

**Lemma II.7.2** (Cycle collections in the sparse realization graph)**.** *Consider the sparse realization graph used in theorem II.7.3: each block is a deterministic directed path to its terminal vertex, each terminal has one local return edge to the start of the same block and one cross edge to the next block, and at most one cross edge is subdivided by deterministic intermediate vertices. Then every directed simple cycle is either one local $q$-cycle inside a single block or the unique global cycle using all $d$ cross edges. Moreover every local cycle shares its terminal vertex with the global cycle. Consequently the vertex-disjoint directed cycle collections are exactly either*

(i) *an arbitrary subset of the $d$ local cycles, or*
(ii) *the singleton global cycle.*

*Proof.* A directed cycle that uses no cross edge is trapped in one block. The only way to leave the terminal and remain in that block is the local return, after which the deterministic path forces the whole local $q$-cycle.

Now let a simple cycle use a cross edge into some block. Once it enters that block, all following edges are forced along the deterministic suffix until the terminal is reached. If the cycle then takes the local return, the deterministic path reaches the previously visited entry vertex before the original cycle has closed; hence a simple cycle using one cross edge cannot use any local return. It must therefore take the cross edge at every terminal it reaches. Since the cross edges advance cyclically through the $d$ blocks, this produces exactly the unique global cross cycle. The subdivision vertices, when present, lie on that global route and introduce no choice.

The global cycle passes through the terminal vertex of every block, and the local cycle in that block also passes through that terminal. Thus no local cycle is vertex-disjoint from the global cycle, while distinct local cycles live in disjoint blocks. This proves the stated classification of cycle collections. $\square$

**Theorem II.7.3** (Sparse stochastic realization of the reduced Ito polynomial)**.** *Let $p/q, r/s$ be a Farey pair of order $n$, labeled by $q \le s$, and let $d = \lfloor n/q \rfloor$. For every $\alpha \in [0,1]$, $\beta = 1-\alpha$, there is a row-stochastic matrix of order*

$$N_0 = \max\{dq, s\} \le n \tag{II.7.2}$$

*whose spectrum contains every root of the reduced Ito polynomial* (II.2.5)*, and hence every nonzero root of* (II.2.4).

*Proof.* Take $d$ blocks

$$B_j = \{v_{j,0}, \ldots, v_{j,q-1}\}.$$

Inside each block put deterministic edges of weight 1, $v_{j,k} \to v_{j,k+1}$ for $k < q-1$. At the terminal vertex $v_{j,q-1}$ put a local return $v_{j,q-1} \to v_{j,0}$ of weight $\beta$ and a cross-block edge of weight $\alpha$ entering block $j+1$, with block indices read cyclically modulo $d$.

Suppose first $s \le dq$. Farey consecutiveness gives $q + s > n \ge dq$, hence $s > (d-1)q$. Put

$$\ell_1 = s - (d-1)q, \qquad \ell_2 = \cdots = \ell_d = q.$$

The cross edge entering block $j$ is, by convention, the edge whose tail is the terminal of block $j-1$. Route it to $v_{j,q-\ell_j}$. The local cycles have length $q$ and weight $\beta$. The unique global cross cycle uses all $d$ cross edges, has weight $\alpha^d$, and visits exactly $\ell_j$ vertices in block $j$, hence has length $s$. By lemma II.7.2, the only cycle-cover collections are arbitrary local-cycle subsets or the singleton global cycle. Therefore the local subsets contribute

$$\sum_{J \subseteq \{1,\ldots,d\}} (-1)^{|J|} \beta^{|J|} t^{dq-q|J|} = (t^q - \beta)^d,$$

and the global cycle contributes $-\alpha^d t^{dq-s}$. By lemma II.7.1,

$$\det(tI - A) = (t^q - \beta)^d - \alpha^d t^{dq-s}. \tag{II.7.3}$$

If $s > dq$, route every cross edge into the first vertex of the next block. Set $K = s - dq > 0$. On one such edge $u \to v$, introduce exactly the $K$ new vertices $w_1, \ldots, w_K$ and replace that edge by the path

$$u \xrightarrow{\alpha} w_1 \xrightarrow{1} \cdots \xrightarrow{1} w_K \xrightarrow{1} v.$$

The global cycle has length $s$ and weight $\alpha^d$. The local cycles remain the $d$ disjoint $q$-cycles of weight $\beta$, and every local-cycle collection leaves all intermediate vertices unused. Using lemmas II.7.1 and II.7.2, the local subsets contribute

$$t^{s-dq} \sum_{J \subseteq \{1,\dots,d\}} (-1)^{|J|} \beta^{|J|} t^{dq-q|J|} = t^{s-dq}(t^q - \beta)^d,$$

while the singleton global cycle contributes $-\alpha^d$. Therefore

$$\det(tI - A) = t^{s-dq}(t^q - \beta)^d - \alpha^d. \tag{II.7.4}$$

By the tail–row convention, every nonterminal block vertex and every subdivision vertex has one outgoing edge of weight 1, while every block terminal has outgoing weight $\alpha + \beta = 1$. Hence $A$ is row-stochastic. Equations (II.7.3) and (II.7.4) are exactly the two reduced forms in (II.2.5). Finally, pad by absorbing states to reach order $n$. □

**Corollary II.7.4** (Attainment of the scalar boundary)**.** *For every open Farey ray $x$, the point $\rho^{(n)}_{f,g}(x)e^{2\pi i x}$ belongs to $\Theta_n$.*

*Proof.* Use (II.2.9) in theorem II.7.3. □

## II.8 Farey refinement and nesting in the order

The Part I monodromy input applies only to non-inherited radial maxima. To close the induction, we prove directly that the candidate radial regions are nested as $n$ increases.

### II.8.1 A log-line lemma

Let $L$ be a line not through the origin. On any interval of angles for which $L$ has a positive radial intersection, write the logarithm of that radius as $\ell(\phi)$. For suitable constants $c, \phi_0$,

$$\ell(\phi) = c - \log\cos(\phi - \phi_0), \qquad \ell''(\phi) = \sec^2(\phi - \phi_0) > 0. \tag{II.8.1}$$

We shall use the following signed form of the chord calculation. Identify complex numbers with vectors in $\mathbb{R}^2$. For

$$X = Pe^{iA}, \qquad Y = Qe^{-iC}, \qquad P, Q > 0, \qquad A, C > 0, \qquad A + C < \pi,$$

define

$$\Delta_{X,Y}(r) := \det(Y - X, r - X) = r(P\sin A + Q\sin C) - PQ\sin(A + C), \qquad r > 0. \tag{II.8.2}$$

Thus the positive-real intercept of the line through $X$ and $Y$ is

$$I(P, Q) = \frac{PQ\sin(A + C)}{P\sin A + Q\sin C}, \tag{II.8.3}$$

and

$$I(P, Q) > 1 \iff \Delta_{X,Y}(1) < 0. \tag{II.8.4}$$

The endpoint monotonicities also have fixed signs:

$$\frac{\partial I}{\partial P} = \frac{Q^2\sin(A + C)\sin C}{(P\sin A + Q\sin C)^2} > 0, \qquad \frac{\partial I}{\partial Q} = \frac{P^2\sin(A + C)\sin A}{(P\sin A + Q\sin C)^2} > 0.$$

All chord comparisons below will be reduced to (II.8.2); no diagrammatic side convention is used.

**Lemma II.8.1** (Mediant chord expansion in both denominator orientations)**.** *Let $n \ge 4$, let $a/b < c/d$ be consecutive in $F_{n-1}$, and suppose the mediant*

$$\frac{a+c}{b+d}$$

*is newly inserted in $F_n$, so $n = b+d$. Fix an interior ray of one of the two new subcells and let $R > 1$ be the reciprocal of the old order-$(n-1)$ candidate radius on that ray. At this same $R$, the positive-real intercept of the order-$n$ reciprocal chord is strictly larger than $1$, the intercept of the old reciprocal chord. At the common Farey endpoints the two candidate outer radii are both $1$ by definition. This holds whether $b < d$ or $d < b$.*

*Proof.* We first treat $b < d$. Put

$$m = \left\lfloor \frac{n-1}{b} \right\rfloor, \qquad U = R^b e^{iA}, \qquad V = R^{d/m} e^{-iB},$$

where

$$A = 2\pi(bx - a), \qquad B = \frac{2\pi}{m}(c - dx).$$

At the old candidate radius, (II.2.10) gives

$$1 = \beta U + \alpha V, \qquad \alpha, \beta \in (0,1).$$

Consequently $1 \in \operatorname{relint}[U, V]$. Let $L$ be the line through $U, V$, and let $\ell$ be its log-radial function on the angular interval $[-B, A]$. Then

$$\ell(0) = 0, \qquad \ell(A) = b \log R, \qquad \ell(-B) = \frac{d}{m} \log R,$$

and (II.8.1) says that $\ell'$ is strictly increasing.

Consider first the left subcell

$$\frac{a}{b} < x < \frac{a+c}{b+d},$$

equivalently $B > A/m$. Suppose initially that $b > 1$. Since $\gcd(b,d) = 1$, one has $b \nmid d$, and hence

$$\left\lfloor \frac{n}{b} \right\rfloor = \left\lfloor \frac{n-1}{b} \right\rfloor = m.$$

The new reciprocal chord has endpoints

$$U, \qquad W = V U^{1/m}, \qquad \arg W = -B + \frac{A}{m}.$$

Put $\phi = -B + A/m \in (-B, 0)$. Strict monotonicity of $\ell'$ gives

$$\begin{aligned} \ell(\phi) - \ell(-B) &= \int_0^{A/m} \ell'(-B+t)\, dt \\ &< \int_0^{A/m} \ell'(mt)\, dt = \frac{1}{m}\ell(A). \end{aligned} \tag{II.8.5}$$

Therefore

$$\log |W| = \ell(-B) + \frac{1}{m}\ell(A) > \ell(\phi).$$

Let

$$V_* = \exp(\ell(\phi)) e^{i\phi} \in [\,V, 1\,], \qquad W = \gamma V_*, \qquad \gamma > 1.$$

Since $U, V_*, 1$ are collinear,

$$\det(V_* - U, 1 - U) = 0.$$

The signed determinant of the new chord at 1 is therefore

$$\begin{aligned}\Delta_{U,W}(1) &= \det(W-U, 1-U)\\ &= (\gamma-1)\det(U,1)\\ &= -(\gamma-1)R^b \sin A < 0.\end{aligned} \tag{II.8.6}$$

By (II.8.4), its positive-real intercept is strictly larger than 1.

The right subcell satisfies

$$\frac{a+c}{b+d} < x < \frac{c}{d}, \qquad \text{equivalently} \qquad A > mB.$$

This calculation is valid also when $b = 1$. Since $b < d$, the new multiplicity at the endpoint of denominator $d$ is one. With the positive-angle endpoint written first, the new rooted endpoints are

$$X = UV^m, \qquad Y = V^m;$$

indeed their arguments are $A - mB > 0$ and $-mB < 0$. The new line is $V^m L$, because $1, U \in L$. Moreover

$$\ell(mB) + m\ell(-B) = m\int_0^B (\ell'(mt) - \ell'(-t))\,dt > 0. \tag{II.8.7}$$

Put

$$Z = \exp(\ell(mB))e^{imB} \in [1, U], \qquad \eta = V^{-m}.$$

Equation (II.8.7) says that $\eta = \gamma Z$ for some $0 < \gamma < 1$. Multiplication by $V^m$ multiplies real determinants by the positive number $|V|^{2m}$, so

$$\begin{aligned}\Delta_{X,Y}(1) &= \det(V^m(1-U), 1-V^mU)\\ &= |V|^{2m}\det(1-U, \eta-U)\\ &= |V|^{2m}(\gamma-1)\det(1-U, U)\\ &= |V|^{2m}(\gamma-1)R^b\sin A < 0.\end{aligned} \tag{II.8.8}$$

Here the third equality uses the collinearity of $1, U, Z$. Again (II.8.4) gives an intercept strictly larger than 1.

It remains to handle the left subcell when $b = 1$. Farey consecutiveness then gives the old cell

$$\frac{0}{1} < \frac{1}{d},$$

and the new left cell has right endpoint $1/(d+1)$. Set

$$\phi_{\text{old}} = \frac{2\pi}{d}, \qquad \phi_{\text{new}} = \frac{2\pi}{d+1}.$$

For $0 < \theta < \phi_{\text{new}}$, the radial intersection of the chord from 1 to $e^{i\phi}$ is

$$r_\phi(\theta) = \frac{\sin\phi}{\sin\theta + \sin(\phi-\theta)} = \frac{\cos(\phi/2)}{\cos(\theta-\phi/2)}.$$

Its logarithmic derivative has the explicit sign

$$\begin{aligned}\frac{\partial}{\partial\phi}\log r_\phi(\theta) &= -\frac{1}{2}\left(\tan\frac{\phi}{2} + \tan\left(\theta - \frac{\phi}{2}\right)\right)\\ &= -\frac{\sin\theta}{2\cos(\phi/2)\cos(\theta-\phi/2)} < 0.\end{aligned} \tag{II.8.9}$$

Hence

$$r_{\phi_{\text{new}}}(\theta) > r_{\phi_{\text{old}}}(\theta).$$

At $R = r_{\phi_{\text{old}}}(\theta)^{-1}$, the intercept of the new reciprocal chord is, by (II.8.3),

$$I_{\text{new}} = \frac{R \sin \phi_{\text{new}}}{\sin \theta + \sin(\phi_{\text{new}} - \theta)} = \frac{r_{\phi_{\text{new}}}(\theta)}{r_{\phi_{\text{old}}}(\theta)} > 1.$$

Finally suppose $d < b$. Complex conjugation sends $[a/b, c/d]$ to the oppositely oriented interval $[-c/d, -a/b]$. Translating both endpoint fractions by the same integer does not change their roots, reciprocal radii, scalar equations, or positive-real chord intercepts. The reflected interval has smaller denominator $d$ at its left endpoint, so the case already proved applies and conjugation gives the two required inequalities in the original interval. Equality $b = d$ would force $b = d = 1$ from $bc - ad = 1$, and then $b + d = 2$, contrary to $n \geq 4$.

At every inherited Farey endpoint the two candidate radii are both 1 by definition. $\square$

**Lemma II.8.2** (Multiplicity padding)**.** *Suppose a Farey cell is unchanged from order $n-1$ to order $n$, but*

$$\left\lfloor \frac{n}{q} \right\rfloor = \left\lfloor \frac{n-1}{q} \right\rfloor + 1 = M.$$

*Let $\lambda = \rho e^{2\pi i x}$ be the order-$(n-1)$ constant-parameter candidate on an open ray of this cell. Orient the cell—conjugating $\lambda$ and reflecting its Farey endpoints when necessary—to obtain $\mu = \rho e^{i\vartheta}$ and $y = \vartheta/(2\pi)$ such that*

$$\frac{p}{q} < y < \frac{r}{s}, \qquad rq - ps = 1, \qquad q \leq s. \tag{II.8.10}$$

*Then the order-$(n-1)$ constant-parameter candidate is a varying-parameter order-$n$ product obtained by adjoining one factor with $\beta_M = 0$, $\alpha_M = 1$. Its factors lie in the common continuous argument interval and satisfy the order-$n$ phase identity. Consequently its radius does not exceed the order-$n$ constant-parameter radius.*

*Proof.* Put $m = M - 1$ and define

$$A = q\vartheta - 2\pi p, \qquad B_m = \frac{2\pi r - s\vartheta}{m}. \tag{II.8.11}$$

If the smaller-denominator endpoint is already on the left, take $\mu = \lambda$ and $y = \vartheta/(2\pi)$. Otherwise replace $\lambda$ by $\overline{\lambda}$ and the cell by its reflected interval; denominators, moduli, and the scalar candidate equation are unchanged. Thus (II.8.10) holds, and the determinant-one calculation used in (II.6.3) gives

$$A > 0, \qquad B_m > 0, \qquad A + B_m < \pi.$$

The order-$(n-1)$ candidate has the constant parameter list $\beta_1 = \cdots = \beta_m = \beta$ and $\alpha_1 = \cdots = \alpha_m = \alpha = 1 - \beta$, where the scalar formulas at multiplicity $m$ are

$$\beta = \frac{\rho^q \sin B_m}{\sin(A + B_m)}, \qquad \alpha = \frac{\rho^{s/m} \sin A}{\sin(A + B_m)}. \tag{II.8.12}$$

Since $\mu^q = \rho^q e^{iA}$, direct resolution into real and imaginary parts gives

$$\mu^q - \beta = \frac{\rho^q \sin A}{\sin(A + B_m)} e^{i(A + B_m)}. \tag{II.8.13}$$

Hence every old factor has its unique argument in the common interval

$$u_1 = \cdots = u_m = A + B_m.$$

In particular, the old phase relation is not an additional assumption:

$$\begin{aligned}(s - mq)\vartheta + \sum_{j=1}^{m} u_j &= (s - mq)\vartheta + m(A + B_m) \\ &= 2\pi(r - mp).\end{aligned} \tag{II.8.14}$$

The old constant-parameter Ito equation is

$$\mu^{s-mq}(\mu^q - \beta)^m = \alpha^m. \tag{II.8.15}$$

Now set

$$\beta_M = 0, \qquad \alpha_M = 1, \qquad u_M = A.$$

Because $M = m + 1$, multiplying (II.8.15) by $\mu^{-q}(\mu^q - 0) = 1$ gives the order-$n$ varying-parameter product

$$\mu^{s-Mq}(\mu^q - \beta)^m(\mu^q - 0) = \alpha^m. \tag{II.8.16}$$

Likewise, (II.8.14) and $A = q\vartheta - 2\pi p$ give

$$\begin{aligned}(s - Mq)\vartheta + \sum_{j=1}^{M} u_j &= 2\pi(r - mp) - q\vartheta + A \\ &= 2\pi(r - Mp).\end{aligned} \tag{II.8.17}$$

Finally, $0 < \beta < 1$ on an open ray. The argument map $t \mapsto \operatorname{Arg}(\mu^q - t)$ is strictly increasing on $[0, 1)$, so with $M_* = \operatorname{Arg}(\mu^q - 1)$ one has

$$u_1 = \cdots = u_m = A + B_m \in (A, M_*), \qquad u_M = A.$$

Thus all factors in (II.8.16) lie in the interval $[A, M_*)$ required by theorem II.6.1, and (II.8.17) is exactly its phase hypothesis at multiplicity $M$. Put

$$B_M = \frac{2\pi r - s\vartheta}{M} = \frac{m}{M} B_m.$$

Applying theorem II.6.1 to (II.8.16)–(II.8.17) gives the explicit order-$n$ scalar sign

$$\rho^{s/M} \sin A + \rho^q \sin B_M - \sin(A + B_M) < 0. \tag{II.8.18}$$

The inequality is strict: the $M$ factor parameters consist of $m$ copies of $\beta \in (0, 1)$ and one copy of 0, so they are not all equal, and the equality condition in theorem II.6.1 cannot hold. The left side of (II.8.18) is strictly increasing as a function of $\rho > 0$ and vanishes at the order-$n$ constant-parameter radius. Thus the old radius is strictly smaller than the new one on every open ray. Conjugating back preserves the modulus. □

For $n \geq 3$, define the candidate outer radius $K_n$ as follows. For $x = \theta/(2\pi)$ in an open Farey cell and $0 < \theta < \pi$, use the unique scalar radius of proposition II.2.3; at a Farey endpoint in $[0, \pi)$ set $K_n = 1$; and set

$$K_n(\pi) = 1. \tag{II.8.19}$$

For $n \geq 4$ this is the continuous closed-cell scalar graph. For $n = 3$ it has the necessary jump $\lim_{\theta \uparrow \pi} K_3(\theta) = 1/2$ but $K_3(\pi) = 1$, representing the outer endpoint $-1$ of the terminal radial segment.

**Lemma II.8.3** (Exhaustive order-refinement split)**.** *Let $n \geq 4$ and let $\theta/(2\pi) \in [0, 1/2]$. Comparing $F^+_{n-1}$ with $F^+_n$, exactly one of the following alternatives applies.*

(i) *The ray is an inherited Farey endpoint; both candidate outer radii are defined to be* 1*.*

(ii) *The ray is a newly inserted endpoint of denominator $n$; the order $n$ candidate outer radius is* 1*, while the order $n-1$ ray is interior to one old cell.*

(iii) *The ray is interior to an old cell which is not split at this order. Then the endpoint pair is unchanged, and for the smaller denominator $q$ the multiplicity changes from $\lfloor (n-1)/q \rfloor$ to $\lfloor n/q \rfloor$, hence either stays the same or increases by exactly one.*

(iv) *The ray is interior to one of the two cells obtained by inserting the mediant $(a+c)/(b+d)$ between consecutive old endpoints $a/b < c/d$; in this case $b + d = n$.*

*There are no further cases.*

*Proof.* The only fractions which can appear in $F_n$ but not in $F_{n-1}$ have reduced denominator exactly $n$. If such a fraction $h/n$ is inserted between consecutive old endpoints $a/b < c/d$, the Farey adjacency criterion lemma II.2.1 gives $bc - ad = 1$ and $b + d \geq n$. Since $h/n$ lies between them, the decomposition argument in the proof of lemma II.2.1 writes

$$(n, h) = m(b, a) + \ell(d, c)$$

with positive integers $m, \ell$. Thus $n = mb + \ell d \geq b + d$. Hence $b + d = n$, $m = \ell = 1$, and $h/n = (a + c)/(b + d)$. This proves that every new interior split is exactly the mediant case and that no other old cell is split.

If the ray itself is a new fraction, it is case (ii). If it is an old endpoint, it is case (i). Otherwise it is interior to an old cell. When that old cell is split, case (iv) applies; when it is not split, the endpoint pair is unchanged. Increasing the order from $n - 1$ to $n$ can change $\lfloor n/q \rfloor$ by at most one, so the unchanged-cell alternatives in case (iii) are exhaustive. The upper-half restriction merely discards the reflected lower-half copies and does not introduce another case. $\square$

**Theorem II.8.4** (Candidate nesting)**.** *With this definition,*

$$K_{n-1}(\theta) \leq K_n(\theta) \tag{II.8.20}$$

*for every $0 \leq \theta \leq \pi$ and every $n \geq 4$; the lower-half-plane version follows by conjugation.*

*Proof.* Fix first an angle $\theta$ which is interior to a cell of both $F^+_{n-1}$ and $F^+_n$. Label the endpoints of its order-$n$ cell by $p/q, r/s$ with $q \leq s$, put

$$D = \left\lfloor \frac{n}{q} \right\rfloor,$$

and let $A, B$ be the positive quantities in (II.2.6). Define the order-$n$ scalar residual

$$\mathcal{F}_{n,\theta}(t) := t^{s/D} \sin A + t^q \sin B - \sin(A + B), \qquad t > 0. \tag{II.8.21}$$

Its derivative has the fixed sign

$$\mathcal{F}'_{n,\theta}(t) = \frac{s}{D} t^{s/D-1} \sin A + q t^{q-1} \sin B > 0. \tag{II.8.22}$$

By proposition II.2.3, $K_n(\theta)$ is the unique zero of $\mathcal{F}_{n,\theta}$.

Set

$$\rho_- = K_{n-1}(\theta), \qquad R = \rho_-^{-1}.$$

By lemma II.8.3, there are three interior-ray possibilities.

If the Farey cell and its multiplicity are unchanged, then its scalar equation is unchanged and

$$\mathcal{F}_{n,\theta}(\rho_-) = 0.$$

If the cell is unchanged but its multiplicity increases, then (II.8.18) gives

$$\mathcal{F}_{n,\theta}(\rho_-) < 0.$$

It remains to consider a cell created by inserting a new denominator-$n$ mediant. The two reciprocal-chord endpoints of the new scalar chord at $\rho_-$ are, after choosing their common orientation,

$$R^q e^{iA}, \qquad R^{s/D} e^{-iB}.$$

Its positive-real intercept is

$$I_R = \frac{R^{q+s/D} \sin(A+B)}{R^q \sin A + R^{s/D} \sin B}.$$

Direct subtraction, without a side convention, gives

$$I_R - 1 = \frac{R^{q+s/D}(-\mathcal{F}_{n,\theta}(\rho_-))}{R^q \sin A + R^{s/D} \sin B}. \tag{II.8.23}$$

By lemma II.8.1, $I_R > 1$; every other factor in (II.8.23) is positive. Hence

$$\mathcal{F}_{n,\theta}(\rho_-) < 0.$$

Thus in all three cases

$$\mathcal{F}_{n,\theta}(\rho_-) \le 0.$$

Together with (II.8.22) and $\mathcal{F}_{n,\theta}(K_n(\theta)) = 0$, this proves

$$K_{n-1}(\theta) = \rho_- \le K_n(\theta)$$

on every ray interior to both Farey decompositions. The inequality is strict whenever the cell is split or its multiplicity increases.

Now let $\theta/(2\pi)$ be a newly inserted fraction. It is interior to one order-$(n-1)$ cell, so

$$K_{n-1}(\theta) < 1$$

by proposition II.2.3, whereas $K_n(\theta) = 1$ by the endpoint definition. At every inherited Farey endpoint below $\pi$, both values are 1. At $\theta = 0$ the same is true, and at $\theta = \pi$ both sides are 1 by (II.8.19). The latter separate clause is essential for $n = 4$, because $\lim_{\theta \uparrow \pi} K_3(\theta) = 1/2$ although $K_3(\pi) = 1$. This proves (II.8.20) everywhere. Conjugation gives the lower-half-plane statement. □

## II.9 Completion of the proof

**Lemma II.9.1** (Compact star-shaped sets with continuous radial function)**.** *Let $S \subset \mathbb{C}$ be compact and star-shaped with respect to the origin, and suppose that*

$$r_S(\phi) := \max\{r \ge 0 : re^{i\phi} \in S\}$$

*is positive and continuous on the circle. Then*

$$S = \{t\, r_S(\phi) e^{i\phi} : 0 \le t \le 1,\ -\pi < \phi \le \pi\}, \qquad \partial S = \{r_S(\phi) e^{i\phi} : -\pi < \phi \le \pi\}. \tag{II.9.1}$$

*Proof.* Compactness makes each maximum attainable, and star-shapedness with respect to the origin then says that the intersection of $S$ with the ray of angle $\phi$ is exactly $\{re^{i\phi} : 0 \le r \le r_S(\phi)\}$. Taking the union over $\phi$ gives the first formula. A point $r_S(\phi)e^{i\phi}$ cannot be interior, since a sufficiently small outward radial displacement would contradict maximality.

Conversely, let $z = re^{i\phi}$ with $0 < r < r_S(\phi)$. Choose $\eta > 0$ with $r + 3\eta < r_S(\phi)$. By continuity, $r_S(\psi) > r + 2\eta$ for all $\psi$ in some neighborhood of $\phi$. A sufficiently small Euclidean ball about $z$ consists of points $w$ with $\arg w$ in that neighborhood and $|w| < r + \eta$; the ray description then puts the whole ball in $S$. Thus every nonzero point strictly below the radial graph is interior. Finally, positivity and compactness of the circle give $c := \min_\phi r_S(\phi) > 0$, so the disk $|z| < c$ lies in $S$ and 0 is interior. No further boundary points remain. □

### II.9.1 The base orders

**Proposition II.9.2** (Orders one, two, and three)**.** *One has*

$$\Theta_1 = \{1\}, \qquad \Theta_2 = [-1, 1],$$

*and, with* $\omega = e^{2\pi i/3}$*,*

$$\Theta_3 = \operatorname{conv}\{1, \omega, \overline{\omega}\} \cup [-1, -1/2]. \tag{II.9.2}$$

*Proof.* For order one the only matrix is [1]. For order two, every stochastic matrix has the form

$$\begin{pmatrix} a & 1-a \\ b & 1-b \end{pmatrix}, \qquad 0 \le a, b \le 1,$$

and its second eigenvalue is $a - b$; hence the first two assertions follow.

Let a $3 \times 3$ stochastic matrix have nonreal eigenvalues $\lambda = x + iy$ and $\overline{\lambda}$. Its spectrum is $1, \lambda, \overline{\lambda}$. Since the diagonal is nonnegative,

$$1 + 2x = \operatorname{tr} A \ge 0,$$

so $x \ge -1/2$. Also

$$(\operatorname{tr} A)^2 \le 3 \sum_i a_{ii}^2 \le 3 \operatorname{tr}(A^2),$$

because every term $a_{ij}a_{ji}$ in $\operatorname{tr}(A^2)$ is nonnegative. Substituting

$$\operatorname{tr} A = 1 + 2x, \qquad \operatorname{tr}(A^2) = 1 + 2(x^2 - y^2)$$

gives

$$3y^2 \le (1 - x)^2.$$

Thus the nonreal region lies in the triangle in (II.9.2); every real stochastic eigenvalue lies in $[-1, 1]$.

The chord $[1, \omega]$ is realized by $(1 - \alpha)I + \alpha C_3$, $0 \le \alpha \le 1$. The terminal sparse realization of theorem II.7.3 has polynomial

$$\lambda^3 - (1 - \alpha)\lambda - \alpha = (\lambda - 1)(\lambda^2 + \lambda + \alpha). \tag{II.9.3}$$

For $0 \le \alpha \le 1/4$, the root $(-1 - \sqrt{1 - 4\alpha})/2$ runs from $-1$ to $-1/2$. For $1/4 < \alpha \le 1$, the upper root is

$$\lambda = -\frac{1}{2} + \frac{i}{2}\sqrt{4\alpha - 1},$$

and runs from $-1/2$ to $\omega$. Conjugation supplies the lower edges, and star-shapedness with respect to the origin fills the triangle and the real segment. This proves (II.9.2).

It remains to identify the two nonreal boundary pieces with the canonical radius $K_3$ used by the induction. On the Farey cell $[0, 1/3]$, the denominator labeling of (II.2.2) is

$$(q, s) = (1, 3), \qquad d = \lfloor 3/1 \rfloor = 3.$$

For $0 < \theta = 2\pi x < 2\pi/3$, (II.2.6) gives

$$A = \theta, \qquad B = \frac{2\pi}{3} - \theta, \qquad A + B = \frac{2\pi}{3}.$$

Thus (II.2.8) is

$$\rho \sin\theta + \rho \sin\left(\frac{2\pi}{3} - \theta\right) = \sin\frac{2\pi}{3}. \tag{II.9.4}$$

With the coefficients from (II.2.9), this equation is exactly $\alpha + \beta = 1$, and direct real and imaginary parts give

$$\rho e^{i\theta} = \beta + \alpha\omega.$$

Indeed, the imaginary part is $\alpha \sin(2\pi/3) = \rho \sin\theta$, while the real part is

$$\frac{\rho \sin(2\pi/3 - \theta) + \rho \sin\theta \cos(2\pi/3)}{\sin(2\pi/3)} = \rho\cos\theta.$$

For $\lambda \neq 0$, the Ito equation (II.2.4) reduces here to $(\lambda - \beta)^3 = \alpha^3$, and its upper branch is precisely $\lambda = \beta + \alpha\omega$. Moreover

$$\frac{d}{d\alpha} \arg(1 + \alpha(\omega - 1)) = \frac{\Im\omega}{|1 + \alpha(\omega - 1)|^2} > 0.$$

Hence the chord meets every ray with angle in $(0, 2\pi/3)$ exactly once; the uniqueness in proposition II.2.3 identifies that modulus with $K_3(\theta)$.

On the terminal cell $[1/3, 1/2]$, denominator order gives

$$(q, s) = (2, 3), \qquad d = \lfloor 3/2 \rfloor = 1.$$

For $2\pi/3 < \theta = 2\pi x < \pi$ one has

$$A = 2(\pi - \theta), \qquad B = 3\theta - 2\pi, \qquad A + B = \theta,$$

and the scalar equation becomes

$$\rho^3 \sin(2(\pi - \theta)) + \rho^2 \sin(3\theta - 2\pi) = \sin\theta. \tag{II.9.5}$$

The nonzero Ito equation is now $\lambda(\lambda^2 - \beta) = \alpha$, equivalently the polynomial (II.9.3). From $\lambda^3 = \beta\lambda + \alpha$, with $\lambda = \rho e^{i\theta}$, imaginary and real parts give

$$\beta = \frac{\rho^2 \sin(3\theta - 2\pi)}{\sin\theta}, \qquad \alpha = \frac{\rho^3 \sin(2(\pi - \theta))}{\sin\theta}.$$

Consequently $\alpha + \beta = 1$ is exactly (II.9.5). Along the upper root $\lambda = -1/2 + iy$,

$$\frac{d}{dy} \arg\left(-\frac{1}{2} + iy\right) = \frac{-1/2}{1/4 + y^2} < 0.$$

As $y$ increases from 0 to $\sqrt{3}/2$, the argument therefore decreases strictly from $\pi$ to $2\pi/3$. This boundary piece meets every ray in $(2\pi/3, \pi)$ exactly once, and proposition II.2.3 again identifies its modulus with $K_3(\theta)$. The real range $0 \leq \alpha \leq 1/4$ supplies the additional segment $[-1, -1/2]$ on the endpoint ray; it is not an open-cell branch.

The trace bounds proved that no order-three point lies beyond these pieces, while the displayed matrices attain them. At the Farey endpoint rays $0, 2\pi/3, \pi$, the outer points are respectively $1, \omega, -1$, and the endpoint convention gives $K_3 = 1$. We have therefore proved the base identity required at order four:

$$R_3(\theta) = K_3(\theta) \qquad (0 \le \theta \le \pi), \tag{II.9.6}$$

where at 0 and $\pi$ the notation $R_3$ denotes the attained outer radial modulus. The jump from the terminal nonreal arc to $-1$ is exactly the exceptional order-three behavior recorded in proposition II.2.4. □

### II.9.2 Induction on the order

*Proof of theorem II.3.1.* The orders $1, 2, 3$ are proposition II.9.2. Assume now that $n \ge 4$ and that the theorem holds at order $n - 1$.

First note the natural nesting

$$\Theta_{n-1} \subseteq \Theta_n : \tag{II.9.7}$$

if $A$ is an $(n-1) \times (n-1)$ row-stochastic matrix, then $A \oplus [1]$ is an $n \times n$ row-stochastic matrix with every eigenvalue of $A$. Consequently

$$R_{n-1}(\theta) \le R_n(\theta) \qquad (0 < \theta < \pi).$$

Fix an angle $\theta$ lying in an open cell of $F_n^+$ and put

$$\lambda_n = R_n(\theta)e^{i\theta}.$$

By corollary II.7.4,

$$K_n(\theta)e^{i\theta} \in \Theta_n, \qquad \text{hence} \qquad 0 < K_n(\theta) \le R_n(\theta).$$

Since $\theta/(2\pi)$ is not a Farey endpoint of order $n$, $e^{i\theta}$ is not a root of unity of order at most $n$. Thus proposition II.4.4 gives

$$0 < R_n(\theta) < 1.$$

Suppose first that $\lambda_n \notin \Theta_{n-1}$. Then $\lambda_n$ satisfies precisely the non-inherited radial-maximum hypotheses (II.4.3). By corollary II.6.2, its modulus is the unique solution of the scalar radius equation on this ray. By definition that solution is $K_n(\theta)$, and hence

$$R_n(\theta) = K_n(\theta).$$

Suppose instead that $\lambda_n \in \Theta_{n-1}$. Its membership and the definition of $R_{n-1}$ give

$$R_n(\theta) \le R_{n-1}(\theta),$$

whereas (II.9.7) gives the reverse inequality. Thus

$$R_n(\theta) = R_{n-1}(\theta).$$

Because an open order-$n$ ray cannot be an inherited Farey endpoint, the induction hypothesis applies to this same ray and gives

$$R_{n-1}(\theta) = K_{n-1}(\theta).$$

Now theorem II.8.4 and attainment give the fully explicit chain

$$R_n(\theta) = K_{n-1}(\theta) \le K_n(\theta) \le R_n(\theta).$$

Hence equality holds throughout, and again

$$R_n(\theta) = K_n(\theta).$$

We have proved this identity on every open cell. If $\theta/(2\pi) = p/q \in F_n^+$ is a Farey endpoint, then $q \le n$ and the cyclic permutation matrix of order $q$, padded to order $n$, realizes $e^{i\theta}$. The disk bound in proposition II.4.1 therefore shows that the outer radial modulus on that endpoint ray is $1 = K_n(\theta)$. Extend $R_n$ to $\theta = 0, \pi$ by these outer radial moduli. Then

$$R_n(\theta) = K_n(\theta) \qquad (0 \le \theta \le \pi). \tag{II.9.8}$$

For $n \ge 4$, proposition II.2.4 now joins the open-cell graphs continuously at their Farey endpoints. Thus

$$\zeta_n(\theta) = K_n(\theta)e^{i\theta}, \qquad 0 \le \theta \le \pi,$$

traces exactly the ordered concatenation of the candidate curves $\Gamma_{f,g}^{(n)}$, from 1 to $-1$. Every point of this chain belongs to $\Theta_n$ by corollary II.7.4 and the preceding endpoint argument. Conversely, the equality $R_n = K_n$ says that no point of $\Theta_n$ lies farther out on any upper-half-plane ray. Conjugation gives the corresponding lower chain.

For completeness, define the full-circle function

$$\widehat{K}_n(\phi) = \begin{cases} K_n(\phi), & 0 \le \phi \le \pi, \\ K_n(-\phi), & -\pi \le \phi \le 0. \end{cases}$$

Conjugation symmetry and (II.9.8) show that this is the radial maximum of $\Theta_n$ on every ray. It is positive by attainment and continuous on the circle by proposition II.2.4; at the identified angles $-\pi$ and $\pi$ both values are 1. The set $\Theta_n$ is compact by proposition II.4.1 and star-shaped with respect to the origin by corollary II.4.3. Therefore lemma II.9.1, applied with $S = \Theta_n$ and $r_S = \widehat{K}_n$, gives

$$\Theta_n = \{t\widehat{K}_n(\phi)e^{i\phi} : 0 \le t \le 1, -\pi < \phi \le \pi\}, \qquad \partial\Theta_n = \{\widehat{K}_n(\phi)e^{i\phi} : -\pi < \phi \le \pi\}.$$

Thus the upper and lower boundary arc chains are exactly the topological boundary, not merely the outer points on individual rays, and the first formula is the claimed radial hull. The unit-circle assertion is proposition II.4.4. The discontinuous terminal graph and the segment $[-1, -1/2]$ occur only at order three and were handled directly in proposition II.9.2; the continuous-radius lemma is not invoked for that exceptional base case. □

## II.10 The complete order-seven example

The upper Farey sequence of order seven is

$$0, \frac{1}{7}, \frac{1}{6}, \frac{1}{5}, \frac{1}{4}, \frac{2}{7}, \frac{1}{3}, \frac{2}{5}, \frac{3}{7}, \frac{1}{2}. \tag{II.10.1}$$

For each cell, the table lists the denominator pair $(q, s)$ with $q \le s$, the multiplicity $d = \lfloor 7/q \rfloor$, the exponent $e = s - dq$, and the reduced Ito polynomial. Throughout $\beta = 1 - \alpha$.

| cell | $(q,s)$ | $d$ | $e$ | reduced Ito polynomial |
|---|---|---|---|---|
| $0 \to 1/7$ | $(1,7)$ | 7 | 0 | $(z-\beta)^7 - \alpha^7$ |
| $1/7 \to 1/6$ | $(6,7)$ | 1 | 1 | $z(z^6-\beta) - \alpha$ |
| $1/6 \to 1/5$ | $(5,6)$ | 1 | 1 | $z(z^5-\beta) - \alpha$ |
| $1/5 \to 1/4$ | $(4,5)$ | 1 | 1 | $z(z^4-\beta) - \alpha$ |
| $1/4 \to 2/7$ | $(4,7)$ | 1 | 3 | $z^3(z^4-\beta) - \alpha$ |
| $2/7 \to 1/3$ | $(3,7)$ | 2 | 1 | $z(z^3-\beta)^2 - \alpha^2$ |
| $1/3 \to 2/5$ | $(3,5)$ | 2 | $-1$ | $(z^3-\beta)^2 - \alpha^2 z$ |
| $2/5 \to 3/7$ | $(5,7)$ | 1 | 2 | $z^2(z^5-\beta) - \alpha$ |
| $3/7 \to 1/2$ | $(2,7)$ | 3 | 1 | $z(z^2-\beta)^3 - \alpha^3$ |

### II.10.1 A worked ray: $x = 3/8$

The ray $x = 3/8$ lies in the cell $1/3 < x < 2/5$. Here

$$q = 3, \qquad s = 5, \qquad d = 2, \qquad e = -1,$$

and

$$A = 2\pi\left(3 \cdot \frac{3}{8} - 1\right) = \frac{\pi}{4}, \qquad B = \pi\left(2 - 5 \cdot \frac{3}{8}\right) = \frac{\pi}{8}.$$

The boundary radius is the unique root of

$$\rho^{5/2} \sin\frac{\pi}{4} + \rho^3 \sin\frac{\pi}{8} = \sin\frac{3\pi}{8}, \tag{II.10.2}$$

namely

$$\begin{aligned} &\rho = 0.940100221928822853\ldots, \\ &\alpha = 0.655850787368397414\ldots, \quad \beta = 0.344149212631602586\ldots. \end{aligned} \tag{II.10.3}$$

Thus

$$\lambda = \rho e^{3\pi i/4} = -0.664751241920849 + 0.664751241920849\, i$$

and

$$(\lambda^3 - \beta)^2 = \alpha^2 \lambda. \tag{II.10.4}$$

A six-state sparse stochastic realization, padded by one absorbing state, is

$$A_7 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \beta & 0 & 0 & \alpha & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & \alpha & 0 & \beta & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{II.10.5}$$

The two local 3-cycles have weight $\beta$, while the unique cross cycle has length five and weight $\alpha^2$. Hence

$$\det(tI - A_7) = (t - 1)\big((t^3 - \beta)^2 - \alpha^2 t\big),$$

which verifies (II.10.4) directly.

### II.10.2 The order-seven region

## II.11 Concluding structural remarks

The proof separates three statements that are often conflated.

First, the boundary equation is not a generic determinant identity for stochastic matrices. For $N \geq 4$, a non-inherited radial maximum first becomes an $N$-critical planar multiplier; only then does critical-polygon monodromy force the varying-parameter product and phase identity. Orders at most three are settled by the direct small-order argument instead.

Second, the Part I monodromy theorem is existential. It selects one of the two adapted complex orientations and supplies the corresponding Farey data. The reflection formulas return those data to the original upper-half-plane point; no uniqueness of invariant polygons or stochastic realizers is used.

Third, the scalar Farey geometry is downstream of the Part I input. Once the product, phase identity, and common continuous argument interval are available, the remainder is strict

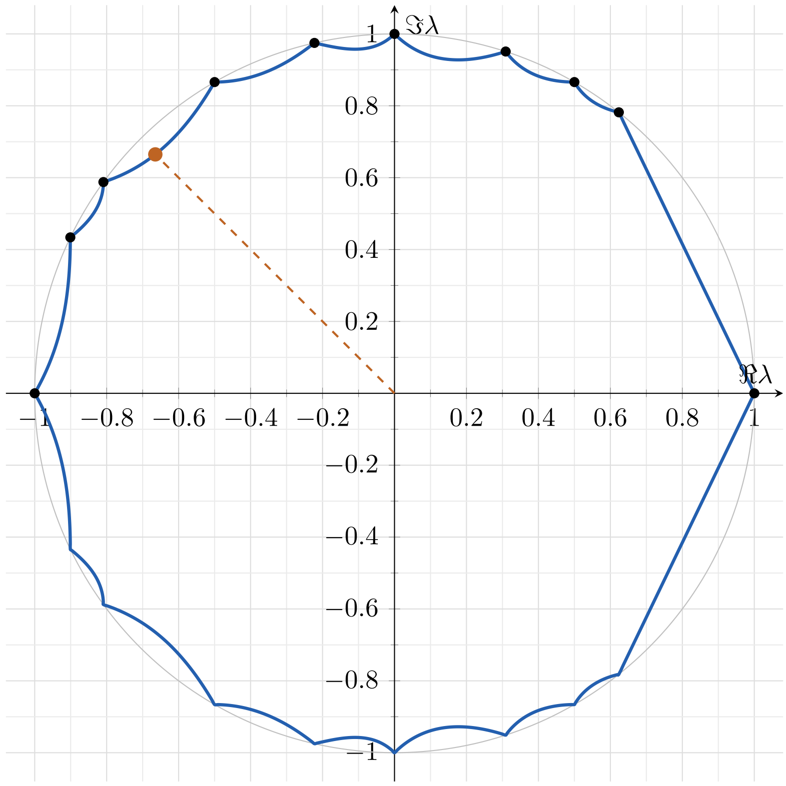


Figure 1: The boundary of $\Theta_7$. The dashed ray is $x = 3/8$, and the marked point is the solution of (II.10.2).

one-dimensional convexity: every nonconstant parameter list lies radially inside the constant-parameter case, and Farey refinement moves the constant-parameter envelope outward. The intrinsic contact and return theory that establishes this input is developed in Part I and culminates in theorem 1.4.

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